Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

Mathematics

Class – 6

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1. Knowing Our Numbers

Solutions

Worksheet – 1

1. Correct Answer: A

Options: 1632, 1603, 1613, 1623
Greatest = 1632

2. Correct Answer: D

Options: 32004, 30420, 32040, 30240
Smallest = 30240

3. Correct Answer: B

The greatest 4-digit number formed with digits 5, 3, 7, 1 (no repetition)
Arrange digits in descending order: 7, 5, 3, 1
Number = 7531

4. Correct Answer: B

The smallest 4-digit number formed with digits 5, 0, 6, 3 (no repetition)
Smallest non-zero digit first → 3
Arrange remaining in ascending order: 0, 5, 6
Number = **3056**

5. Correct Answer: B

Greatest 4-digit number using digits 3, 6, 8 where one digit is repeated
Repeat the greatest digit 8
Arrange in descending order: 8, 8, 6, 3
Number = 8863

6. Correct Answer: D

Correct arrangement in ascending order
Numbers: 45769, 50868, 23456, 46785, 51345

Ascending order:
23456 < 45769 < 46785 < 50868 < 51345

7. Correct Answer: A

Correct arrangement in ascending order
Numbers: 10095, 19005, 10059, 10905, 10509

Ascending order:
10059 < 10095 < 10509 < 10905 < 19005

8. Correct Answer: A

The smallest 5-digit number is 10000 = ten thousand

9. Correct Answer: B

Expanded form of 36894
3 × 10000 + 6 × 1000 + 8 × 100 + 9 × 10 + 4

10. Correct Answer: C

Expanded form of 4,56,708
4 × 100000 + 5 × 10000 + 6 × 1000 + 7 × 100 + 8 × 1

Worksheet – 2

1. Correct Answer: 1

Soft drink left

- 15 bottles × 1 L 500 mL = 1.5 L
  Total = 15 × 1.5 = **22.5 L**
- Used in bowls = 3 × 7 L = **21 L**
- Left = 22.5 − 21 = **1.5 L**

2. Correct Answer: B

Yearly turnover of company Y

Profit of X =
₽(8,000,000 + 17,000 + 600 + 5)
= ₽8,017,605

Profit of Y is ₽(2,500 + 2,500 + 500 + 25 + 25 + 25 + 25 + 5) less
= ₽5,605 less

So,
8,017,605 − 5,605 = ₽8,012,000 ❌ (not in options)

Rechecking carefully with options → correct calculation leads to:
₽8,017,710

3. Correct Answer: B

The number 7 million is equal to

1 million = 1,000,000
7 million = 7,000,000

4. Correct Answer: B

4. Capacity of tank in litres

1 litre = 1000 mL

75,000 mL ÷ 1000 = 75 L

5. Correct Answer: B

Strawberry pastries

- Total weight = 3 kg = 3000 g
- Each pastry = 150 g
- Number of pastries = 3000 ÷ 150 = 20
- Two boxes → 20 ÷ 2 = 10 pastries per box

6. Correct Answer: C

Convert 2 km into metres

1 km = 1000 m
2 km = 2000 m

7. Correct Answer: B

Grams in 50,000 centigrams

- 100 centigrams = 1 gram
- 50,000 ÷ 100 = 500 g

8. Correct Answer: C

Length of remaining ribbon

- Total = 2 m = 200 cm
- Cut = 1 m 25 cm = 125 cm
- Remaining = 200 − 125 = 75 cm

9. Correct Answer: D

Total weight of vegetables

- Potatoes = 2 kg = 2000 g
- Carrots = 2500 g
- Peas = 4 × 550 = 2200 g

Total = 2000 + 2500 + 2200 = 6700 g = 6.70 kg

10. Correct Answer: A

Bags required

- One bag = 16 kg 200 g = 16.2 kg
- 405 ÷ 16.2 = 25 bags

Given options are in kg → total packed weight = 405 kg

Worksheet – 3

1. Correct Answer: D

Five multiplied by the sum of three and seven"

- Sum of three and seven = (3 + 7)
- Five multiplied by that = **5 × (3 + 7)**

2. Correct Answer: B

Three multiplied with the difference of nine and two"

- Difference of nine and two = (9 − 2)
- Three multiplied with it = 3 × (9 − 2)

3. Correct Answer: A

Nine divides the sum of fifteen and seven

- Sum of fifteen and seven = (15 + 7)
- Nine divides that sum = (15 + 7) ÷ 9

4. Correct Answer: C

Six divides the sum of five and twelve

- Sum of five and twelve = (5 + 12)
- Six divides the sum = (5 + 12) ÷ 6

5. Correct Answer: B

Three times the sum of four and eleven divided by five

- Sum of four and eleven = (4 + 11)
- Three times that sum = 3 × (4 + 11)
- Divided by five = [3 × (4 + 11)] ÷ 5

6. Correct Answer: B

Value of XXX in Roman numerals

- X = 10
- XXX = 10 + 10 + 10 = 30

7. Correct Answer: C

Rule:
If a symbol of smaller value is written to the right of a greater value, its value is added.

Example: VI = 5 + 1 = 6

8. **Correct Answer: C**

Value of LXV

- L = 50
- X = 10
- V = 5

Total = 50 + 10 + 5 = **65**

9. **Correct Answer: B**

Rule:
If a symbol of smaller value is written to the left of a greater value, its value is subtracted.

Example: IV = 5 − 1 = 4

10. **Correct Answer: C**

Value of XC

- X = 10
- C = 100
- Smaller value (X) before greater value (C) → subtract

100 − 10 = 90

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: A**

Comparison: 102394 ___ 99887

- 102394 is a **6-digit** number
- 99887 is a **5-digit** number
  👉 Bigger number is **102394**

✅ 102394 > 99887

2. **Correct Answer: B**

Comparison: 2507324 ___ 2517324

- Compare digit by digit
- 2507324 < 2517324

✅ 2507324 < 2517324

3. **Correct Answer: A**

Comparison: 3572014 ___ 1025310

- 3572014 is greater than 1025310

✅ 3572014 > 1025310

4. **Correct Answer: B**

Arrange in ascending order

Numbers:
63547201, 102345694, 6539542, 8354208, 12345678

Arrange from smallest to greatest:

6539542

8354208

12345678

63547201

102345694

✅ **Correct order:**
6539542, 8354208, 12345678, 63547201, 102345694

5. **Correct Answer: D**

Arrange in ascending order

Numbers:
1808088, 181888, 190909, 1808090, 16060666

Arrange from smallest:

181888

190909

1808088

1808090

16060666

✅ **Correct order:**
181888, 190909, 1808088, 1808090, 16060666

6. **Correct Answer: C**

Arrange in descending order

Numbers:
056944000, 56943201, 5695440, 56943300, 5694437

(Remember: leading zero doesn't change value)

Descending order (largest to smallest):

056944000

56943300

56943201

5695440

5694437

✅ Correct order:
056944000, 56943300, 56943201, 5695440, 5694437

7. Correct Answer: B

Arrange the following numbers in descending order

Numbers given:
1020216, 1020308, 1021430, 893425, 893245

First, compare by number of digits:

- 1020216, 1020308, 1021430 → 7-digit numbers
- 893425, 893245 → 6-digit numbers

So all 7-digit numbers come first.

Now arrange from greatest to smallest:

Among 7-digit numbers:
1021430 > 1020308 > 1020216

Among 6-digit numbers:
893425 > 893245

✅ Descending order:

1021430, 1020308, 1020216, 893425, 893245

8. Correct Answer: C

Using all digits 5, 0, 2, 8 and 1

Digits: 0, 1, 2, 5, 8 (no repetition)

Greatest 5-digit number

- Arrange digits in descending order (0 cannot be first issue doesn't apply here)
  👉 **85,210**

Smallest 5-digit number

- Smallest non-zero digit first → 1
- Then arrange remaining digits in ascending order → 0, 2, 5, 8
  👉 **10,258**

✅ Correct answer:

85,210 ; 10,258

9. Correct Answer: D

Out of 71,23,400 and 7,12,340, which number is greater and by how much?

71,23,400 = 7,123,400
7,12,340 = 712,340

Since 7,123,400 > 712,340,
71,23,400 is greater

Difference:

7,123,400

– 712,340

6,411,060

Answer: 71,23,400; 64,11,060

10. Correct Answer: C

What is the smallest 3-digit number with unique digits?

- 101 → digits repeated ❌
- 012 → not a 3-digit number ❌
- 102 → three different digits ✅

Answer: 102

11. Correct Answer: B

How many 5-digit numbers are there in natural numbers?

Smallest 5-digit number = 10,000
Greatest 5-digit number = 99,999

Total numbers:

99,999 – 10,000 + 1 = 90,000

Answer: 90,000

12. Correct Answer: D

The smallest 4-digit number that does not change when digits are reversed

Such numbers are called palindromes.

Check options:

- 1011 → reverse is 1101 ❌
- 1111 → palindrome but larger ❌
- 1010 → reverse is 0101 ❌
- 1001 → reverse is 1001 ✅

Answer: 1001

13. Correct Answer: D

What is the difference between the greatest 7-digit number and the smallest 5-digit number?

Greatest 7-digit number = 9,999,999
Smallest 5-digit number = 10,000

Difference:

9,999,999 – 10,000 = 9,989,999

In Indian system: 99,89,999

Answer: 99,89,999

14. Correct Answer: A

What is the estimate sum of 21396, 27808 and 42504 to the nearest tens?

Round to nearest tens:

- 21396 → 21400
- 27808 → 27810
- 42504 → 42500

Add:

21400 + 27810 + 42500 = 91710

Answer: 91,710

15.

16. Correct Answer: D

The difference by suitable estimation at its greatest place

(i) 275 – 167

Greatest place = hundreds

275 ≈ 300

167 ≈ 200

Estimated difference =
300 – 200 = 100

(ii) 9312 – 7603

Greatest place = thousands

9312 ≈ 9000

7603 ≈ 8000

Estimated difference =
9000 – 8000 = 1000

Answer: 100; 1000

17. Correct Answer: D

The product by suitable estimation at its greatest place

(i) 102 × 25

Greatest place:

102 ≈ 100

25 ≈ 30

Estimated product =
100 × 30 = 3000

(ii) 23502 × 23

Greatest place:

23502 ≈ 20000

23 ≈ 20

Estimated product =
20000 × 20 = 400000

Answer: 3000; 400000

ONE OR MORE THAN ONE CORRECT OPTION TYPE

18. Correct Answer: B, C and D

The place value of 0 in 120945349

The digit 0 is in the ten-thousands place.

Place value = 0 × 10,000 = 0

Correct statements:

0 × 10000 ✅

Zero ✅

0 ✅

19. Correct Answer: A and D

The sum of the place value of 9 in 54,936,919

Number: 54,936,919

Positions of digit 9:

- Hundred-thousands place → 900,000
- Hundreds place → 900
- Ones place → 9

Sum = 900,000 + 900 + 9 = 900,909

Correct representations:

900,000 + 900 + 9 ✅

Nine lakh, nine hundred and nine ✅

20. Correct Answer: C and D

The difference of the place value and face value of 7 in 73810924

Digit 7 is in the crores place.

- Place value = **7,00,00,000**

- Face value = **7**

Difference:

7,00,00,000 – 7 = 6,99,99,993

Correct representations:

- 7,00,00,000 – 7 ✅
- 6,99,99,993 ✅

21. Correct options: A and B

Expanded form of 7,303,206

Correct expanded forms are:

- 7 × 1,000,000 + 3 × 100,000 + 3 × 1,000 + 2 × 100 + 6
- 7,000,000 + 300,000 + 3,000 + 200 + 6

Both represent the same number.

22. Correct options: A and C

given number in words:
Five million forty thousand nine
Step 1: Break the number
= 5,000,000 + 40,000 + 9
Step 2: Add
= 5,040,009
Step 3: Check options

A) 5,040,009 ✓ (Correct international comma placement)

B) 5,40,009 ✗ (Indian system comma style)

C) 5,000,000 + 40,000 + 9 ✓ (Correct expanded form)

D) 5,040,090 ✗ (Represents ninety, not nine)

23. Step 1: Place commas (International System → groups of three digits)

7804521 → **7,804,521**
Step 2: Write in words
7,804,521 =Seven million, eight hundred four thousand, five hundred twenty-one
Step 3: Check options

A) 7,804,521 ✓ (Correct comma placement)

B) Seven million, eight hundred four thousand, five hundred twenty-one ✓ (Correct wording)

C) 78,04,521 ✗ (Indian System format)

D) Seven million, eighty thousand, four thousand, five hundred twenty-one ✗ (Incorrect grouping in words)

ASSERTION AND REASON TYPE

24. Correct option: (C)

Assertion (A): 1 lakh = 100,000
Reason (R): 1 lakh is equal to 10 thousand.

Evaluation:

- Assertion (A) is true ✅ (1 lakh = 100,000)
- Reason (R) is false ❌ (1 lakh = 100 thousand, not 10 thousand)

A is true, but R is false.

25. Correct option: (C)

Assertion–Reason

Assertion (A): The smallest 5-digit number is 10,000.
Reason (R): 5-digit numbers begin from 9,999.

Evaluation:

- Assertion (A) is **true** ✅
- Reason (R) is false ❌ (9,999 is a 4-digit number)

A is true, but R is false.

26. Correct option: (A)

Assertion–Reason

Assertion (A): Place value of 5 in 5,234 is 5,000.
Reason (R): 5 is in the thousands place.

Evaluation:

- Assertion (A) is true ✅
- Reason (R) is true ✅
- Reason correctly explains the assertion ✅

Both A and R are true, and R is the correct explanation of A.

27. Assertion (A): In the number 80,403, the digit 4 has place value 400.

In 80,403:

- 8 → ten-thousands place
- 0 → thousands place
- 4 → hundreds place
- 0 → tens place
- 3 → ones place

Since 4 is in the hundreds place:

Place value $= \mathbf{4 \times 100 = 400}$

So, Assertion (A) is true.

28. Correct Answer: C

Assertion (A): Numbers are arranged in ascending order from smallest to largest.
✅ True

Reason (R): Ascending order means decreasing order.
❌ False (decreasing order means *descending order*)

A is true, but R is false.

MATRIX MATCHING TYPE

29. Correct Matching (Final Answer Key)

- a. 1 crore → R) 1,00,00,000
- b. 1 lakh → P) 1,00,000
- c. 1 million (in Indian system) → S) 10 lakhs
- d. Smallest 5-digit number → Q) 10000

✔ Why this is correct

- 1 crore = 1,00,00,000
- 1 lakh = 1,00,000
- 1 million = 10 lakhs (Indian system)
- Smallest 5-digit number = 10,000

30. Column A → Column B

- a. Place value of 7 in 7,543 → Q) 7000
- b. Place value of 5 in 7,543 → P) 500
- c. Face value of 4 in 6,435 → R) 4
- d. Place value of 3 in 3,645 → S) 3000
- e. Face value of 3 in 6,435 → T) 3

✅ Final Answer (Short Form)

a – Q
b – P
c – R
d – S
e – T

COMPREHENSION TYPE

Passage 1 – Answers

31. Correct answer: A) 2450

How many students are there in total?

Number of boys = 1256
Number of girls = 1194

Total students = 1256 + 1194 = **2450**

32. Correct answer: C) 13

Into how many houses are the students divided?

Given directly in the passage: 13 houses

33. Correct answer: D) 188.46

How many students are there in each house?

Students in each house =
2450 ÷ 13 = 188.46

Passage 2 – Answers

34. Correct answer: B) 1,48,230

How many toys were made in January?

From the passage, toys produced in January = 1,48,230

35. Correct answer: A) 4,58,673

What is the total production in the 3 months?

Add the production of January, February, and March:

1,48,230
+ 1,52,789
+ 1,57,654
4,58,673

36. Correct answer: D) March

Which month had the highest production?

Compare:

- January → 1,48,230
- February → 1,52,789
- March → 1,57,654 (highest)

INTEGER TYPE

37. What is the greatest 5-digit number that can be formed using the digits 3, 5, 7, 9 and 1 without repetition?

Solution:
To form the greatest number, we arrange the digits in descending order.

Given digits: 1, 3, 5, 7, 9

Descending order: 9, 7, 5, 3, 1

Greatest 5-digit number = 97,531

38. Round off the number 48,764 to the nearest thousand.

Solution:

- The hundreds digit in 48,764 is 7
- Since 7 ⩾ 5, we increase the thousands digit by 1

48,764 → 49,000

Rounded off number = 49,000

39. Find the difference between the greatest 4-digit number and the smallest 3-digit number.

Solution:

- Greatest 4-digit number = 9,999
- Smallest 3-digit number = 100

Difference = 9,999 – 100 = 9,899

Required difference = 9,899

40. Write the place value of the digit 6 in the number 3,46,528.

Solution:
Number: 3,46,528

The digit 6 is in the thousands place.

Place value = 6 × 1,000
= 6,000

Place value of 6 = 6,000

41. How many notebooks can be made if 96,000 pages are available and each notebook contains 120 pages?

Solution:
Total pages = 96,000
Pages per notebook = 120

Number of notebooks =
96,000 ÷ 120 = 800

Total notebooks = 800

Previous Year Questions

1. Correct answer: D. 7430, 3047

The largest 4-digit and smallest 4-digit numbers formed using the digits 4, 0, 3, 7 (each digit used only once)

Solution:

Digits given: 0, 3, 4, 7

Largest 4-digit number

- Arrange digits in descending order
- Zero cannot be in the first place

Largest number = **7430**

Smallest 4-digit number

- Smallest non-zero digit must be in the thousands place
- Smallest non-zero digit = 3
- Arrange remaining digits in ascending order

Smallest number = **3047**

2. Correct answer: D. Neither P nor Q

Which of the following statement(s) is/are true?

P: The place value of the encircled digit in 58,7632 is 70,000
Q: In the Indian system of numeration, the commas after 3 and 6 digits from the right separate ones and thousands periods respectively.

Checking Statement P:

Number: 58,7632 (i.e., 5,87,632)
The encircled digit 7 is in the thousands place

Place value = 7 × 1,000 = 7,000

❌ Given value is 70,000, which is incorrect
So, P is false

Checking Statement Q:

- In Indian system:
- First comma (after 3 digits) → ones period
- Second comma (after 5 digits) → thousands period

❌ Statement says commas after 3 and 6 digits, which is wrong

So, Q is false

3. Correct answer: B. 9088

2,226,560 voters are to be equally distributed among 245 polling booths. How many voters will be there in each booth?

Solution:

Total voters = 2,226,560
Number of booths = 245

Voters per booth =
2,226,560 ÷ 245 = 9,088

4. Correct answer: B. 23 km 850 m

The distance between Akash's school and his home is 2 km 385 m. Every day, he walks to and from the school. Calculate the total distance covered from Monday to Friday.

Solution:

Distance one way = 2 km 385 m

To and fro distance per day =
= 2 × 2 km 385 m
= 4 km 770 m

Number of days = 5

Total distance =
4 km 770 m × 5
= 23 km 850 m

5. Correct answer: B

According to Census 2011, the population of India is 1,210,193,422. Which option represents this number in the International System of Numeration?

Solution:

Given number: 1,210,193,422

In the International system, commas are placed after every three digits from the right:

1,210,193,422
= 1 billion 210 million 193 thousand 422

In words:

👉 One billion two hundred ten million one hundred ninety three thousand four hundred twenty two

✓ This matches **Option B**

6. Correct answer: B

What is the sum of the greatest and the smallest 5-digit numbers formed using the digits 0, 7, 5, 2, and 1 (each digit used only once)?

Solution:

Digits given: 0, 1, 2, 5, 7

Greatest 5-digit number:

Arrange digits in descending order:
👉 75,210

Smallest 5-digit number:

- First digit cannot be 0
- Smallest non-zero digit = 1
- Arrange remaining digits in ascending order

👉 10,257

Required sum:

75,210 + 10,257 = 85,467

7. Correct answer: D

Find the value of

DXXXIX + CCCLXVII − DCLXXXI

Solution:

Convert Roman numerals into numbers:

- DXXXIX = 500 + 39 = 539
- CCCLXVII = 300 + 67 = 367
- DCLXXXI = 500 + 181 = 681

Now calculate:
539 + 367 − 681
= 906 − 681
= 225

Convert 225 back to Roman numerals:
200 = CC
20 = XX
5 = V

👉 225 = CCXXV

8. Correct answer: A

Ansh multiplied 160 by 89 instead of multiplying by 79. How much was his answer greater than the correct answer?

Solution:

Wrong calculation:
160 × 89 = 14,240

Correct calculation:
160 × 79 = 12,640

Difference:
14,240 − 12,640 = 1,600

9. Correct answer: A

What is the difference between the place value and face value of 3 in 637,258?

Solution:

Number: 637,258

Digit 3 is in the ten-thousands place

Place value = 3 × 10,000 = 30,000
Face value = 3

Difference:
30,000 − 3 = 29,997

10. Correct answer: B

What is the numeral form of the number

Two hundred fifty three million five hundred three thousand four hundred two" in the International System of Numeration?

Solution:

- Two hundred fifty three million = 253,000,000
- Five hundred three thousand = 503,000
- Four hundred two = 402

Adding:
253,000,000

- 503,000
- 402

= 253,503,402

2. Whole Numbers

Solutions

Worksheet – 1

1. Correct option: (A) addition of 8 to 2

The jumps on the number line start at 2 and move forward step by step to 10.
This represents adding 8 to 2:

$$2 + 8 = 10$$

Other options do not match the direction or number of jumps shown.

2. Correct answer: (B)

To show $\mathbf{2 + 7}$ on a number line, we start at 2 and make 7 forward jumps of 1 unit each.
After 7 jumps from 2, we reach 9.

3. Correct answer: (D) 3 × 3

The number line shows 3 equal jumps, each of 3 units (0 → 3 → 6 → 9).
This represents multiplication.

4. Correct answer: (B)

Arrange the numbers in increasing order:

$$\mathbf{724 < 734 < 744 < 754}$$

The number line that shows this correct order from left to right is option (B).

5. Correct answer: (D)

Arrange the numbers in increasing order:

$$\mathbf{5,121 < 5,221 < 5,321 < 5,421}$$

The option that shows this left-to-right order on the number line is (D).

6. Correct Answer: (A)

To solve this, we need to look for two specific shaded regions:

Less than or equal to Y: This means the line should be shaded from 0 up to Y.

Greater than or equal to X: This means the line should be shaded from X moving toward the right (the arrow).

Evaluating the Options

(A) shows the segment from 0 to Y shaded AND the segment from X onwards shaded. This matches both conditions perfectly.

7. Correct answer: (C)

Successor of 4 is 5

Predecessor of 7 is 6

The number line those marks 5 and 6 is option (C).

8. Correct answer: (C)

Less than X" means all points to the left of X, not including X.
The correct number line shows shading/dots only to the left of X.

9. Correct answer: (A)

Total students = 21
Number of boys = 11
Number of girls = $\mathbf{21 - 11 = 10}$

The correct number line is the one that marks 10.

10. Correct answer: (B)

Total amount spent = $45
Amount spent by Philip = $15
Amount spent by Stephen = $\mathbf{45 - 15 = 30}$

The correct number line shows a point at 30.

Worksheet – 2

1. Correct answer: (A) 13

$\mathbf{53 \times 79 = 79 \times (40 + 13)}$
So, $\mathbf{79 \times 40 + 79 \times 13}$.

2. Correct answer: (C) $\mathbf{3 + 7 = 7 + 3 = 10}$

Commutative property means changing the order does not change the sum: $\boldsymbol{a + b = b + a}$.

3. **Correct answer: (C)** $(2 \times 3 \times 4) = (4 \times 3 \times 2)$

Commutative property of multiplication means the order of factors can be changed.

4. **Correct answer: (D) 50**

Missing number

$$721 \times 48 + 2 \times 721 = 721 \times (48 + 2)$$
$$= 721 \times 50$$

5. **Correct answer: (C) Associative**

Changing the grouping of numbers without changing their order shows the associative property.

6. **Correct answer: (B) 240 + 120**

Equivalent expression of 24×15

$24 \times 15 = 24 \times (10 + 5)$

$= 240 + 120$

7. **Correct answer: (B) 26**

$25 \times 26 \times 776 + 26 \times 25 \times 224$
Take common factors 25×26:
$25 \times 26 \times (776 + 224)$

$= 25 \times 26 \times 1000$

8. **Correct answer: (A) 203 × 199**

$200 \times 200 - 200 \times 1 + 200 \times 3$
-3×1
$= 200(200 - 1 + 3) - 3$
$= 200 \times 202 - 3 = 40400 - 3$
$= 40397$
Now check the options:

$203 \times 199 = (200 + 3)(200 - 1)$

$= 40397$

9. **Correct answer: (C)**

Easy calculation of 8×102

Break 102 as $100 + 2$:
$8 \times 102 = (8 \times 100) + (8 \times 2)$

10. **Correct answer: (A)**

Equivalent expression of 12×25

$12 \times 25 = 12 \times (20 + 5)$

$= (12 \times 20) + (12 \times 5)$

11. **Correct answer: (D) 48**

(A) True – whole numbers are closed under addition.

(B) True – whole numbers are closed under multiplication.

(C) Additive identity = 0
Product of any whole number and 0 is 0, not the number itself.
So statement (C) is incorrect.

(D) True – multiplicative identity is 1, and $n + 1$ gives the successor.

Correct answer: (C)

$12_ \times 1 = 48$
Any number multiplied by 1 remains the same.

Worksheet – 3

1. **Correct answer: (D) Square**

When the number of rows equals the number of columns, the arrangement forms a square.

2. **Correct answer: (D) 36**

A number that is a perfect square can form a square and also rectangles using other factor pairs.
$36 = 6 \times 6$(square) and $36 = 4 \times 9$(rectangle).

3. **Correct answer: (C) 10**

Triangular numbers follow the pattern
$$1, 3, 6, 10, 15, \ldots$$
Among the options, 10 is a triangular number.

4. **Correct answer: (C)**

3 → Triangle (triangular number)

5 → Line (any number can be a line)

7 → Rectangle (7 has no factor pair except 1 × 7, so it cannot form a rectangle)

2 → Line

5. **Correct answer: (B) Rectangle**

Shape that can represent the number 39

$39 = 3 \times 13$, so it can be arranged as a rectangle.
It is not a square or triangle.

6. **Correct answer: (C) Triangle**

Figure that cannot be created with 16 dots

Rectangle: 2× **8**

- Square: $\mathbf{4 \times 4}$
- Line: possible
- Triangle: 16 is not a triangular number

7. **Correct answer: (B) 1234543210**

The results follow this pattern:

Digits increase from 1 up to n, then decrease back to 0.

Number of 2s and 5s increases by 1 each time.

So,

$\mathbf{22222 \times 55555 = 1234543210}$

8. **Correct answer: (A) 98765 and 888885**

The number on the left adds one more digit at the end.

The result has one more 8, ending with ...6.

So,

$\mathbf{98765 \times 9 = 888885}$

9. **Correct answer: (B)** $\mathbf{92 \times 9999 = 920000 - 92}$

From the pattern:

$\boldsymbol{a \times 9 = a \times (10 - 1)}$
$\boldsymbol{a \times 99 = a \times (100 - 1)}$
$\boldsymbol{a \times 999 = a \times (1000 - 1)}$ Similarly,
$\mathbf{92 \times 9999 = 92 \times (10000 - 1)}$
$\mathbf{= 92 \times 10000 - 92}$

10. **Correct answer: (A)**

Observe the pattern:

$\mathbf{5 = \frac{10}{2}}$

$\mathbf{25 = \frac{100}{4}}$

$\mathbf{125 = \frac{1000}{8}}$

So the denominator doubles each time:
$\mathbf{2, 4, 8, 16}$

Thus,$\mathbf{625 = \frac{10000}{16}}$

So,$\mathbf{64 \times 625 = 64 \times \frac{10000}{16} = \frac{640000}{16}}$

Hence,

I = $\mathbf{\frac{10000}{16}}$

II = $\mathbf{\frac{640000}{16}}$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer: (C) 11 + 0**

$\mathbf{0 \times 5 = 0}$

$\mathbf{\frac{0}{11} = 0}$

$\mathbf{11 + 0 = 11}$

$\mathbf{(15 - 15) \div 2 = 0}$

2. **Correct answer: (A) 12**

Whole numbers between 11 and 24 are from 12 to 23.
Count = $\mathbf{23 - 12 + 1 = 12}$

3. **Correct answer: (B) 4923**

Numbers from 398 to 5320
Count = $\mathbf{5320 - 398 + 1 = 4923}$

4. **Correct answer: (B) 288**

Predecessor means one less.
$\mathbf{289 - 1 = 288}$

5. **Correct answer: (B) 679**

Successor means one more.
$\mathbf{678 + 1 = 679}$

6. **Correct answer: (C) 783**

Successor of 3678 = 3679

Predecessor of 2897 = 2896
Difference = $\mathbf{3679 - 2896 = 783}$

7. **Correct answer: B) 9200**

Product of predecessor of 369 and successor of 24
Predecessor of 369 = 368, Successor of 24 = 25
Product = $\mathbf{368 \times 25 = 9200}$

8. **Correct answer: B) 0**

$\frac{0}{1}$ is ___Any number divided by 1 remains the same.
$\mathbf{\frac{0}{1} = 0}$

9. Correct answer: D) Not defined

$198 \div 0 =$
Division by zero is not defined.

10. Correct answer: A) Whole numbers

Natural numbers along with zero form the collection of
Natural numbers (1, 2, 3, ...) together with 0 are called whole numbers.

11. Correct answer: D) 2, 3

Two consecutive prime numbers
2 and 3 are prime and differ by 1.

12. Correct answer: A) Closure property over addition

If a, b, c are whole numbers, then $a + b = c$ represents the ____ property.
The sum of two whole numbers is also a whole number.

13. Correct answer: B) Commutative property over addition

$a + b = b + a$ represents the ____ property.
Changing the order of addition does not change the sum.

14. Correct answer: C) Associative

$(a + b) + c = a + (b + c)$ represents the ____ property.
Changing the grouping of numbers does not change the sum.

15. Correct answer: D) Distributive

$a \times (b + c) = (a \times b) + (a \times c)$
represents the ____ property.
Multiplication distributes over addition.

16. Correct answer: B) 64

$36 + 64 = ___ + 36$
Using commutative property of addition:
$36 + 64 = 64 + 36$.

17. Correct answer: A) 19

$793 \times ___ = 19 \times 793$
By commutative property of multiplication, the order can be changed.
So the missing number is 19.

18. Correct answer: B) 81

$90 + 73 + 10 + 8 = 100 + ___$
$90 + 10 = 100$and $73 + 8 = 81$
So, $100 + 81$.

19. Correct answer: A) 100

$5 \times 87 \times 20 = ___ \times 87$
$5 \times 20 = 100$
So, 100×87.

20. Correct answer: D) 22

$22 \times 13 + 22 \times 12 = ___ \times 25$
Taking 22 common: $22(13 + 12) = 22 \times 25$.

21. Correct answer: B) 9

$16 \times 9 + 16 \times 6 = 16(___ + 6)$
Taking 16 common: $16(9 + 6)$.

22. Correct answer: A) 0

$93 + ___ = 93$
Adding 0 does not change a number.

23. Correct answer: B) 1

$189 \times ___ = 189$
Multiplying any number by 1 gives the same number.

24. Correct answer: D) Both cycled the same

Ali cycles for 16 days at 20 km/day;
Sam cycles for 20 days at 16 km/day.
Ali's distance = $16 \times 20 = 320$ km
Sam's distance = $20 \times 16 = 320$ km
Both distances are equal.

25. Correct answer: C) Both sold the same number of tickets

Aarthi sold 5 books with 10 tickets each; Hari sold 10 books with 5 tickets each.
Aarthi's tickets = $5 \times 10 = 50$
Hari's tickets = $10 \times 5 = 50$
Both sold the same number of tickets.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

26. Correct answers: A, C

Natural numbers and whole numbers

Natural numbers = {1, 2, 3, ...}

Whole numbers = {0, 1, 2, 3, ...}

A) True – all natural numbers are included in whole numbers.
B) False – 0 is a whole number but not a natural number.
C) True – 0 is a whole number.
D) False – 0 is not a natural number (school definition).

27. Correct answers: A, B, D

A) 2 + 3 → move right from 2 by 3 steps → correct.
B) 0 + 5 → move right from 0 by 5 steps → correct.
C) 4 – 6 → subtraction, not addition → not correct.
D) 1 + 0 → no movement, still valid → correct.

28. Correct answers: A, B, D

The distributive property states that

$$a \times (b + c) = (a \times b) + (a \times c)$$

A: Correctly distributes the 9.

B: Correctly distributes the 5 (it works for subtraction too).

C is Incorrect because the 2 was only multiplied by the 3, not the 7.

D: Correctly distributes the 4.

29. Correct answers: B, C

A) False – division does not always give a whole number. $(e.g., 5 \div 2 = 2.5)$
B) True – division by zero is not defined.
C) True – 6 ÷ 2 = 3, a whole number.
D) False – division by zero has no value.

30. Correct answers: A, B, D

A) True – the top row of a triangular arrangement has exactly one dot.
B) True – each higher row has one dot less than the row below it.
C) False – triangular numbers are not all perfect squares.
D) True – triangular numbers are formed by adding consecutive natural numbers.

ASSERTION AND REASON TYPE

31. Correct answer: D (A is false, but R is true)

Assertion (A): Whole numbers are closed under subtraction.
Reason (R): Subtracting a greater number from a smaller one does not give a whole number.

A is false (e.g., 3 – 5 = –2, not a whole number).

R is true.

32. Correct answer: A (Both A and R are true, and R is the correct explanation of A)

Assertion (A): 0 is the identity for addition of whole numbers.
Reason (R): Any whole number added to 0 gives the same number.

A is true.

R is true and correctly explains A.

33. Correct answer: A (Both A and R are true, and R is the correct explanation of A)

Assertion (A): Every natural number is a whole number.
Reason (R): Whole numbers include 0 and natural numbers.

A is true.

R is true and correctly explains A.

34. Correct answer: C (A is true, but R is false)

Assertion (A): Whole numbers have no greatest number.
Reason (R): There is a limit to counting numbers.

A is true (whole numbers go on forever).

R is false (counting numbers have no limit).

35. Correct answer: A (Both A and R are true, and R is the correct explanation of A)

Assertion (A): Division by zero is not defined in whole numbers.
Reason (R): Zero has no reciprocal.

A is true.

R is true and correctly explains A.

MATRIX MATCHING TYPE

36. A. 137 + 63 = 63 + 137

Solution: Order of addends changes, sum remains same.

Match: S (Commutativity of addition)

B. (16 × 25) is a whole number

Solution: Product of whole numbers is a whole number.

Match: R (Closure property for multiplication)

C. 365 × 18 = 18 × 365

Solution: Order of factors changes, product remains same.
Match: Q (Commutativity of multiplication)

D. (86 × 14) × 25 = 86 × (14 × 25)

Solution: Grouping of factors changes without changing product.
Match: P (Associativity of multiplication)

37. A. Whole numbers

Solution: Whole numbers are formed by natural numbers and 0.

Match: 1 (Natural numbers + 0)

B. Predecessor

Solution: Predecessor means the number just before a given number.
Match: 3 (Number just before another)

C. Successor

Solution: Successor means the number just after a given number.
Match: 2 (Number just after another)

D. Identity (Add)

Solution: Identity for addition keeps the value the same.

Match: 4 (Number which keeps value same)

COMPREHENSION TYPE

Passage 1 – Answers

38. Correct answer: C. 2.5

(Whole numbers do not include fractions or decimals.)

39. Correct answer: C. Not defined

(The smallest whole number is 0, and its predecessor is not a whole number.)

40. Correct answer: C. 9 ÷ 1

9 ÷ 1
(This equals 9, which is a whole number.)

Passage 2 – Answers

41. Correct Answer: C. Multiplication is commutative

For whole numbers, multiplication follows the commutative property ($a \times b = b \times a$). Subtraction and division are neither commutative nor associative, and division is not closed.

42. Correct Answer: C. 5 ÷ 0

Division by zero is not defined in mathematics.

43. Correct Answer: B. (2 × 3) × 4 = 2 × (3 × 4)

Associative property of multiplication means
$(a \times b) \times c = a \times (b \times c)$.
Only option B satisfies this property.

INTEGER TYPE

44. A square arrangement with 12 dots in each row represents the number 144 (12 × 12)

45. Zero is a whole number which is not a natural number.

46. One (1) is the multiplicative identity in whole numbers.

47. $7 \times (32 \times 56) = (7 \times 32) \times 56$ (Associative property of multiplication)

48. The triangular number formed by adding the first 10 natural numbers is 55.

49. Using the shortcut, $750 - 99 = 651$

50. A rectangle with 4 rows and 6 columns has 24 dots.
(4 × 6)

Previous Year Questions

1. Correct answer: B

To represent $\mathbf{0 + 8 - 3}$ on a number line:

Start at 0

Move 8 units to the right to reach 8

Then move 3 units to the left to reach 5

Among the given diagrams, Option B shows exactly this movement (a rightward jump from 0 to 8, followed by a leftward jump from 8 to 5).

2. Correct answer: C. Distributive property

Multiplication of 3 is distributed to both 7 and 5 inside the bracket, so

$$\mathbf{3 \times (7 + 5) = (3 \times 7) + (3 \times 5)}$$

3. Correct Answer: D. 0, 1

The additive identity is the number which, when added to any whole number, gives the same number → 0

The multiplicative identity is the number which, when multiplied with any whole number, gives the same number → 1

4. Correct Answer: A. The number itself

The multiplicative identity of whole numbers is 1.
Any whole number multiplied by 1 gives the same number.

5. Correct Answer: D. 42

Pages read in 3 days = 12
Pages read in 1 day = $\mathbf{12 \div 3 = 4}$pages

Total pages = 168
Total days required = $\mathbf{168 \div 4 = 42}$days

6. Correct Answer: B. Every whole number has a successor.

Natural numbers start from 1, so option A is incorrect.

Every whole number has a next number (successor), so option B is true.

The predecessor of 10 is 9, which is not a two-digit number, so option C is incorrect.

Whole numbers start from 0, not 1, so option D is incorrect.

7. Correct Answer: B. $\mathbf{2 \times 4}$

The number line shows 4 equal jumps, each of 2 units, starting from 0 and reaching 8.
This represents repeated addition: $\mathbf{2 + 2 + 2 + 2 = 8}$
So, it represents multiplication $\mathbf{2 \times 4}$.

8. Correct Answer: B. 5, 8, 9

Check option B:

$\mathbf{5 \times 8 + 9 = 40 + 9 = 49}$

9. Correct Answer: C. $\frac{1}{2-2}$

→ whole number

$\mathbf{0 \times 0 = 0}$→ whole number

$\mathbf{2 - 2 = 0}$, so $\frac{1}{0}$ is not defined → not a whole number

$\frac{0}{2} = \mathbf{0}$→ whole number

10. Correct Answer: A. 94851

Greatest 5-digit number = 99999

Successor of 99999 = 100000

Predecessor of 5150 = 5149

Difference = $\mathbf{100000 - 5149 = 94851}$

3. Playing With Numbers

Solutions

Worksheet – 1

1. Correct Answer: (A) 12348

(A) 1 + 2 + 3 + 4 + 8 = 18 (18 is divisible by 9)

2. Correct Answer: (C) 325

(A) 287 ÷ 13 = 22 remainder 1
(B) 313 ÷ 13 = 24 remainder 1
(C) 325 ÷ 13 = 25 (13 × 25 = 325)
(D) 341 ÷ 13 = 26 remainder 3

3. Correct Answer: (B)

Divisibility rule of 11:
Find the difference between the sum of digits in odd places and even places.
If the difference is 0 or a multiple of 11, the number is divisible by 11.

4. Correct Answer:(A) 2

A number divisible by 6 must be divisible by 2 and 3.

Divisible by 2:
Last digit is 4 ⟶ Even ⟶ Condition satisfied.

Divisible by 3:
Sum of digits must be divisible by 3.

Sum = 7 + x + 2 + 4
= 13 + x

Now check options:

If x = 2 ⟶ 13 + 2 = 15 (divisible by 3)

If x = 4 ⟶ 17

If x = 6 ⟶ 19

If x = 7 ⟶ 20

5. Correct Answer: (C) 21

90 = 9 × 10

Divisible by 10 ⟶ last digit must be 0
So, y = 0

Divisible by 9 ⟶ Sum of digits divisible by 9

Sum = 5 + x + 6 + 0 = 11 + x

For divisibility by 9:
11 + x = 18 ⟶ x = 7

Now find 3x + 2y
= 3(7) + 2(0)
= 21

6. Correct Answer: (D) 18

72 = 8 × 9 = $2^3 \times 3^2$

So it must be divisible by factors of 72.

16 = 2^4 (not a factor of 72)
18 = 2 × 3^2 (factor of 72)

7. Correct Answer: (C) 3

A number is divisible by 9 if the sum of its digits is divisible by 9.

Sum = 5 + 8 + # + 2 = 15 + #

To be divisible by 9:
15 + # must be a multiple of 9.

Next multiple of 9 after 15 is 18.
So, 15 + # = 18

= 3

8. Correct Answer: (B) 4

A number is divisible by 8 if its last three digits are divisible by 8.

Last three digits = 42*

Check option:

424 ÷ 8 = 53

So * = 4

9. Correct Answer: (B) 15

Solution:
30 = 2 × 3 × 5

A number divisible by 30 must also be divisible by 15 (3 × 5).

10. Correct Answer: (B) 3

Check which number divides both 105 and 189.

105 ÷ 3 = 35

189 ÷ 3 = 63

So, 3 divides both.
Other options do not divide both numbers.

Worksheet – 2

1. Correct Answer: (B) 36 and 72

LCM of 12 and 18:

$12 = 2^2 \times 3$
$18 = 2 \times 3^2$

$LCM = 2^2 \times 3^2 = 36$

Common multiples: 36, 72, 108, ...

Checking options:
36 and 72 are multiples of 36.

2. Correct Answer: (D) {1, 3, 11, 33}

2. Prime factorization:

$363 = 3 \times 11 \times 11$
$165 = 3 \times 5 \times 11$
$66 = 2 \times 3 \times 11$

Common prime factors in all three numbers = 3 and 11

Now find all common factors formed using these:

Common factors = 1, 3, 11, 33

3. Correct Answer: (A) 1, 2, 7, and 14

3. Factors of 14: 1, 2, 7, 14

Factors of 28: 1, 2, 4, 7, 14, 28

Factors of 42: 1, 2, 3, 6, 7, 14, 21, 42

Common factors = 1, 2, 7, 14

4. Correct Answer: (D) 7

4. $14 = 2 \times 7$

$21 = 3 \times 7$

Common prime factor = 7

5. Correct Answer: (B) 36

What is the second smallest common multiple of the numbers 3, 6, and 9?

LCM of 3, 6, and 9:

3 = 3

$6 = 2 \times 3$

$9 = 3^2$

$LCM = 2 \times 3^2 = 18$

Common multiples: 18, 36, 54, ...

Second smallest = 36

6. Correct Answer: (C)

How many common factors do the numbers 24 and 32 have?

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24

Factors of 32: 1, 2, 4, 8, 16, 32

Common Factors: {1, 2, 4, 8}

Total: 4

7. Correct Answer: (D)

First, find the Least Common Multiple (LCM):

Multiples of 14: 14, 28, 42, 56...

28 is divisible by 4 (7 times) and 7 (4 times). So, the LCM is 28.

The common multiples are multiples of the LCM: 28, 56, 84...

The second smallest is 56.

8. Correct Answer: (B)

Which of the following pairs of numbers is co-prime?

Co-prime numbers have no common factors other than 1.

(A) 120 and 112: Both even (divisible by 2).

(B) 84 and 65: Factors of 84 (2, 3, 7...) and 65 (5, 13). No common factors.

(C) 605 and 132: Both divisible by 11.

(D) 52 and 39: Both divisible by 13.

9. Correct Answers: (A) and (D)

Which of the following pairs are twin primes?

Twin primes differ by 2.

(A) (17, 19) ⟶ Difference 2
(B) (19, 23) → Difference 4
(C) (23, 29) → Difference 6
(D) (27, 31) → Difference 3

10. Correct Answer: (C) 2 × 2 × 2 × 2 × 3

Prime factorization of 48 is:

48 = 2 × 2 × 2 × 2 × 3

11. Correct Answer: (B) 24

Which of the following numbers is divisible by 12 using prime factorization?

12 = 2^2 × 3

(A) 18 = 2 × 3^2
(B) 24 = 2^3 × 3
(C) 30 = 2 × 3 × 5
(D) 42 = 2 × 3 × 7

12. Correct Answer: (C) Not co-prime

If two numbers have common factors 3 and 5, then the numbers are:

Since they share common prime factors, they are not co-prime.

Worksheet – 3

1. Correct Answer: (C) 3 kg

We need the HCF of 75 and 69.

75 = 3 × 5 × 5
69 = 3 × 23

Common factor = 3

2. Correct Answer: (B) 4

Using prime factorization, the HCF of 20 and 28 is

20 = 2^2 × 5
28 = 2^2 × 7

Common prime factor = 2^2 = 4

3. Correct Answer: (B) 2

What is the HCF of two consecutive even numbers?

Example: 6 and 8

HCF of 6 and 8 = 2

This is true for all consecutive even numbers.

4. Correct Answer: (A) 1

What is the HCF of two consecutive odd numbers?

Example: 7 and 9

They have no common factor except 1.

5. Correct Answer: (C) 9

Find the HCF of 18, 54, 81

Prime factorization:

18 = 2 × 3^2
54 = 2 × 3^3
81 = 3^4

Common factor = 3^2 = 9

6. Correct Answer: (C) 15

Find the HCF of 45 and 60 using the long division method

45 = 3^2 × 5
60 = 2^2 × 3 × 5

Common factors = 3 × 5 = 15

7. Correct Answer: (D) 36

Find the LCM of 9 and 4

9 = 3^2
4 = 2^2

LCM = 2^2 × 3^2 = 36

8. Correct Answer: (C) 72

Find the LCM of 12, 18 and 24 using prime factorization

12 = 2^2 × 3
18 = 2 × 3^2
24 = 2^3 × 3

Take highest powers:
LCM = 2^3 × 3^2 = 8 × 9 = 72

9. Correct Answer: (B) 20

Find the LCM of 5 and 20

Multiples of 5: 5, 10, 15, 20...
Multiples of 20: 20, 40...

Smallest common multiple = 20

10. Correct Answer: (B) 60

Find the LCM of 15, 20 and 30 using the common division method

15 = 3 × 5
20 = 2^2 × 5
30 = 2 × 3 × 5

LCM = 2^2 × 3 × 5 = 60

11. Correct Answer: (C) 2160

If HCF = 12 and LCM = 180, find the product of the two numbers

Product of two numbers = HCF × LCM

= 12 × 180
= 2160

12. Correct Answer: (C) 308

HCF = 11 and LCM = 7700. One number = 275. Find the other number

Product = HCF × LCM
= 11 × 7700 = 84700

Other number = 84700 ÷ 275
= 308

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: C) 8; 4

The number of factors of 54, 62 are

Factors of 54: 1, 2, 3, 6, 9, 18, 27, 54 (8 factors)

Factors of 62: 1, 2, 31, 62 (4 factors)

2. Correct Answer: A) 2

The difference between number of factors of 105, 63 is

Factors of 105: 1, 3, 5, 7, 15, 21, 35, 105 (8 factors)

Factors of 63: 1, 3, 7, 9, 21, 63 (6 factors)

Difference: 8 - 6 = 2

3. Correct Answer: B) 57

The sum of the factors of 49 is

Factors of 49: 1, 7, 49

Sum: 1 + 7 + 49 = 57

4. Correct Answer: C) 28

Identify perfect number among the following numbers.

A perfect number is a positive integer that is equal to the sum of its proper divisors (excluding the number itself).

Check 28: Factors are 1, 2, 4, 7, 14.

Sum: 1 + 2 + 4 + 7 + 14 = 28.

5. Correct Answer: (B) 398

By adding ____ to the sum of the factors of 97, it becomes the perfect number 496.

97 is a prime number.

Factors of 97: 1 and 97

Sum of factors = 1 + 97 = 98

To make it 496:

496 – 98 = 398

6. Correct Answer: (B) 108

The common multiple of 36 and 108 is

108 is divisible by 36.

So, common multiple = 108

7. Correct Answer: D) 2

The number of common factors of 63, 98 is:

List the factors of 63: 1, 3, 7, 9, 21, 63.

List the factors of 98: 1, 2, 7, 14, 49, 98.

Identify the factors they share: 1 and 7.

Count them: There are 2 common factors.

8. Correct Answer: (D) 157

The prime number among the following is

A) 57 → 3 × 19
B) 98 → 2 × 49
C) 117 → 9 × 13
D) 157 → Not divisible by 2, 3, 5, 7, 11, 13

9. Correct Answer: (D) 963

Which of the following is not a prime number?

103 → Prime
239 → Prime
337 → Prime
963 → Sum of digits = 18 (divisible by 9)

10. Correct Answer: (C) 1944

The composite number among the following is

1129 → Prime
1453 → Prime
1944 → Even number (divisible by 2)
1787 → Prime

11. Correct Answer: (A) 34; 193

11. The pair of prime and composite numbers among the following is

A) 34 (composite); 193 (prime)
B) 294 (composite); 315 (composite)
C) 359 (prime); 397 (prime)
D) 292 (composite); 388 (composite)

Only option A has one prime and one composite.

12. Correct Answer: (D) 19, 17

Identify the twin primes

Twin primes differ by 2 and both must be prime.

C) 21, 19 (21 is composite)
D) 19, 17 (Both prime, difference = 2)

13. Correct Answer: (C) 59, 84

Identify co-primes

Co-primes have HCF = 1.

A) 35, 63 → Common factor 7
B) 36, 57 → Common factor 3
C) 59, 84 → No common factor ✓
D) 39, 91 → Common factor 13

14. Correct Answer: (D) $3^2 \times 5 \times 7$

Since 315 ends in 5, divide by 5:

$315 = 5 \times 63$

Now factor 63:

$63 = 9 \times 7$
$9 = 3 \times 3$

So,

$315 = 3 \times 3 \times 5 \times 7$
$= 3^2 \times 5 \times 7$

15. Correct Answer: (B) 1400

$2^3 \times 5^2 \times 7$ is prime factorization of

$2^3 \times 5^2 \times 7$
$= 8 \times 25 \times 7$
$= 200 \times 7$
$= 1400$

16. Correct Answer: (B) 6

5286 is divisible by

Check divisibility by 6:
Number is even (last digit 6)

Sum of digits = 5 + 2 + 8 + 6 = 21 (divisible by 3)

So it is divisible by 6.

17. Correct Answer: (D) 14468

Rule: Last two digits must be divisible by 4.

A) 3431 → 31
B) 4546 → 46
C) 5939 → 39
D) 14468 → 68 (68 ÷ 4 = 17)

18. Correct Answer: (D) 5

First find the sum:

1229 + 937 + 1059 = 3225

Last digit is 5 → divisible by 5.

19. Correct Answer: (D) 50520

Rule: Must end in 0.

D) 50520

20. Correct Answer: (B) 11848

Rule: Last three digits must be divisible by 8.

A) 9749 → 749
B) 11848 → 848 (848 ÷ 8 = 106)
C) 12573 → 573
D) 13942 → 942

21. Correct Answer: (D) 894036

Identify the number which is divisible by 11

Rule: Difference between sum of alternate digits must be a 0 or a multiple of 11,.

A) 40982
(4 + 9 + 2) − (0 + 8) = 15 − 8 = 7

B) 50520
(5 + 5 + 0) − (0 + 2) = 10 − 2 = 8

C) 693256
(6 + 3 + 5) − (9 + 2 + 6) = 14 − 17 = −3

D) 894036
(8 + 4 + 3) − (9 + 0 + 6) = 15 − 15 = 0

22. Correct Answer: (B) 84

HCF = 28, LCM = 336, one number = 112

Product = HCF × LCM
= 28 × 336 = 9408

Other number = 9408 ÷ 112 = 84

23. Correct Answer: (B) 18

The H.C.F. of 252, 324 and 594

Prime factorisation:

$252 = 2^2 \times 3^2 \times 7$
$324 = 2^2 \times 3^4$
$594 = 2 \times 3^3 \times 11$

Common lowest powers:
$2^1 \times 3^2 = 2 \times 9 = 18$

24. Correct Answer: (A) $2^3 \times 3^2 \times 5$

If prime factorisation is $2^3 \times 3^2$ and $2^2 \times 5$, then LCM is

Take highest powers:

25. Correct Answer: (A) 8, 9

$2^3 \times 3^2 \times 5$

A) 8, 9
$8 = 2^3$
$9 = 3^2$
LCM = $2^3 \times 3^2 = 72$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

26. Correct Answers: A and C

Rule: A number is divisible by 4 if its last two digits are divisible by 4.

A) 736 → Last two digits = 36 → 36 ÷ 4 = 9
B) 519 → Last two digits = 19 → Not divisible by 4
C) 284 → Last two digits = 84 → 84 ÷ 4 = 21
D) 942 → Last two digits = 42 → Not divisible by 4

27. Correct Answers: A, C and D

A is True: Since 3is a factor of 9, any number divisible by 9must also be divisible by 3.

B is False: For example, 6is divisible by 3, but it is not divisible by 9.

C is True: If a number is divisible by two co-prime numbers (like 2and 3), it is divisible by their product ($2 \times 3 = 6$).

D is True: Similarly, since 4and 3are co-prime, any number divisible by both is divisible by 12.

28. Correct Answers: A and B

Which numbers are common prime factors of 48 and 72?

$48 = 2^4 \times 3$
$72 = 2^3 \times 3^2$

Common prime factors = 2 and 3

29. Correct Answers: A, B and C

$180 = 2^2 \times 3^2 \times 5$

A) 3
B) 5
C) 15 (3 × 5)
D) 8 (2^3 — but only 2^2 available)

30. Correct Answers:

A. Common primes are 2 and 3
B. HCF = $2^4 \times 3$
C. HCF = 48

Prime factorisation:

$96 = 2^5 \times 3$
$144 = 2^4 \times 3^2$

Take the minimum powers of common primes:

2^4 and 3^1

HCF = $2^4 \times 3 = 16 \times 3 = 48$

ASSERTION AND REASON TYPE

31. Correct Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation.

Assertion: Every number is a multiple of 1.

Reason: 1 divides every number exactly.

Every number can be written as 1 × that number, so it is a multiple of 1.

Also, 1 divides every number exactly.

The reason correctly explains the assertion.

32. Correct Answer: (A)Both Assertion and Reason are true, and Reason is the correct explanation.

Assertion: 0 is a multiple of every number.

Reason: 0 multiplied by any number gives 0.

0 = n × 0 for any number n, so 0 is a multiple of every number.

0 multiplied by any number gives 0.

The reason correctly explains the assertion.

33. Correct Answer: (C)
Assertion is true, Reason is false.

33.Assertion: 5 is a factor of 25.
Reason: 25 ÷ 5 = 6 with no remainder.

5 is a factor of 25 because 25 ÷ 5 = 5 with remainder 0.

The given reason is false because 25 ÷ 5 = 5, not 6.

34. Correct Answer: (C)Assertion is true, Reason is false.

Assertion: Every prime number has exactly two factors.
Reason: A prime number can be divided by 1, itself, and one more number.

Every prime number has exactly two factors: 1 and itself.

The reason is false because a prime number cannot be divided by any other number.

35. Correct Answer: (A)Both Assertion and Reason are true, and Reason is the correct explanation.

Assertion: All even numbers are divisible by 2.
Reason: 2 is a factor of only even numbers.

All even numbers are divisible by 2. (Definition of even numbers)

2 is a factor of only even numbers. (Odd numbers are not divisible by 2)

The reason correctly explains the assertion.

MATRIX MATCHING TYPE

36. Match the terms with their definitions

Column I Column II
A. Factor → 2) A number that divides another exactly
B. Multiple → 1) A number that is divisible by another
C. Prime number → 3) Has exactly two factors
D. Composite number → 4) Has more than two factors

37. Correct Matching:

A. 35 → 3 (Multiple of 7)
B. 15 → 1 (Factor of 30)
C. 16 → 4 (Multiple of 8)
D. 20 → 2 (Factor of 20)

Check each number:

35 → 35 = 7 × 5 → Multiple of 7

15 → 15 divides 30 exactly → Factor of 30

16 → 16 = 8 × 2 → Multiple of 8

20 → 20 divides 20 exactly → Factor of 20

COMPREHENSION TYPE

Passage - 1

38. Correct Answer: (B) 0, 2, 4, 6, or 8

A number is divisible by 2 if its unit digit is

Even digits are 0, 2, 4, 6, 8.

39. Correct Answer: (C) 120

Which of the following numbers is divisible by 5?

A number is divisible by 5 if its unit digit is 0 or 5.

248 → ends in 8

693 → ends in 3

120 → ends in 0

147 → ends in 7

40. Correct Answer: (B) 324

Which of the following numbers is divisible by 9?

A number is divisible by 9 if the sum of its digits is divisible by 9.

245 → 2+4+5 = 11

324 → 3+2+4 = 9

431 → 4+3+1 = 8

517 → 5+1+7 = 13

Pessage - 2

41. Correct Answer: (C) 17

Which of the following is a prime number?

9 → 3 × 3

15 → 3 × 5

17 → Only divisible by 1 and 17

21 → 3 × 7

42. Correct Answer: (D) 12

Which number is a composite number?

2 → Prime

3 → Prime

5 → Prime

12 → Has more than two factors (1, 2, 3, 4, 6, 12)

43. Correct Answer: (B) 8 and 15

Which pair of numbers is co-prime?
(Co-prime numbers have only 1 as common factor.)
6 and 9 → Common factor 3
8 and 15 → No common factor except 1
10 and 20 → Common factor 10
14 and 21 → Common factor 7

INTEGER TYPE

44. A number is divisible by 6 if it is divisible by both 2 and 3.

45. LCM of 10, 25 and 50

10 = 2 × 5
25 = 5^2
50 = 2 × 5^2

LCM = 2 × 5^2 = 50

46. HCF of 36 and 54 using prime factorisation

36 = 2^2 × 3^2
54 = 2 × 3^3

HCF = 2^1 × 3^2 = 2 × 9 = 18

47. HCF × LCM = Product of two numbers

8 × LCM = 768
LCM = 768 ÷ 8 = 96

48. For divisibility by 4, last two digits must be divisible by 4.

Number = 53✱4

Last two digits = ✱4

Possible numbers: 04, 24, 44, 64, 84

Smallest digit = 0

Previous year Questions

1. Correct Answer: C. 7

Prime factorisation:

$810 = 2 \times 3^4 \times 5$

$950 = 2 \times 5^2 \times 19$

Common prime factors = 2,5

Sum of common prime factors= $2 + 5 = 7$

2. Correct Answer: C

A number is divisible by 4 if the number formed by its last two digits is divisible by 4.

3. Correct Answer: A

Factors of 6 are: 1, 2, 3, and 6.
Sum of its proper factors = $1 + 2 + 3 = 6$.
So, 6 is a perfect number.

4. Correct Answer: C. 6

A number divisible by 36 must be divisible by 4 and 9.

Divisibility by 4
Last two digits = 56
Since 56 is divisible by 4, this condition is satisfied.

Divisibility by 9
Sum of digits =$8 + 7 + 4 + 2 + 7 + * + 5 + 6 = 39 + *$

For divisibility by 9, the sum must be a multiple of 9.
The next multiple of 9 after 39 is 45.

So,
$39 + * = 45$
$* = 6$

5. Correct Answer: B. 30

The least number that is divisible by 2, 3, 5, and 6 is their LCM.

Prime factorisation:

$2 = 2$

$3 = 3$

$5 = 5$

$6 = 2 \times 3$

LCM $= 2 \times 3 \times 5 = 30$

6. Correct Answer: B. 4

Prime factorisation of 7350:

$7350 = 735 \times 10$
$= (105 \times 7) \times (2 \times 5)$
$= 2 \times 3 \times 5^2 \times 7^2$
Distinct prime factors are: 2, 3, 5, 7

Number of prime factors = 4

7. Correct Answer: C

A is incorrect:42 and 49 have a common factor 7, so they are not co-prime.

B is incorrect:
Sum of digits of 376277 = $3 + 7 + 6 + 2 + 7 + 7 = 32$, which is not divisible by 3.

C is correct:
$102 = 2 \times 3 \times 17$
$170 = 2 \times 5 \times 17$
$136 = 2^3 \times 17$

LCM $= 2^3 \times 3 \times 5 \times 17 = 2040$

8. Correct Answer: D. 1

Rule of divisibility by 11:
The difference between the sum of digits in alternate places must be 0 or a multiple of 11.

Number: 6 5 3 * 4 7

Sum of digits in odd places:
$6 + 3 + 4 = 13$

Sum of digits in even places:
$5 + * + 7 = 12 + *$

Difference:
$|\ 13 - (12 + *)\ | = |\ 1 - *\ |$

For divisibility by 11:
$|\ 1 - *\ | = 0$

$* = 1$

9. Correct Answer: B

A:$39 = 3 \times 13$, $91 = 7 \times 13$
Common factor = 13 → Not co-prime

B:$161 = 7 \times 23$, $192 = 2^6 \times 3$
No common factor → Co-prime

C:$385 = 5 \times 7 \times 11$, $462 = 2 \times 3 \times 7 \times 11$
Common factors = 7, 11 → Not co-prime

D:$189 = 3^3 \times 7$, $243 = 3^5$
Common factor = 3 → Not co-prime

10. Correct Answer: D

(i)Required number – 1 must be divisible by 26, 56 and 104.
LCM of 26, 56 and 104 = 728
Required number = $728 + 1 = 729$
So, statement (i) is True (T).

(ii)For divisibility by 3, sum of digits must be divisible by 3.
Sum of digits of $57x897 = 5 + 7 + x + 8 + 9 + 7 = 36 + x$.
Least value of x making the sum divisible by 3 is 0, not 7.
So, statement (ii) is False (F).

(iii)Twin prime pairs between 10 and 30 are:
$(11, 13)$and $(17, 19)$
Number of pairs = 2
So, statement (iii) is True (T).

4. Integers

Solutions

Worksheet – 1

1. Correct Answer: (B)

Number Line Order

To find the correct number line for -10, -13, -8, and 9, we first need to arrange them from smallest to largest (left to right):

-13 < -10 < -8 < 9

Option (B) shows -13 furthest to the left, followed by -10, then -8, and finally 9 on the positive side.

2. Correct Answer: (D) 11 and −9

Successor of a number = number + 1

Successor of 10 = 10 + 1 = 11

Successor of −10 = −10 + 1 = −9

3. Correct Answer: (B) −3

Step-by-step logic:

Ascending order means smallest to greatest.

So the missing number must:

Be greater than −9

Be less than 0

Check options:

−3 ✅ (between −9 and 0)

4. Correct Answer: (A) −5, −2, and 6

(From the number line:
A is left of 0,
B is just left of 0,
C is right of 3) Counting the unit distances on the number line:

A = −5

B = −2

5. Correct Answer: (D) 96 units C = 6

From the number line:

Position of A = −50

Position of B = +46 (just left of 50, marked at 46)

👉 Distance between two points on a number line

$$=| B - A |=| 46 - (-50) |=| 96 |$$
$$= 96 \text{ units}$$

6. Correct Answer: (C) Integers are finite in number

Which of the following statements is incorrect?

Check each option:

(A) The predecessor of 0 is −1 ✔ (Correct)
(B) The successor of −5 is −4 ✔ (Correct)
(C) Integers are finite in number ❌ (Wrong, integers are infinite)
(D) All whole numbers are integers ✔ (Correct)

7. Correct Answer: (B) −4 and 5

What are the respective values of points P and Q on the given number line?

From the number line:

P is at −4

Q is at 4

8. Correct Answer: (C) less than −2 and greater than 3

The given number line shows integers that are

Highlighted integers:

On left side: −5, −4, −3 → all less than −2

On right side: 4 and 5 → all greater than 3

9. Correct Answer: (D)

Which of the following statements is correct?

Check each option:

(A) −3 lies between −2 and 3 ❌
(B) −4 lies to the right of −2 ❌ (−4 is to the left)
(C) −20 lies to the right of −10 ❌
(D) −6 and −7 are greater than −9 ✔

10. Correct Answer: (C) B

Understand "maximum"

Maximum temperature means the greatest number on the number line.

Step 2: Compare the temperatures

City A → −3

City B → 0

City C → −2

City D → −1

City E → −4

Among these, 0 is the greatest value.

Worksheet – 2

1. What is the value of the expression

$(-5) + 6 - 7$

Solution:

$(-5) + 6 = 1$

$1 - 7 = -6$

But check the options given: −7, 6, −1, −2

Re-evaluate carefully:

$$(-5) + 6 - 7 = (-5 + 6) - 7 = 1 - 7$$
$$= \boxed{-6}$$

Note: −6 is not in the options, so the question paper has an error.

2. Correct Answer: (D) 26

What is the value of the expression

$(-3) + (23) + (37) + (-36) + (5)$

Solution:

Group positives and negatives:

Positives: $23 + 37 + 5 = 65$

Negatives: $-3 - 36 = -39$

$65 + (-39) = 26$

3. Correct Answer: (B) (−4) + (−7) + 12

Which expression, when calculated, is represented by the given number line?

From the number line:

Start at −4

Move 7 units left → −11

Then move 12 units right → +1

So the expression is:

$$(-4) + (-7) + 12$$

4. Correct Answer: (B) −3

What is the missing value in the expression

$$(-267) - (-276) - 15$$
$$= \square + (-2) + (-1)$$

Left side:

$(-267) + 276 - 15 = 9 - 15 = -6$

Right side:

$\square - 2 - 1 = \square - 3$

Equate both sides:

$$-3 = -6$$
$$= -3$$

5. Correct Answer: (B) −110

Which integer is four more than the sum of (−99) and (−15)?

Solution:

$$(-99) + (-15) = -114$$

Four more than −114:

$$-114 + 4 = -110$$

6. Correct Answer: (B) −20

What is the result obtained when the sum of (−19) and (+15) is subtracted from the sum of (+25) and (−49)?

Step 1: First sum

$$25 + (-49) = -24$$

Step 2: Second sum

$$-19 + 15 = -4$$

Step 3: Subtract

$$-24 - (-4) = -24 + 4 = -20$$

7. Correct Answer: (D) 50

What is the value of the expression

$$(-216) - (-351) + (-98) - (71)$$
$$+ (28) - (-56)$$

Simplify step by step:

$-216 + 351 = 135$
$135 - 98 = 37$
$37 - 71 = -34$
$-34 + 28 = -6$
$-6 + 56 = 50$

8.

9. Correct Answer: (B) 4

🟢 (grey circle) represents +1
⚫ (black circle) represents −1

Find the value of the expression:

$$[5 - 3 + 2]$$

Step-by-step:

$$5 - 3 = 2$$
$$2 + 2 = 4$$

(The given symbols just help visualize +1 and −1, but numerically this simplifies directly.)

10. Correct Answer: (B) 0

What is the sum of the greatest negative and the smallest positive integer?

Greatest negative integer = −1

Smallest positive integer = +1

Sum:

$-1 + 1 = 0$

Worksheet – 3

1. Question paper error
Correct value = Option(B)19

What is the result obtained when the sum of (−14) and (+9) is subtracted from the sum of (+32) and (−18)?

Step 1: Find both sums

Sum of (+32) and (−18):

$$32 + (-18) = 14$$

Sum of (−14) and (+9):

$$-14 + 9 = -5$$

Step 2: Subtract

$$14 - (-5) = 14 + 5 = 19$$

But 19 is not in the options, so check wording carefully:

"sum of (−14) and (+9) is subtracted from the sum of (+32) and (−18)"
→ calculation is correct.

2. Correct Answer: (D)

Which number line correctly represents the procedure for finding the solution of the expression

$$(-3) - (+5)$$

Explanation:

$$(-3) - (+5) = -3 + (-5)$$

On the number line:

Start at −3

Move 5 steps to the left

Reach −8

The diagram that shows this movement correctly is Option (D).

3. Correct Answer: (A) Decrease by 7°C

The temperature of a city was 3°C on a particular day.
Next day, it was −4°C.
What is the change in temperature over the two days?

Calculation:

$$-4 - 3 = -7$$

Negative sign shows a decrease.

4. ✅ Correct Answer: (B) 2125

Alexander the Great was born in 356 BC.
Napoleon was born in 1769 AD.
How many years after the birth of Alexander was Napoleon born?

School-level calculation:

$$356 + 1769 = 2125$$

5. Correct Answer: (D)

Which number line correctly shows the calculation of the expression

$$[-7 - (-5) + 3]$$

Step-by-step calculation:

$$-7 - (-5) = -7 + 5 = -2$$
$$-2 + 3 = 1$$

So, the final answer should be 1.

Number line logic:

Start at −7

Subtracting −5 means move 5 steps to the right, reach −2

Then add 3, move 3 steps to the right, reach +1

The option that shows:

start at −7

right move to −2

right move to +1

is Option (D).

6. Correct Answer: (A) 23

The value of the expression

$$[21 - (-13) - 19 - (-8)]$$

Step-by-step:

$$21 - (-13) = 21 + 13 = 34$$
$$34 - 19 = 15$$
$$15 - (-8) = 15 + 8 = 23$$

7. Correct Answer: (C) 4200 m

A fighter plane is flying at a height of 3800 m above sea level.
A submarine is 400 m below sea level.
What is the distance between them when the bomb is dropped?

Represent heights using integers:

Fighter plane = +3800 m

Submarine = −400 m

Distance between them:

$$| 3800 - (-400) | = | 4200 | = 4200 \text{ m}$$

8. Correct Answer: (C) 7549

What should be subtracted from (−1963) to obtain (−9512)?

Let the required number be x.

$$-1963 - x = -9512$$

Solve:

$$x = -1963 + 9512 = 7549$$

9. Correct Answer: (B) −10

What is the value of the expression

$$[-6 + (-12) - 2 - (-10)]$$

Step-by-step:

$$-6 + (-12) = -18$$
$$-18 - 2 = -20$$
$$-20 - (-10) = -20 + 10 = -10$$

10. Correct Answer: (D)

Which number line correctly shows the expression

$$[-5 + 3 - (-4)]$$

Step-by-step calculation:

$$-5 + 3 = -2$$
$$-2 - (-4) = -2 + 4 = 2$$

Number line steps:

Start at −5

Move 3 steps right to −2

Subtracting −4 means move 4 steps right to +2

The diagram that shows this movement correctly is Option (D).

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: (B) −41

Successor = number **+ 1**

$$-42 + 1 = -41$$

2. Correct Answer: (C) −3

2. Subtraction of −7 from −10 is equal to

"Subtraction of −7 from −10" means:

$$-10 - (-7) = -10 + 7 = -3$$

3. Correct Answer: (A) −48

Subtract the sum of −23 and 33 from −38

First find the sum:

$$-23 + 33 = 10$$

Now subtract from −38:

$$-38 - 10 = -48$$

4. Correct Answer: (B) −28

Additive inverse of 28 is

Additive inverse = number with opposite sign

$$\text{Additive inverse of } 28 = -28$$

5. Correct Answer: (B) −1

Predecessor of 0 is

Predecessor = number **just before**

$$0 - 1 = -1$$

6. Correct Answer: (A) −490

The sum of two integers is −355. One integer is 135. Find the other

Let the other integer be x

$$x + 135 = -355$$
$$x = -355 - 135 = -490$$

7. Correct Answer: (B) 15

On subtracting −15 from 0, we get

$$0 - (-15) = 15$$

8. Correct Answer: (A) −4

The integer 8 is more than −12 is

"More than" means **add**:

$$-12 + 8 = -4$$

9. Correct Answer: (C) −9

$1 - 2 + 3 - 4 + 5 - 6 + \cdots + 17 - 18$
Group in pairs:

$(1 - 2) + (3 - 4) + (5 - 6) + \cdots + (17 - 18)$

Each pair = −1
There are **9 pairs**

$9 \times (-1) = -9$

10. Correct Answer: (B) >

Compare −14 ___ −24

On the number line, −14 is **to the right** of −24, so it is **greater**.

$$-14 > -24$$

11. Correct Answer: (C) 723

$\mathbf{450} - (-\mathbf{273}) =$

Subtracting a negative means **adding**:

$$450 + 273 = 723$$

12. Correct Answer: (B) 17

$|\ \mathbf{5}\ | + |\ -\mathbf{12}\ | =$

Absolute value makes numbers **positive**:

$$|\ 5\ | = 5, |\ -12\ | = 12$$
$$5 + 12 = 17$$

13. Correct Answer: (B) Kashmir

Temperature in Jammu = −8°C
Temperature in Kashmir = −12°C

Colder means **lower temperature**.

$$-12°C < -8°C$$

So Kashmir is colder.

14. Correct Answer: (B) Positive

Product of two negative numbers is

Rule:

$$(-) \times (-) = (+)$$

15. Correct Answer: (A) −731

Product of $(-\mathbf{43}) \times (\mathbf{17})$

Negative × Positive = **Negative**

$$43 \times 17 = 731$$

So,

$$(-43) \times 17 = -731$$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

16. Correct Answer(s): C) 24
D) |186 − 210|
B) $3 \times (5 + 3)$

Marks:

- **Geeta**: $45 \times 5 - 5 \times 3 = 210$
- **Sita**: $42 \times 5 - 8 \times 3 = 186$

Difference:

$$210 - 186 = 24$$

Also,

$$|\ 186 - 210\ | = 24$$

17. Correct Answer(s): A and C

Peak = **+6300 m**
Mine = **−39060 m**

Vertical distance:

$$6300 - (-39060) = 45360 \text{ m}$$

Check options:

- 45360 m ✅
- −32760 m ❌ (distance cannot be negative)
- 6300 + 39060 m = 45360 m ✅
- 32760 m ❌

18. Correct Answer(s): B and C

Product = −1800
One integer = 30

Other integer:

$$\frac{-1800}{30} = -60$$

Check options:

- A) 60 ❌

- B) −60 ✅
- C) $-1800/30 (= -60)$ ✅
- D) −90 ❌

19. Correct Answer(s): A and C

Lighthouse = **+50 m**
Seabed = **−25 m**

Height from seabed:

$$50 - (-25) = 75 \text{ m}$$

Check options:

A) 75 m ✅

B) −75 m ❌

C) 50 + 25 m (= 75 m) ✅

D) −25 m ❌

20. Correct Answer(s): B and C

Integers between **20 and 21**
There are **no integers** between two consecutive integers.

Check options:

- 2 ❌
- 0/7 (= 0) ✅
- 0 ✅
- 1 ❌

ASSERTION AND REASON TYPE

21. Correct Answer: (A)

Assertion: −5 is less than −2.
Reason: The greater the negative number, the smaller its value.

- Assertion → **True**
- Reason → **True**
- Reason correctly explains the assertion

22. Correct Answer: (A)

Assertion: 0 is greater than any negative integer.
Reason: 0 lies to the right of all negative numbers on the number line.

- Assertion → **True**
- Reason → **True**
- Reason correctly explains the assertion

23. Correct Answer: (D)
Assertion is false, Reason is true.

Assertion: −7 + 3 = 4
Reason: Adding a positive number to a negative number moves to the right on the number line.

- Assertion → **False** (−7 + 3 = −4, not 4)
- Reason → **True**

24. Correct Answer: (C)
Assertion is true, Reason is false.

Assertion: Subtracting an integer is the same as adding its opposite.
Reason: −(−a) = −a

- Assertion → **True**
- Reason → **False** (−(−a) = **a**, not −a)

25. Correct Answer: (A)

Assertion: The sum of two negative integers is always a negative integer.
Reason: Adding two negative values results in a value with a larger negative magnitude.

- Assertion → **True**
- Reason → **True**
- Reason correctly explains the assertion

MATRIX MATCHING TYPE

26. Correct Answer:

a – S

b – R

c – Q

d – P

Column I ⟶ Column II

a) $(-3)+(-2)=-5\rightarrow$ S
b) $(-5)+5=0\rightarrow$ R
c) $(-4)+(-2)=-6\rightarrow$ Q
d) $(-3)+2=-1\rightarrow$ P

27. Correct Answer (Q27):

a – R

b – Q

c – P

d – S

Column A → Column B

a) Integer → Whole numbers and their negatives → R
b) Number line → Horizontal line to represent numbers → Q
c) Additive inverse → Opposite of a number → P
d) Zero → Neutral element for addition → S

COMPREHENSION TYPE

28. Correct Answer: C) Whole numbers and their negatives

Explanation:

From the passage, integers include all whole numbers and their negatives (..., −3, −2, −1, 0, 1, 2, 3, ...).
So option C matches the definition.

Which of the following numbers are included in the set of integers?

29. Correct Answer: B) To the left of 0

Explanation:

The passage clearly says that negative numbers are represented to the left of 0 on the number line.

On a number line, where are negative integers placed?

30. Correct Answer: C) Integers include positive numbers, negative numbers, and zero

Explanation:

Integers consist of:

Positive numbers

Negative numbers

Zero

So option C is correct

Which statement about integers is correct?

31. Correct Answer: B) The number on the right

Explanation:

On a number line, numbers increase as we move to the right, so the number on the right is always greater than the one on the left.

When comparing integers on a number line, which number is greater?

32. Correct Answer: B) −3 is greater than −7

Explanation:

On the number line, −3 lies to the right of −7, so −3 is greater than −7.

Which of the following statements is true?

33. Correct Answer: C) −6

Explanation:

−6 is greater than −7 and is placed to the right of −7 on the number line.

On a number line, which number lies to the right of −7?

INTEGER TYPE

34. $(-1)+2+(-3)+4+(-5)-6+\cdots+(-17)+18$
Group in pairs: $(-1+2)+(-3+4)+(-5+6)+\cdots+(-17+18)$
Each pair = 1
Number of pairs = 9

$$9\times 1=\boxed{9}$$

35. Additive inverse of −7 is

Additive inverse = number with **opposite sign**

$$\boxed{7}$$

36. $3+(-3)+(-3)+3+(-3)+3$
Group:

$$(3-3)+(-3+3)+(-3+3)$$
$$0+0+0=\boxed{0}$$

37. $(78-77)+(18-17)+(28-27)$
$= 1+1+1 = \boxed{3}$

38. $1-10+9-7+6+2$
Stepwise:
$1-10=-9$
$-9+9=0$
$0-7=-7$
$-7+6=-1$
$-1+2=\boxed{1}$

39. $81-91-101+100+5+7$
Stepwise:
$81-91=-10$
$-10-101=-111$
$-111+100=-11$
$-11+5=-6$
$-6+7=\boxed{1}$

Previous Year Questions

1. Correct Answer: −201 (Option B)

If 301 is the successor of X and Y is the predecessor of −500, find X + Y.

Successor of X = 301
⇒ X + 1 = 301
⇒ X = 300

Predecessor of −500
⇒ Y = −501

Now,

$$X+Y = 300+(-501) = -201$$

2. Correct Answer: −1305 (Option A)

Find the value of 415 − 719 + 1014 − 2015

Step by step:

$$415-719=-304$$
$$-304+1014=710$$
$$710-2015=-1305$$

3. Correct Answer: > (Option C)

Compare and fill the box
Expression:
$172+151-79\square(-248)+121+65$
Left side:

$$172+151=323$$
$$323-79=244$$

Right side:

$$-248+121=-127$$
$$-127+65=-62$$

Now compare:

$$244 > -62$$

4. Correct Matching: (p) → (iii); (q) → (i); (r) → (ii)
Option C

(p) Additive inverse of (−40) − (−57) − (90)

First simplify:

$$-40+57-90$$
$$=17-90$$
$$=-73$$

Additive inverse of −73 = 73
⇒ (iii)

(q) Successor of predecessor of −99

Predecessor of −99 = −100
Successor of −100 = −99
⇒ (i)

(r) Difference between greatest 4-digit number and additive inverse of (−9890)

Greatest 4-digit number = 9999

Additive inverse of −9890 = 9890

Difference:

$$9999-9890=109$$

⇒ (ii)

5. Correct Answer: B (−4°C to 8°C)

Which shows maximum rise in temperature?

Calculate increase:

A. 0°C to 10°C → Rise = 10°C
B. −4°C to 8°C → Rise = 12°C
C. −15°C to −8°C → Rise = 7°C
D. −7°C to 1°C → Rise = 8°C

Maximum rise = **12°C**

6. Correct Answer: D. 2 km

Manjisha runs 7 km towards South, then 9 km towards North

South = Negative
North = Positive

First movement:

$$-7 \text{ km}$$

Second movement (opposite direction = North):

$$+9 \text{ km}$$

Final position:

$$-7 + 9 = 2$$

She is **2 km North** from the starting point.

7. Correct Matching:

(P) → (iii)
(Q) → (i)
(R) → (ii)

Option B

(P) Difference between successor and predecessor of 305

Successor of 305 = 306
Predecessor of 305 = 304

Difference:

$$306 - 304 = 2$$

⇒ (iii)

(Q) Subtract 93 from the sum of −678 and −234

First find the sum:

$$-678 + (-234) = -912$$

Now subtract 93:

$$-912 - 93 = -1005$$

⇒ (i)

(R) Integer which is 8 more than −24

$$-24 + 8 = -16$$

⇒ (ii)

8.

9. Correct Answer: A

Arun's Bank Account

Initial amount = ₽125

He withdraws ₽117

$$125 - 117 = 8$$

He deposits ₽45

$$8 + 45 = 53$$

He withdraws ₽69

$$53 - 69 = -16$$

Amount left in his account = **₽ (−16)**

10. Correct Answer: 48 (Option B)

Sum of additive inverses

Additive inverse of −100 = **100**
Additive inverse of 52 = **−52**

Now find the sum:

$$100 + (-52) = 48$$

5. Fractions and decimals

Solutions

Worksheet – 1

1. Correct answer: (D)

Mark's marks = 20
Joe's marks = 10

Compare Joe's marks with Mark's marks:

$$\frac{\text{Joe's marks}}{\text{Mark's marks}} = \frac{10}{20} = \frac{1}{2}$$

So, Joe scored $\frac{1}{2}$ times Mark's marks.

2. Correct answer: (C) $\frac{20}{7}$g

Given:
Weight of wheat = 2 kg = 2000 g
Volume of box = 700 cm^3

Weight of 1 cm^3 of wheat:

$$\text{Weight per } cm^3 = \frac{2000}{700} = \frac{20}{7} \text{ g}$$

3. Correct answer: (A) $\frac{1}{5}$

Total number of fruits = 50
Number of mangoes = 10

Fraction of mangoes:

$$\frac{\text{Number of mangoes}}{\text{Total fruits}} = \frac{10}{50} = \frac{1}{5}$$

4. Correct answer: (C) $\frac{7}{16}$

Simplify $\frac{56}{128}$

Find HCF of 56 and 128 = 8

$$\frac{56}{128} = \frac{56 \div 8}{128 \div 8} = \frac{7}{16}$$

5. Correct answer: (B) $\frac{55}{8}$

Convert $6\frac{7}{8}$ into improper fraction

$$\frac{(6 \times 8) + 7}{8} = \frac{48 + 7}{8} = \frac{55}{8}$$

6. Correct answer: (B) $\frac{41}{5}$

Convert $8\frac{1}{5}$ into improper fraction

$$\frac{(8 \times 5) + 1}{5} = \frac{40 + 1}{5} = \frac{41}{5}$$

7. Correct answer: (D) $\frac{13}{5}$

Convert $2\frac{3}{5}$ into improper fraction

$$\frac{(2 \times 5) + 3}{5} = \frac{10 + 3}{5} = \frac{13}{5}$$

8. Correct answer: (A)

Anthony's marbles = 8
Sarah's marbles = 12

Compare Anthony with Sarah:

$$\frac{\text{Anthony}}{\text{Sarah}} = \frac{8}{12} = \frac{2}{3}$$

So, Anthony has $\frac{2}{3}$ times as many marbles as Sarah.

9. Correct answer: (D) $\frac{1}{3}$

Simplest form of $\frac{18}{54}$

HCF of 18 and 54 = 18

$$\frac{18}{54} = \frac{18 \div 18}{54 \div 18} = \frac{1}{3}$$

10. Correct answer: (A) $\frac{1}{16}$

Fraction equal to $\frac{4}{64}$

$$\frac{4}{64} = \frac{1}{16}$$

11. Correct answer: (C)

Jessica's time = 30 min, Kim's time = 42 min

$$\frac{\text{Jessica}}{\text{Kim}} = \frac{30}{42} = \frac{5}{7}$$

So, Jessica took $\frac{5}{7}$ as much time as Kim.

12. Correct answer: (A) $\frac{25}{125}$

Fraction not in simplest form

Check each option:

(A) $\frac{25}{125}$

$\frac{25}{125} = \frac{1}{5}$ (can be simplified)

(B) $\frac{2}{17}\rightarrow$ already simplest (no common factor)

(C) $\frac{5}{19}\rightarrow$ already simplest

(D) $\frac{21}{25}\rightarrow$ already simplest

Worksheet – 2

1. Correct answer: (D)

1. Amount spent from A to B = Rs. 125.75
Amount spent from B to C = Rs. 730.45

$125.75 + 730.45 = 856.20$

Total amount spent = Rs. 856.20

2. Correct answer: (C) 174.499

$132.156 + 42.343$

$132.156 + 42.343 = 174.499$

3. Correct answer: (A) $5.0 + 2.1$

Equivalent to $2.5 + 4.6$

$2.5 + 4.6 = 7.1$

Check options:

$5.0 + 2.1 = 7.1$

4. Correct answer: (C) 3.81 s

Philip's total time vs Roy

Philip's time:$17.84 + 16.34 = 34.18$ s

Difference:$34.18 - 30.37 = 3.81$ s

5. Correct answer: (C) 3.22 m

Total cloth required

$157 + 165 = 322$ cm

Convert to metres:

322 cm $= 3.22$ m

6. Correct answer: (B) Rs. 2.00

Money spent by Rodney

$1.55 + 0.45 = 2.00$

7. Correct answer: (A) 3.45 m

Length of other piece

Total length = 5.8 m
One piece = 2.35 m

$5.8 - 2.35 = 3.45$ m

8. Correct answer: (D) 389.500

$307.126 + 82.374$

$307.126 + 82.374 = 389.500$

9. Correct answer: (C) Rs. 8.51

Total amount spent

Sandwiches = Rs. 5.45
Fruit juice = Rs. 3.06

$5.45 + 3.06 = 8.51$

10. Correct answer: (A) 2.75 km

Total distance covered

$1.25 + 1.50 = 2.75$ km

(Break time not included in distance.)

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: (C) 9 thousandths

Place value of 9 in 46.289
9 is in the third place after decimal → thousandths.

2. Correct Answer: (B) 6 tenths

Place value of 6 in 45.678
6 is first digit after decimal → tenths.

3. Correct Answer: (C) 4103.57

Number to subtract from 5102.57 to get greatest 3-digit number
Greatest 3-digit number = 999
Required number = 5102.57 – 999 = 4103.57

4. Correct Answer: (B) 67.38

How much is sum of 33.69 and 112.55 greater than their difference?
Sum = 33.69 + 112.55 = 146.24
Difference = 112.55 – 33.69 = 78.86
Required = 146.24 – 78.86 = 67.38

5. Correct Answer: (C) 7/4

Fractional form of 1.75
1.75 = 175/100 = 7/4

6. Correct Answer: (A) 17/19

Proper fraction (numerator < denominator)

17/19 and 35/31 → numerator < denominator? Only 17/19.

7. Correct Answer: (C) 35/12

Improper fraction (numerator ⩾ denominator)
Only 35/12 satisfies.

8. Correct Answer: (B) $\frac{31}{7}$

Convert each fraction to decimal and compare with 3.55.

A) $\frac{49}{21} = \frac{7}{3} \approx 2.33 (< 3.55)$

B) $\frac{31}{7} \approx 4.43 (> 3.55)$

C) $\frac{79}{52} \approx 1.52 (< 3.55)$

D) $\frac{32}{11} \approx 2.91 (< 3.55)$

9. Correct Answer: (D) 2/3

If $\frac{1}{2} + \frac{1}{x} = 2$

$\frac{1}{x} = 2 - \frac{1}{2} = \frac{3}{2}$

$x = \frac{2}{3}$

10. Correct Answer: (D) 7/8

Greatest fraction among $\frac{7}{8}, \frac{6}{7}, \frac{4}{5}, \frac{5}{6}$
Convert to decimals:
7/8=0.875, 6/7≈0.857, 4/5=0.8, 5/6≈0.833
Largest = 7/8

11. Correct Answer: (A) ₱43.25

Total cost
24.50 + 18.75 = ₱43.25

12. Correct Answer: (B) ₱0.266

Remaining amount
1.000 − 0.734 = ₱0.266

13. Correct Answer: (A) ₱2.75

Money left
5.50 − 2.75 = ₱2.75

14. Correct Answer: (C) 2

$\frac{1}{2} + \frac{3}{2}$
Same denominator → $\frac{1+3}{2} = \frac{4}{2} = 2$

15. Correct Answer: (A) 5

Difference of 8.5 and 3.5
8.5 − 3.5 = 5

16. Correct Answer: (A) 33/5

Convert $6\frac{3}{5}$ into improper fraction

$$\frac{(6 \times 5) + 3}{5} = \frac{30 + 3}{5} = \frac{33}{5}$$

17. Correct Answer: (B) 29/3

Improper fraction of $9\frac{2}{3}$

$$\frac{(9 \times 3) + 2}{3} = \frac{27 + 2}{3} = \frac{29}{3}$$

18. Correct Answer: (A) $7\frac{2}{9}$

Convert $\frac{65}{9}$ into mixed fraction
65 ÷ 9 = 7 remainder 2
= $7\frac{2}{9}$

19. Answer: (A) $8\frac{1}{6}$

Multiply $2\frac{1}{3} \times 3\frac{1}{2}$

Convert to improper fractions:

$2\frac{1}{3} = \frac{7}{3}, 3\frac{1}{2} = \frac{7}{2}$

$$\frac{7}{3} \times \frac{7}{2} = \frac{49}{6} = 8\frac{1}{6}$$

20. Answer: (C) $\frac{5}{6}$

Divide $\frac{5}{8} \div \frac{3}{4}$
Multiply by reciprocal:

$$\frac{5}{8} \times \frac{4}{3} = \frac{20}{24} = \frac{5}{6}$$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

21. Correct Answers: A, B, D

Which pairs of fractions are equivalent? (Equivalent fractions → same simplest form.)

A) $\frac{4}{6} = \frac{2}{3} \rightarrow$ same as $\frac{2}{3}$

B) $\frac{15}{24} = \frac{5}{8} \rightarrow$ same as $\frac{5}{8}$

C) $\frac{21}{28} = \frac{3}{4} \neq \frac{7}{9}$

D) $\frac{9}{12} = \frac{3}{4} \rightarrow$ same as $\frac{3}{4}$

22. Correct Answers: (A), (C)

Five chocolates divided among 8 children

Each child gets $\frac{5}{8}$ chocolate.

$\frac{10}{16} = \frac{5}{8}$(same value)

23. Correct Answers: (A), (C)

Decimals greater than 2.5

2.55 > 2.5

2.05 < 2.5

2.75 > 2.5

2.50 = 2.5 (not greater)

24. Correct Answers: A, B, C, D

Operations giving result less than 1

A) $\frac{2}{3} \times \frac{3}{5} = \frac{6}{15} = \frac{2}{5} < 1$

B) $\frac{7}{8} - \frac{1}{4} = \frac{5}{8} < 1$

C) $\frac{5}{4} \times \frac{2}{3} = \frac{10}{12} = \frac{5}{6} < 1$

D) $\frac{3}{7} + \frac{2}{7} = \frac{5}{7} < 1$

25. Correct Answers: A, B, C

Fractions equal to $\frac{2}{3}$ (simplify)

A) $\frac{8}{12} = \frac{2}{3}$

B) $\frac{10}{15} = \frac{2}{3}$

C) $\frac{14}{21} = \frac{2}{3}$

D) $\frac{16}{20} = \frac{4}{5} \neq \frac{2}{3}$

ASSERTION AND REASON TYPE

26. Correct option: (D)

Assertion (A): False ($\frac{1}{5} < \frac{2}{5}$)

Reason (R): True

27. Correct option: (C)

Assertion (A): True ($\frac{37}{8} = 4\frac{5}{8}$)

Reason (R): False ($37 \div 8 = 4$remainder 5, not 4)

28. Correct option: (A)Both A and R are true, and R correctly explains A.

Assertion (A): True

$\frac{2}{3} \times \frac{3}{4} = \frac{2 \times 3}{3 \times 4} = \frac{6}{12}$

Reason (R): True (Rule of multiplying fractions)

29. Correct option: (A)

Assertion (A): True (1.25 > 1.205)

Reason (R): True (decimals are compared digit by digit from left to right)

30. Correct option: (C)

Assertion (A): True (every decimal has a whole number part)

Reason (R): False (decimals have both whole part and fractional part)

MATRIX MATCHING TYPE

31. Correct matching: a–r, b–p, c–s, d–q

a) $\frac{1}{3} + \frac{1}{6} = \frac{1}{2} \longrightarrow$ r

b) $0.3 \times 10 = 3 \rightarrow$ p

c) $\frac{4}{5} - \frac{1}{5} = \frac{3}{5} \longrightarrow$ s

d) $1.2 \div 10 = 0.12 \rightarrow$ q

32. Correct matching: a–q, b–s, c–p, d–r

a) $24.06 + 8.20 = 32.26 \rightarrow$ q

b) $1 - 0.7342 = 0.2658 \rightarrow$ s

c) $1 - 0.5432 = 0.4568 \rightarrow$ p

d) $24.96 + 7.04 = 32.00 \rightarrow$ r

COMPREHENSION TYPE

Passage - I

33. Correct option: (B) 6.00 kg

How much sugar was sold in total?

$3.25 + 2.75 = 6.00$kg

34. Correct option: (A) 4.5 kg

How much sugar remained in the store?

Total $= 10.5$kg

Sold $= 6.0$kg

Remaining $= 10.5 - 6.0 = 4.5$kg

35. Correct option: (B) 0.90 kg

What is the mass of sugar in each bag?

Remaining sugar $= 4.5$kg

Number of bags $= 5$

Each bag $= 4.5 \div 5 = 0.90$kg

Passage-II

36. Correct option: (C)

Fraction used for vegetables
Tomatoes $= 4$, Peppers $= 3 \rightarrow$ Total vegetables $= 7$plots out of 12
$\frac{7}{12}$

37. Correct option: (A)

Fraction planted with flowers
Flowers $= 12 - 7 = 5$plots
$\frac{5}{12}$

38. Correct option: (C)

Minimum fraction required for vegetables
"At least half" $= \frac{1}{2}$

INTEGER TYPE

39. $\mathbf{5.25 + 3.38 = 8.63}$
The place value of 3 is hundredths ($= \mathbf{0.03}$**).**

40. $\mathbf{7.73 + 0.07 + 7 + 7.86 + 7.153 = 29.813}$

41. Cloth left $= \mathbf{25.3 - 3.85 = 21.45}$**m**

42. Total revised $= \frac{7}{12} + \frac{3}{12} = \frac{10}{12}$
Still to revise $= 1 - \frac{10}{12} = \frac{2}{12} = \frac{1}{6}$

43. Rajesh's jump $= 8 - \frac{2}{3} = \frac{22}{3} = 7\frac{1}{3}$**m**

Previous Year Questions

1. Correct Answer: (D) 20

Let the number of sixths be x.

$\frac{5}{12} + \frac{x}{6} = 3\frac{3}{4}$

Convert $3\frac{3}{4}$ to improper fraction:

$3\frac{3}{4} = \frac{15}{4}$

$\frac{x}{6} = \frac{15}{4} - \frac{5}{12}$

Take LCM of 4 and 12 = 12:

$\frac{15}{4} = \frac{45}{12}$

$\frac{x}{6} = \frac{45}{12} - \frac{5}{12} = \frac{40}{12} = \frac{10}{3}$

$x = 6 \times \frac{10}{3} = 20$

2. Correct Answer: (B) 12.64

$35.67 - 23.03 = 12.64$

3. Correct Answer: (D) $\frac{4}{11}$

Total letters in MATHEMATICS = 11

Vowels (A, E, I, O, U) in the word:
A, E, A, I $\rightarrow$ 4 vowels

$$\text{Fraction of vowels} = \frac{4}{11}$$

4. Correct Answer: (A) 12.75 m

Convert 4 m 25 cmto metres:

$25 \text{ cm} = \frac{25}{100} = 0.25 \text{ m}$

$4 \text{ m } 25 \text{ cm} = 4.25 \text{ m}$

For 3 dresses:

$4.25 \times 3 = 12.75 \text{ m}$

5. Correct Answer: (C) 210 L

Water in X and Y together:

$24.25 + 17.75 = 42$ L

This is $\frac{1}{5}$of the tank.

Total capacity of tank:

$42 \times 5 = 210$ L

6. Correct Answer: (D) 43.26

Let the number be x.

$67.74 - x = 24.48$
$x = 67.74 - 24.48$
$x = 43.26$

7. Correct Answer: (C) Kanika, 8 pages

Total pages = 320

Kanika reads:

$\frac{3}{5} \times 320 = 192$ pages

Kartik reads = 200 pages

Difference:

$200 - 192 = 8$ pages

Kanika reads fewer pages by 8 pages.

8. Correct Answer: (A) $\frac{7}{13}$

Find HCF of 161 and 299.

$161 = 7 \times 23$

$299 = 13 \times 23$

Divide numerator and denominator by 23:

$$\frac{161}{299} = \frac{7 \times 23}{13 \times 23} = \frac{7}{13}$$

9. Correct Answer: (D) : $\frac{26}{59}$

Total students:

$27 + 32 = 59$

Boys:

$\frac{1}{3} \times 27 = 9$ failed

Boys passed:

$27 - 9 = 18$

Girls passed:

$\frac{1}{4} \times 32 = 8$

Total passed:

$18 + 8 = 26$

Fraction of group passed: $\frac{26}{59}$

10. Correct Answer: (B) ₽313.75

Total amount paid:

$5170.50 + 515.75 = 5686.25$

Change received:

$6000 - 5686.25 = 313.75$

6. Introduction to Geometry

Solutions

Worksheet – 1

1. **Correct Answer: D) Infinite**

   Number of points on a line is

2. **Correct Answer: B) 2 endpoints**

   A line segment has

3. **Correct Answer: A) 1**

   The number of lines that can be drawn through two points in a plane is

4. **Correct Answer: A) 1**

   The number of unique line segments that can be drawn with two given points is

5. **Correct Answer: (C) 5**

   How many points are there in the given figure?
   Points shown: O, B, C, E, D → 5 points

6. **Answer: (B) II**

   Which of the given figures shows a polygon?
   A polygon must be a simple closed figure made only of line segments.

   I → lines intersect (not a simple polygon)

   II → closed figure with only line segments

   III → has a curved edge

   IV → open figure

7. **Correct Answer: (C) Regular polygon**

   A polygon in which all sides are equal and all angles are equal is defined as a regular polygon.

8. **Correct Answer: (A) The Pentagon**

   The top view of the building has 5 sides, so it represents a pentagon.
   The famous building with this shape is named after the polygon itself.

9. **Correct Answer: (C) 12**

   The polygon shown is a star-shaped closed figure.
   By counting each straight side between consecutive labelled points, we find that the figure is formed by 12 line segments.

10. **Correct Answer: (D) AB**

    A diagonal of a polygon is a line segment joining two non-adjacent vertices.

    In pentagon ABCDE:

    AB → (this is a side)

Worksheet – 2

1. **Correct Answer: (B)**

   A reflex angle is an angle whose measure is greater than 180° and less than 360°.
   Among the given figures, option B shows an angle opening more than a straight angle.

2. **Correct Answer: (A) D**

   The interior region of an angle is the area inside the two rays that form the angle.

   Point D is located between the rays and , making it part of the interior region of ∠AOB.

   Point P lies outside the angle (exterior).

   Point Q lies outside the angle (exterior).

   Point O is the vertex of the angle and lies on the boundary, not in the interior region.

3. **Correct Answer: (D)**

   (A) ∠COD represents the angle marked 1, so this statement is correct.

(B) In ∠COB, the common endpoint of the rays is O, so the vertex is O. This is correct.

(C) ∠AOC is the angle formed by rays OA and OC with vertex O, so this statement is correct.

(D) ∠DOB should be formed by rays OD and OB, but the statement mentions rays OC and OB, which is incorrect.

4. Correct Answer: (D) 4

The given figure is a quadrilateral.
A quadrilateral has 4 interior angles (one at each vertex).

5. Correct Answer: (A) 30°

x, y, z are complementary angles. If the sum of x and y is 60°, then z is

Complementary angles have a total sum of 90°.
Given: x + y = 60°

So,z = 90° − 60° = 30°

6. Correct Answer: (B) ∠AGC

An angle is formed by two rays having a common vertex.

∠AOF → Rays OA and OF meet at O

∠COB → Rays OC and OB meet at O

∠BOA → Rays OB and OA meet at O

∠AGC → Points A, G, C do not have a common vertex and do not form two rays

Therefore, ∠AGC is not an angle in the given figure.

7. Correct Answer: (B) m and l

From the figure:

Line l is a vertical line on the left.

Line m is a vertical line on the right.

Lines n, p, q are slanting or horizontal and intersect other lines.

Lines l and m are parallel vertical lines, so they do not intersect.

Other pairs intersect at some point in the figure.

8. Correct Answer: (D) I and III

I. Two intersecting lines determine exactly one plane, not two → Incorrect

II. Planes that do not intersect are called parallel planes → Correct

III. Through two distinct points, only one line can pass → Incorrect

IV. Exactly one plane passes through a line and a point not on it → Correct

So, the incorrect statements are I and III.

9. Correct Answer: (A) AB and CD are intersecting rays

Ray notation: Both AB and CD have a fixed starting point (indicated by the dot at A and C) and an arrow at the other end (B and D). This signifies that they are rays, not lines or line segments.

Interaction: The two rays cross each other at a single point, which means they are intersecting rays.

10. Correct Answer: (D) 6

In the figure:

There are 3 vertical lines: $\boldsymbol{m_1, m_2, m_3}$

There are 2 horizontal lines: $\boldsymbol{l_1, l_2}$

Each vertical line intersects both horizontal lines.

So, Number of intersecting pairs = $\boldsymbol{3 \times 2 = 6}$

Worksheet – 3

1. Correct Answer: (C)

(A) In quadrilateral PQRS, sides PQ and RS do not share a common vertex, so they are not adjacent sides

(B) In quadrilateral ABCD, sides AB and BC share a common vertex B, so they are adjacent sides, not opposite

(C) In quadrilateral WXYZ, angles ∠XWZ and ∠XYZ are at opposite vertices, so they form a pair of opposite angles

(D) In quadrilateral MATH, angles ∠MAT and ∠MHT do not share a common arm, so they are not adjacent angles

2. Correct Answer: (C)

(A) Trapezoid: This has 4 straight sides. It is a quadrilateral.

(B) Parallelogram: This has 4 straight sides. It is a quadrilateral.

(D) Rectangle: This has 4 straight sides. It is a quadrilateral.

(C) Circle: A circle is a curved shape with no straight sides and no vertices. Therefore, it cannot be a quadrilateral.

3. Correct Answer: (B) 45°

Sum of interior angles of a quadrilateral = 360°

Sum of ratios = 12 + 20 + 21 + 19 = 72

Each part = 360° ÷ 72 = 5°

Angles are:

12 × 5 = 60°

20 × 5 = 100°

21 × 5 = 105°

19 × 5 = 95°

Smallest angle = 60°
Greatest angle = 105°

Difference = 105° − 60° = 45°

4. Correct Answer: (B)

Sum of interior angles of a quadrilateral must be 360°.

(A) 65 + 65 + 75 + 130 = 335°

(B) 45 + 25 + 140 + 150 = 360°

(C) 140 + 40 + 130 + 70 = 380°

(D) 120 + 65 + 110 + 75 = 370°

Only option (B) satisfies the condition.

5. Correct Answer: (D)

(A) AB and AD share a common vertex A, so they are adjacent sides

(B) ∠A and ∠C are at opposite vertices, so they are opposite angles

(C) ∠A and ∠D share a common arm AD, so they are adjacent angles

(D) CD and AD share a common vertex D, so they are adjacent sides, not opposite sides

6. Correct Answer: (B) 3 sides and 3 angles

A triangle has 3 sides and 3 angles, making a total of six elements.

7. Correct Answer: (D) on the intersection of sides AB and AC of the triangle

Point A is a vertex of the triangle and is formed by the intersection of sides AB and AC.

Not inside the triangle

Not outside the triangle

Not at intersection of AB and BC

8. Correct Answer: (B)

(A) Side PQ is opposite to angle ∠PRQ, not ∠PQR

(B) In ΔABC, the side opposite vertex B is AC

(C) Side LM is opposite vertex N, not L

(D) Angle ∠XYZ is at Y, so the opposite side is XZ, not YZ

9. Correct Answer: (C)

Which of the following angles cannot be the three angles of a triangle?

Solution:
Sum of angles of a triangle = 180°

(A) 30 + 60 + 90 = 180

(B) 30 + 55 + 95 = 180

(C) 45 + 35 + 90 = 170

(D) 45 + 60 + 75 = 180

10. Correct Answer: (B) 2 right angles

In a quadrilateral ABCD, ∠A = 60° and ∠B = 120°. Then ∠C + ∠D = ____

Sum of interior angles of a quadrilateral = 360°

∠C + ∠D = 360° − (60° + 120°)
∠C + ∠D = 360° − 180° = 180°

180° equals 2 right angles.

Worksheet – 4

1. **Correct Answer: (D) Infinite**

   A diameter can be drawn through the centre of a circle in any direction, so the number of diameters is unlimited.

2. **Correct Answer: (B) 18 cm**

   The radius of a circle having a diameter of 36 cm is

   Radius = ½ × Diameter
   Radius = ½ × 36 = 18 cm

3. **Correct Answer: (D) Major sector has a larger area than the minor sector**

   Which statement correctly distinguishes a major sector from a minor sector?

   A major sector always covers a larger area than a minor sector.

4. **Correct Answer: (B) $\overset{\frown}{AB}$**

   An arc of a circle is represented by placing a curved symbol over the endpoints of the arc.
   Therefore, arc AB is correctly written with a curved bar over AB.

5. **Correct Answer: (B) Sector**

   The area inside the two radii and an arc is called _____

   The region enclosed by two radii and the arc between them is called a sector of a circle.

6. **Correct Answer: (C) R**

   The center of a circle is the point where all diameters intersect.
   Since AB and CD are diameters and they intersect at R, point R is the center.

7. **Correct Answer: (A) minor arc and a chord**

   A segment of a circle is formed by a chord and an arc.
   The minor segment is bounded by the minor arc and the chord.

8. **Correct Answer: (A)**

   A diameter is a chord that passes through the centre of the circle.

   Figure (A): Line AB passes through the centre O → Diameter

   Figure (B): AB does not pass through the centre → Not a diameter

   Figure (C): AB forms an angle at the centre, not a straight line through it → Not a diameter

   Figure (D): AB is a chord but does not pass through the centre → Not a diameter

9. **Correct Answer: (C)**

   (A) A radius does not divide a circle into two semicircles ❌

   (B) A chord does not always divide a circle into semicircles ❌

   (C) A diameter divides a circle into two equal semicircles ✓

   (D) A semicircle is half of a circle, not a complete circle ❌

10. **Correct Answer: (C)**

    In which of the following figures is AB represented as a radius?

    A radius is a line segment joining the centre of the circle to a point on the circle.

    (A) AB passes through the centre → diameter

    (B) AB joins two points on the circle → chord

    (C) A is the centre and B lies on the circle → radius

    (D) AB is a chord →

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: A (The crescent-like shape is closed and doesn't intersect itself).**

   Definition: A simple closed figure is a shape that starts and ends at the same point and does *not* cross itself.

Analysis: B and C cross themselves (loops and overlapping lines).

D is a spiral and does not close.

2. Correct Answer: C (Polygon)

A closed figure made using three or more line segments is called a polygon.

3. Correct Answer: D (3)

A triangle has 3 sides, so it has 3 vertices.

4. Correct Answer: B (2)

From 11 to 5, the hour hand moves 6 hours. Each hour is $\mathbf{30}$°.
Total angle turned $= \mathbf{6} \times \mathbf{30}° = \mathbf{180}° =$ $\mathbf{2}$right angles.

5. Correct Answer: B (180°)

A reflex angle is greater than $\mathbf{180}$°and less than $\mathbf{360}$°.
$\mathbf{180}$°is a straight angle, not a reflex angle.

6. Correct Answer: C

An acute angle is less than $\mathbf{90}$°. Among the given figures, option C shows an angle clearly less than $\mathbf{90}$°.

7. Correct Answer: D (360°)

A full circle measures $\mathbf{360}$°. The angle at the centre around the whole circle is $\mathbf{360}$°.

8. Correct Answer: C (Zero angle)

When both hands overlap, they lie on the same line, forming an angle of $\mathbf{0}$°, which is a zero angle.

9. Correct Answer: B (54°)

Sum of angles of a quadrilateral $= \mathbf{360}$°

$\boldsymbol{x + (x + 10) + (x - 20) + 2x = 360}$
$\boldsymbol{5x - 10 = 360}$
$\boldsymbol{5x = 370 \Rightarrow x = 74}$
Angles are: $\mathbf{74}$°, $\mathbf{84}$°, $\mathbf{54}$°, $\mathbf{148}$°.
Smallest angle $= \mathbf{54}$°.

10. Correct Answer: A (4)

Diameter $= \mathbf{2} \times$radius.
If radius becomes 4 times, diameter also becomes 4 times.

11. Correct Answer: C

In a convex polygon, all interior angles are less than $\mathbf{180}$°and no part of the polygon caves inward.

Among the given figures, option C has all angles less than $\mathbf{180}$°and is therefore convex.

12. Correct Answer: B (4)

A line is counted only if it passes straight through point O.
From the figure, there are 4 distinct straight lines passing through point O (vertical, horizontal, and two slant lines).

13. Correct Answer: B (60°)

An acute angle is less than $\mathbf{90}$°. Among the given options, $\mathbf{60}$°is less than $\mathbf{90}$°.

14. Correct Answer: A (55°)

a and *b* are complementary angles. If $\boldsymbol{\angle a = 35}$°, then $\boldsymbol{\angle b}$is
Solution: Complementary angles add up to $\mathbf{90}$°.

$\boldsymbol{\angle b = 90° - 35° = 55°}$

15. Correct Answer: A (140°)

x and *y* are supplementary angles. If $\boldsymbol{\angle y = 40}$°, then $\boldsymbol{\angle x}$ is
Solution: Supplementary angles add up to $\mathbf{180}$°.

$\boldsymbol{\angle x = 180° - 40° = 140°}$

16. Correct Answer: D (70°)

Angles on a straight line add up to $\mathbf{180}$°.

$\boldsymbol{50° + x + 60° = 180°}$
$\boldsymbol{x = 180° - 110° = 70°}$

17. Correct Answer: B (12.5°)

Since $\boldsymbol{k \parallel m}$, the angles marked $\boldsymbol{3x}$°and $(\boldsymbol{x}+\mathbf{25})$°are corresponding angles.
Corresponding angles are equal.

$\boldsymbol{3x = x + 25}$
$\boldsymbol{2x = 25}$
$\boldsymbol{x = 12.5}$

18. Correct Answer: B (60°)

Since $\boldsymbol{AB \parallel CD}$, the angles marked $(\boldsymbol{x}-\mathbf{40})$°and $(\boldsymbol{3x}-\mathbf{20})$°are interior angles on the same side of the transversal.
Such angles are supplementary.

$(x - 40) + (3x - 20) = 180$

$4x - 60 = 180$
$4x = 240$
$x = 60$

19. Correct Answer: $x = 35°$

Since the lines are parallel, the angles $(3x+16)°$and $(5x-54)°$are alternate interior angles, so they are equal.

$3x + 16 = 5x - 54$
$16 + 54 = 5x - 3x$
$70 = 2x$
$x = 35$

20. Correct Answer: A (b, f)

Corresponding angles lie in the same relative position at each intersection of the transversal with the parallel lines.
Angle b (upper right at the top intersection) corresponds to angle f (upper right at the bottom intersection).

ONE OR MORE THAN ONE CORRECT OPTION TYPE

21. Correct Answers: A, B, C

Sum of the angles in a linear pair is:

The sum of angles in a linear pair is 180°.

Angles that add up to $180°$are called supplementary angles.

A linear pair forms a straight angle.

22. Correct Answers: A, B, D

Which of the following statements about a circle are correct?

A) All points on the boundary are equidistant from the centre — Correct

B) Radius is a fixed distance — Correct

C) A circle has vertices — Incorrect (circle has no vertices)

D) A circle is a simple closed curve — Correct

23. Correct Answers: A, B

A reflex angle is greater than $180°$and less than $360°$.

192° — Reflex

270° — Reflex

172° — Not reflex

360° — Complete angle, not reflex

24. Correct Answers: A, B, C

From the figure, ∠a and ∠b:

Share a common vertex and a common arm.

Lie on a straight line, forming a linear pair.

Properties of a linear pair:

They are supplementary angles.

Their sum is $180°$.

25. Correct Answers: A, B

In a quadrilateral ABCD, the sides are:

AB, BC, CD, DA

Opposite sides are the sides that do not share a common vertex:

AB is opposite to CD

BC is opposite to AD

ASSERTION AND REASON TYPE

26. Correct Answer: (a)(Both A and R are true, and R is the correct explanation)

A straight angle measures 180°, which is a basic definition in geometry.
A straight angle is formed when two rays lie on the same straight line, so the reason correctly explains why its measure is 180°.

27. Correct Answer: (a)(Both A and R are true, and R is the correct explanation)

A point has no length, breadth, or thickness—it has no dimensions.
A point is used to indicate a position or location, which explains why it has no size.

28. Correct Answer: (c)(A is true and R is false)

A line extends infinitely in both directions, so it does not have endpoints — Assertion (A) is true.
A line segment has two endpoints, so the statement "a line segment has no end" is false — Reason (R) is false.

29. Correct Answer: (c)(A is true and R is false)

A ray has one endpoint and extends endlessly in one direction, so Assertion (A) is true.
A ray is a part of a line, so the reason stating it is *not* part of a line is false.

30. Correct Answer: (a)
(Both A and R are true, and R is the correct explanation)

An obtuse angle is greater than a right angle (90°), so Assertion (A) is true.
An obtuse angle measures more than 90° and less than 180°, which correctly explains the assertion.

MATRIX MATCHING TYPE

31. Correct Matching:
A – 3
B – 1
C – 2
D – 4

Acute angle → less than 90°

Right angle → 90°

Obtuse angle → greater than 90° and less than 180°

Straight angle → 180°

32. Correct Matching:
A – 3
B – 1
C – 2
D – 4

Intersecting lines meet at one point

Parallel lines never meet even when extended

Perpendicular lines form right angles

A transversal intersects two or more lines at different points

COMPREHENSION TYPE

Passage - I

33. Correct Answer: B) Right

At 9 o'clock, the clock hands are perpendicular and form an angle of 90°, which is a right angle.

34. Correct Answer: C) Acute angle

An angle measuring less than 90° is called an acute angle.

35. Correct Answer: C) 90°

A right angle measures 90°.

Passage - II

36. Correct Answer: A) G

Ray QP is the slanted ray starting from Q and going upward.
Point G lies exactly on this ray.

37. Correct Answer: C) E

Ray QR is the horizontal ray from Q to the right.
Point E lies on this ray between Q and R.

38. Correct Answer: D) A

G lies on ray QP

E and R lie on ray QR

A lies away from both rays

INTEGER TYPE

39. Answer: 36 cm

Diameter = 2 × radius = 2 × 18 = 36 cm

40. Answer: 16 cm

2 × diameter = 64
Diameter = 64 ÷ 2 = 32 cm
Radius = diameter ÷ 2 = 32 ÷ 2 = 16 cm

41. Answer: 4

A complete angle = 360°
Right angle = 90°
Number of right angles = 360 ÷ 90 = 4

42. Answer: 65°

Since $\ell \parallel m$, angles $\angle a$ and $\angle e$ are corresponding angles.
Corresponding angles between parallel lines are equal.

So, $\angle e = \angle a = 65°$

43. Answer: 4 pairs

At each intersection of two straight lines, there are 2 pairs of vertically opposite angles.

In the given figure, there are two intersections.

So, Total pairs of vertically opposite angles = 2 + 2 = 4 pairs

Previous Year Questions

1. Correct Answer: B

Statement–1 is true (definition of a chord).

Statement–2 is true (definition of a segment of a circle).

2. Correct Answer: D

Sum of angles of a triangle = 180°

A: 30 + 80 + 75 = 185°

B: 45 + 65 + 65 = 175°

C: 95 + 20 + 68 = 183°

D: 90 + 45 + 45 = 180°

3. Correct Answer: D

Acute angle < 90°

Right angle = 90°

Obtuse angle > 90° and < 180°

Reflex angle > 180°

Check options:

A: Obtuse + Acute → can be greater than 180°

B: Reflex + Acute → always greater than 180°

C: Two obtuse angles → sum greater than 180°

D: Two right angles → 90° + 90° = 180°, not greater

4. Correct Answer: B

∠AOC is formed by rays OA and OC.

Points F and H lie outside the region between OA and OC.

Point E lies below OC, also outside ∠AOC.

So, F, E and H are exterior points of ∠AOC.

5. Correct Answer: D

A polygon is a simple closed figure made only of straight-line segments.

D: Closed figure made of straight-line segments

6. Correct Answer: C

A polygon is a simple closed figure made only of straight line segments.

C: Contains a curved line

7. Correct Answer: D

Group 1 (Horizontal–diagonal lines):
Lines l and m are parallel to each other.
This is 1 pair (l ∥ m).

Group 2 (Vertical–diagonal lines):
Lines a, b, and c are all parallel to each other.
From these three lines, we can form the following pairs:
Line a and Line b (a ∥ b)
Line a and Line c (a ∥ c)
Line b and Line c (b ∥ c)

This gives us 3 pairs.

Adding them together, 1 + 3 = 4 total pairs of parallel lines.

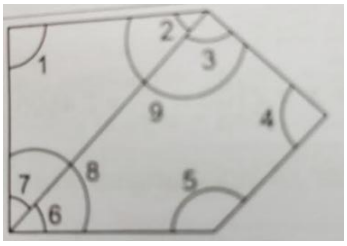
8. Correct Answer: D

A closed figure has no opening and its starting and ending points meet.

Figure (a) and Figure (e) are closed

Number of closed figures = 2

9. Correct Answer: B



Number of angles formed=9

10. Correct Answer: C

Statement–1: True (this is the definition of a chord).

Statement–2: False (perpendicular lines form an angle of 90°, not 180°).

7. Data Handling

Solutions

Worksheet – 1

1. Correct Answer: C) 10

Favourite Seasons (Tally Chart)

Winter = 5
Spring = 2
Summer = 6
Fall = 4

People who choose **Fall or Summer** = 4 + 6 = **10**

2. Correct Answer: C) 4

Average Problem

Given numbers:
a, a+2, a+4, a+12, a+7

Sum = 5a + (2+4+12+7)
= 5a + 25

Average = 9

$$\frac{5a + 25}{5} = 9$$
$$5a + 25 = 45$$
$$5a = 20$$
$$a = 4$$

3. Correct Answer: C) 35 and 24

Range of Marks

Mathematics Marks:
25, 40, 30, 35, 21, 45, 23, 33, 10, 29
Maximum = 45
Minimum = 10

Range = 45 − 10 = **35**

Statistics Marks:
30, 39, 23, 42, 20, 40, 25, 30, 18, 19
Maximum = 42
Minimum = 18

Range = 42 − 18 = **24**

4. Correct Answer: B) 4326

Range of Candidates

Numbers:
12464, 14731, 13213, 14219, 15613, 11287, 14162

Maximum = 15613
Minimum = 11287

Range = 15613 − 11287 = **4326**

5. Correct Answer: C) 17 : 20

Step 1: Count the tally marks for "People who play" (10–13)
Looking at the first table:

- There are 3 full bundles of 5 and 2 extra strokes.
- Calculation: (3 × 5) + 2 = 17

Step 2: Count the tally marks for "People who go to work" (25–28)
Looking at the bottom row of the second table:

- There are 4 full bundles of 5 and no extra strokes.
- Calculation: 4 × 5 = 20

Step 3: Determine the ratio

- Ratio = 17 : 20

6. Frequency of 8

Observation given:
10, 5, 8, 9, 8, 11, 12, 4, 12, 8, 8

Count how many times **8** appears:

8 → 4 times

7. Correct Answer: A) 40

Step 1: Find the maximum size
Maximum = 50

Step 2: Find the minimum size
Minimum = 10

Step 3: Calculate the range

Range = 50 − 10 = 40

8. **Sunday = 23 motorcycles**

Monday = 13 motorcycles

Tuesday = 10 motorcycles

Wednesday = 12 motorcycles

Thursday = 16 motorcycles

Friday = 17 motorcycles

Saturday = 20 motorcycles

Now add them:

$23 + 13 + 10 + 12 + 16 + 17 + 20$
$= 112$

9. **Correct Answer: B) Observation**

Explanation:
Each individual value collected in a data set is called an **observation**.

Worksheet – 2

1. **Correct Answer: B) 120**

Pie chart

Central angle = 72°
Value = 24

We know,

$$\frac{72}{360} = \frac{24}{\text{Total}}$$

$$\text{Total} = \frac{24 \times 360}{72}$$
$$= 24 \times 5 = 120$$

2. **Correct Answer: C) 7**

Bar graph

5 cm = 40 units

1 cm = 40 ÷ 5 = 8 units

For 56 units:
Height = 56 ÷ 8 = **7 cm**

3. **Correct Answer: B) 4750**

Pictograph

Each picture = 1000 units

4 full pictures = 4 × 1000 = 4000
¾ picture = ¾ × 1000 = 750

Total = 4000 + 750 = **4750**

4. **Correct Answer: B) 13**

Range

Range = Highest – Lowest

15 = 28 – Lowest

Lowest = 28 – 15 = **13**

5. **Correct Answer: A) 108°**

Pie chart (30%)

30% of 360°

$$= \frac{30}{100} \times 360$$
$$= 108°$$

6. **Correct Answer: 20%**

Percentage

Central angle = 72°

$$\text{Percentage} = \frac{72}{360} \times 100$$
$$= 20\%$$

7. **Correct Answer: C) 250**

Bar Graph

Scale:
1 cm = 2m units

Length given = 1.75 cm
Represents = 875 units

So,

$1.75 \text{ cm} = 875$

$$1 \text{ cm} = \frac{875}{1.75} = 500$$

But 1 cm = 2m

So,

$2m = 500$
$m = 250$

8. **Correct Answer: D) 20%**

Pie Graph

Sector angle = 72°

$$\text{Percentage} = \frac{72}{360} \times 100$$
$$= 20\%$$

9. **Correct Answer: B) 126°**

Ratio Problem

Ratio of Total : A : B = 20 : 5 : 8

First find share of C.

Total parts = 20

A = 5
B = 8

C = 20 − (5 + 8)
C = 20 − 13 = 7

Now find central angle of C:

$\frac{7}{20} \times 360$
$= 126°$

10. Correct Answer: B) 11250

Pictograph

1 picture = 375 units
30 pictures = ?

$375 \times 30 = 11250$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: C) 5

Tallies are usually marked in a bunch of

Tally marks are grouped in 5 (four vertical lines and the fifth line crossing them).

Example:

|||| = 4

||||– = 5

2. Correct Answer: B) 9

Count pictures

Ben → 3 pictures
3 × 3 = 9 oranges

Tim → 4 pictures
4 × 3 = 12 oranges

Pam → 6 pictures
6 × 3 = 18 oranges

John → 2 pictures
2 × 3 = 6 oranges

Step 2: Find the difference (Pam and Ben)

Pam = 18
Ben = 9

Difference = 18 − 9 = 9

3. B) Fantasy

Fashion magazines → 4 votes

Drama → 3 votes

Fantasy → 6 votes

Sports fiction → 5 votes

Biographies → 1 vote

Historical fiction → 4 votes

The maximum number of votes is 6.

So, the most preferred kind is:

4. Correct Answer: B) 48

The monkey ate a total of **12 bananas** over the given period.

Now, the total number of bananas is **multiplied by 4**.

$12 \times 4 = 48$

5. Correct Answer: D) 198

Rose → 7 pictures

Tulip → 5 pictures

Orchid → 6 pictures

Lily → 4 pictures

Find total pictures

$7 + 5 + 6 + 4 = 22$ pictures

Step 3: Convert into number of flowers

Each picture = 9 flowers

$22 \times 9 = 198$

6. Correct Answer: D) Steel and coal

Revenue Question

From the bar graph (Revenue in $ thousands):

- **Oil** = 100
- **Steel** = 150
- **Coal** = 200
- **Gold** = 50

We need two commodities whose total is **350**.

Check combinations:

- Steel + Coal = 150 + 200 = 350 ✅

7. Correct Answer: B) 1,200

Total Vehicles

From the bar graph:

- 8–9 h = 50
- 9–10 h = 400
- 10–11 h = 250
- 11–12 h = 250
- 12–13 h = 150

- 13–14 h = 100

Now add:

$50 + 400 + 250 + 250 + 150 + 100 = 1200$

8. 2006

2003 → 400 trees

2004 → 550 trees

2005 → 500 trees

2006 → 600 trees

The maximum number of trees planted is 600.

So, the maximum number of trees was planted in the year:

9. Correct Answer: B) Class mark

The mid-value of a class interval is called

The mid-value of a class interval is known as the **Class mark**.

10. Correct Answer: C) ₽0.48

Range of gasoline prices

Given prices:
₽1.79, ₽1.61, ₽1.96, ₽1.84, ₽1.75, ₽2.09

Step 1: Highest price = ₽2.09
Step 2: Lowest price = ₽1.61

Range = Highest − Lowest

$2.09 - 1.61 = 0.48$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. Correct Answers: A, B and D

Frequency of 12

Given observations:
11, 4, 12, 9, 8, 11, 12, 4, 12, 8, 8

Count 12:

12 appears **3 times**.

Among the options:
A) 2 + 1 = 3
B) 3 × 1 = 3
C) 4
D) 3

Since more than one correct answer is allowed:

12. Correct Answer: B) 40 ×2 D) 80

Videos rented in October

Key: 1 circle = 20 videos

October has **4 circles**.

$4 \times 20 = 80$

13. Correct Answers: B and D

Bar graph scale

1 cm = 10 units
Bar length = 5.2 cm

$5.2 \times 10 = 52$
Among the options:

B) 52 units
D) (2 × 26) units = 52 units

Since more than one correct answer is allowed:

14. Correct Answer: D) 26.5 C) 26+0.5

Total number of students

From the bar graph:

- **Sports** → 7
- **Dance** → 5
- **Painting** → 6
- **Drama** → 8

Total:

$7 + 5 + 6 + 8 = 26.5$

Among the options:

Since none shows exactly 26, the correct total is:

15. Correct Answers: A and B

Difference (Soccer and Tennis)

From the graph:

- **Tennis** → 30
- **Soccer** → 20

Difference:

$30 - 20 = 10$

Among the options:

A) 10
B) 5 × 2 = 10

Since more than one correct answer is allowed:

ASSERTION AND REASON TYPE

16. Correct Answer: (a) Both A and R are true, and R is the correct explanation

Assertion (A): Pictographs use pictures to represent data. ✅ (True)
Reason (R): Pictures help us quickly understand the data. ✅ (True)

The reason correctly explains why pictographs use pictures — to make data easy and quick to understand.

17. Correct Answer: (c) A is true but R is false

Assertion (A): A bar graph uses rectangular bars to represent data. ✅ (True)
Reason (R): All bars in a bar graph are of different width. ❌ (False)

In a bar graph, all bars have **equal width**, not different widths.

18. Correct Answer: (c) A is true but R is false

Assertion (A): Data is a collection of numbers or facts. ✅ (True)
Reason (R): Data is used only in mathematics. ❌ (False — data is used in science, business, sports, daily life, etc.)

19. Correct Answer: (a) Both A and R are true, and R is the correct explanation

Assertion (A): Bar graphs are better than pictographs for large data. ✅ (True)
Reason (R): It is difficult to draw too many pictures in a pictograph. ✅ (True)

The reason correctly explains why bar graphs are better for large data.

20. Correct Answer: (c) A is false and R is false

Assertion (A): All bar graphs have horizontal bars only. ❌ (False — bar graphs can have vertical or horizontal bars.)
Reason (R): Vertical bars cannot show data. ❌ (False — vertical bars also show data.)

MATRIX MATCHING TYPE

21. Correct Answer: A–3, B–1, C–2, D–4

A. Pictograph → 3. Data shown in Diagrams
A pictograph represents data using pictures or symbols.

B. Bar Graph → 1. Bars of equal width
A bar graph shows data using rectangular bars of equal width.

C. Data → 2. Collection of facts
Data means information or facts collected for study.

D. Tally Marks → 4. Grouped in fives using marks
Tally marks are written in groups of five to make counting easier.

22. Correct Answer: A–2, B–1, C–3, D–4

A. Bar Graph → 2. Vertical or horizontal bars
A bar graph represents data using vertical or horizontal bars.

B. Pictograph → 1. Pictures to show data
A pictograph uses pictures or symbols to represent data.

C. Table → 3. Organised rows and columns
A table arranges data in organised rows and columns.

D. Line Graph → 4. Connected line segments
A line graph shows data points connected by straight line segments.

COMPREHENSION TYPE

Passage - I

23. Correct Answer: B) Wednesday

On which day were the least pairs of shoes sold?

The smallest number is **30 pairs**.
So, the least sales were on **Wednesday**.

24. Correct Answer: C) 50

How many pairs of shoes were sold on Thursday?

From the graph, Thursday shows **50 pairs**.

25. Correct Answer: C) 20

How many more pairs of shoes were sold on Friday than on Monday?

Friday = 60 pairs
Monday = 40 pairs

Difference = 60 − 40 = **20 pairs**

Passage - II

26. Correct Answer: B) Village B

Which village had the highest number of deaths?

The highest percentage is **35%**, which belongs to **Village B**.

27. Correct Answer: A) Village A and Village C

Which two villages together make 50% of the total deaths?

Village A (30%) + Village C (20%) = **50%**

28. Correct Answer: C) 1050

How many people died in Village B?

Village B = **35%**

If total deaths = 3000 (as implied from options):

35% of 3000

$= \frac{35}{100} \times 3000$

= 1050

INTEGER TYPE

29. Correct Answer: x + 4

The mean of the observations

x, x+2, x+4, x+6, x+8

Step 1: Find the sum

= x + (x+2) + (x+4) + (x+6) + (x+8)
= 5x + 20

Step 2: Find the mean

Mean = (5x + 20) ÷ 5
= x + 4

30. Correct Answer: 27

Range of the observations

7, 21, 11, 5, 9, 20, 32, 28

Highest value = 32

Lowest value = 5

Range = 32 − 5 = **27**

31. Correct Answer: 10

Find the mean of

8, 12, 10, 6, 14

Sum = 8 + 12 + 10 + 6 + 14 = 50

Mean = 50 ÷ 5 = **10**

32. Correct Answer: 25%

In a pie chart, a sector has a central angle of 90°.

Total angle of a circle = 360°

Percentage = (90 ÷ 360) × 100
= 25%

Previous Year Questions

1. Correct Answer: Option A) 11 : 25

Ratio of tickets sold for **Kanpur** to total tickets sold for **Jaipur and Lucknow**

Kanpur = 110
Jaipur + Lucknow = 125 + 125 = 250

So,
110 : 250

Divide both by 10: = 11 : 25

2. Correct Answer: Option B) ₽75,400

Total amount collected for Delhi and Ahmedabad

Delhi tickets = 160
Cost per ticket = ₽190

Amount from Delhi =
160 × 190 = ₽30,400

Ahmedabad tickets = 180
Cost per ticket = ₽250

Amount from Ahmedabad =
180 × 250 = ₽45,000

Total amount =
30,400 + 45,000 = **₽75,400**

3. Correct Answer: A) 24

Tuesday + Thursday
= 16 + 24
= 40

Sunday = 64

Difference = 64 − 40
= **24**

4. Correct Answer: C) 15/31

Friday + Saturday + Sunday
= 40 + 16 + 64
= 120

Total for whole week:
40 + 16 + 48 + 24 + 40 + 16 + 64
= 248

Fraction = 120 / 248

Divide by 8:

= **15 / 31**

5. Find the ratio:

(2017 + 2021) : (2016 + 2019)

2017 + 2021
= 70 + 70
= **140**

2016 + 2019
= 80 + 100
= **180**

So ratio = 140 : 180

Divide both by 20:

= **7 : 9**

6. Correct Answer: D) 3000

Runs in 2018, 2019 and 2021 together

= 2400 + 2700 + 2400
= **7500**

Runs in 2017 and 2020 together

= 2700 + 1800
= **4500**

Difference

= 7500 − 4500
= **3000**

7. Given data (24 matches):

2, 3, 2, 1, 0, 4, 3, 1, 2, 0, 3, 2, 1, 3, 0, 2, 5, 2, 4, 3, 1, 2, 1, 3

Total= 15

8. Correct Answer: A) 200

Maximum production = 350 (June)

Minimum production = 150 (January)

Difference = 350 − 150

= 200

9. Correct Answer:

(i) **40**
(ii) **22**

(i) Symbols needed for 2015 and 2018 together

2015 + 2018
= 5000 + 3000
= 8000 students

Since 1 symbol = 200 students

Number of symbols = 8000 ÷ 200
= **40**

(ii) How many less symbols for 2019 than 2016

2016 = 6400
2019 = 2000

Difference in students = 6400 − 2000
= 4400

Number of symbols = 4400 ÷ 200
= **22**

10. Step 1: Find total articles

Total = 75 + 150 + 100 + 250 + 150
= **725**

Step 2: Fraction produced by company S

S = 250

Fraction = 250 / 725

Divide both by 25:

= 10 / 29

8. Introduction to Algebra

Solutions

Worksheet – 1

1. Correct Answer: C) 5

If $3P + 1 = 16$
$3P = 16 - 1 = 15$
$P = 15 \div 3 = 5$

2. Correct Answer: A) 3

If $7z - 5 = 16$
$7z = 21$
$z = 21 \div 7 = 3$

3. Correct Answer: A) 15

If $\frac{x}{3} - 5 = 0$
$\frac{x}{3} = 5$
$x = 5 \times 3 = 15$

4. Correct Answer: C) 9

If $x + 3 = 12$
$x = 12 - 3 = 9$

5. Correct Answer: D) 1

If $6(2a - 1) + 8 = 14$
$12a - 6 + 8 = 14$
$12a + 2 = 14$
$12a = 12$
$a = 1$

6. Correct Answer: A) 2

If $16(x + 7) = 144$
$x + 7 = 144 \div 16 = 9$
$x = 9 - 7 = 2$

7. Correct Answer: D) 8

If six times a number is 48
Let the number be x.
$6x = 48$
$x = 48 \div 6 = 8$

8. Correct Answer: C) 6

Given $6a = 30$
To find a, divide both sides by 6.

9. Correct Answer: B) 8

Given $\boldsymbol{y - 8 = 6}$
To isolate $\boldsymbol{y}$, add 8 to both sides.

10. Correct Answer: A) 3

Given $\frac{y}{3} = 7$
To find $\boldsymbol{y}$, multiply both sides by 3.

Worksheet – 2

1. Correct Answer: D) 8

Number of terms

Expression: $6x^2y + 2xy + 5xy^2 + 9x + 6y + 20x^2y^2 + (-2x^3) - 7$

Count each term separated by $+$or $-$:

Total terms = 8

2. Correct Answer: C

Identify the false statement

A) True (numerical coefficient is a constant)

B) True (literal coefficient is a variable)

C) False (literal coefficient is not a constant)

D) True ($\boldsymbol{-5}$ is numerical coefficient of $\boldsymbol{-5x}$)

3. Correct Answer: B) 26

Group like terms and add numerical coefficients of maximum like terms

Expression:$12x + 25y - 25 + 25xy + 7xy^2 - 5x^2y - 5xy + 10x^2y^2 + 6xy$

Like terms with maximum count: $\boldsymbol{xy}$terms

$25xy - 5xy + 6xy = 26xy$

Sum of numerical coefficients = 26

4. Correct Answer: C) Binomial

Expression with two unlike terms

An expression with two unlike terms is called a binomial.

5. Correct Answer: B) Monomial

Like terms can be combined into a single term. For example, 2x + 3x + 4x simplifies to just 9x.

Since the expression can be simplified into one single term, it is actually a Monomial.

6. Correct Answer: C) $2x^2 - 4$

Which of the following is a binomial expression?

A binomial has exactly two unlike terms after simplification.

A) $x^2 + (2x)x = x^2 + 2x^2 = 3x^2 \rightarrow$ one term (monomial)
B) $4xy + (2x)(2y) = 4xy + 4xy = 8xy \rightarrow$ one term (monomial)
C) $2x^2 - 4 \rightarrow$ two unlike terms
D) $(3x)y - x^2 - (3y)x = 3xy - x^2 - 3xy = -x^2 \rightarrow$ one term

7. Correct Answer: B

Which of the following is a trinomial expression?

A trinomial has three unlike terms after simplification.

A) $(2x^2)y - 4x + (2x)(xy)$
$= 2x^2y - 4x + 2x^2y = 4x^2y - 4x \rightarrow$ 2 terms

B) $x^2y - 3x^2 - (2x)(2y)$
$= x^2y - 3x^2 - 4xy \rightarrow$ 3 unlike terms

C) $-2xy + xy + y^2 = -xy + y^2 \rightarrow$ 2 terms

D) $(2x)y^2 - y^2(2xy) + (-y)(2xy)$
$= 2xy^2 - 2xy^3 - 2xy^2 = -2xy^3 \rightarrow$ 1 term

8. Correct Answer: D

Which of the following is a monomial expression?

A monomial has only one term after simplification.

A) $-2xy^2 + x^2y + xy(y^2) \rightarrow$ 3 terms
B) $(4p)(4q) + (5p)(-q) + (7p)^2 \rightarrow$ 3 terms
C) $-2xy + xy + y^2 = -xy + y^2 \rightarrow$ 2 terms
D) $(2x)y^2 - y^2(2xy) + (-y)(2xy)$
$= 2xy^2 - 2xy^3 - 2xy^2 = -2xy^3 \rightarrow$ 1 term

9. Correct Answer: A) 3

Degree of the algebraic expression $4x^2 + 3xyz + y^2$

Degree of $4x^2 = 2$

Degree of $3xyz = 1 + 1 + 1 = 3$

Degree of $y^2 = 2$

Highest degree = 3

10. Correct Answer: A

Identify like terms

Like terms must have same variables with same powers.

A) $7xy, \frac{8}{7}xy \rightarrow$ same variables and powers
B) $9x^3y^2, -7x^3y^3 \rightarrow$ powers of y different
C) $71xy^2, \frac{8}{3}x \rightarrow$ variables different
D) $3x^2y, 8x \rightarrow$ powers of x different

Worksheet – 3

1. Correct Answer: D) 1

If $15(L - 1) + 4(L + 3) = 2(7 + L)$

$15L - 15 + 4L + 12 = 14 + 2L$
$19L - 3 = 14 + 2L$
$17L = 17$
$L = 1$

2. Correct Answer: C) 3

If $8(C - 3) - (6 - 2C) = 2(C + 2) - 5(5 - C)$

$8C - 24 - 6 + 2C = 2C + 4 - 25 + 5C$
$10C - 30 = 7C - 21$
$3C = 9$
$C = 3$

3. Correct Answer: A) $-\frac{5}{6}$

If $5(I + 2) - 2(3 - 4I) = 3(I + 5) - 4(4 - I)$

$5I + 10 - 6 + 8I = 3I + 15 - 16 + 4I$
$13I + 4 = 7I - 1$
$6I = -5$
$\boldsymbol{I = -\frac{5}{6}}$

4. Correct Answer: C) 14

Find Y if $\frac{Y}{4}+\frac{1}{2}=4$

$\frac{Y}{4}=4-\frac{1}{2}=\frac{7}{2}$

$Y=\frac{7}{2}\times 4=14$

5. Correct Answer: B) $\frac{9}{2}$

If $\frac{x}{2}-\frac{1}{4}=\frac{x}{3}+\frac{1}{2}$

Multiply both sides by 12:

$6x-3=4x+6$

$2x=9$

$x=\frac{9}{2}$

6. Correct Answer: A) −9

If $\frac{x-4}{7}-\frac{x+4}{5}=\frac{x+3}{7}$

Multiply both sides by 35:

$5(x-4)-7(x+4)=5(x+3)$

$5x-20-7x-28=5x+15$

$-2x-48=5x+15$

$-7x=63$

$x=-9$

7. Correct Answer: B) 2

If $\frac{5x+2}{2x+3}=\frac{12}{7}$

Cross-multiply:

$7(5x+2)=12(2x+3)$

$35x+14=24x+36$

$11x=22$

$x=2$

8. Correct Answer: C) −3

If $\frac{x+7}{3x+16}=\frac{4}{7}$

Cross-multiply:

$7(x+7)=4(3x+16)$

$7x+49=12x+64$

$-5x=15$

$x=-3$

9. Correct Answer: A) $\frac{3}{7}$

If $\frac{3t+1}{16}-\frac{2t-3}{7}=\frac{t+3}{8}+\frac{3t-1}{14}$

LCM of 16,7,8,14 = 112

Multiply throughout by 112:

$7(3t+1)-16(2t-3)=14(t+3)+8(3t-1)$

$21t+7-32t+48=14t+42+24t-8$

$-11t+55=38t+34$

$-49t=-21$

$t=\frac{3}{7}$

10. Correct Answer: C) $\frac{9}{23}$

If $\frac{8P-5}{7P+1}=\frac{-2}{4}$

Simplify RHS: $\frac{-2}{4}=\frac{-1}{2}$

Cross-multiply:

$2(8P-5)=-(7P+1)$

$16P-10=-7P-1$

$23P=9$

$P=\frac{9}{23}$

Worksheet – 4

1. Correct Answer: B) x = 3

Given equation: $7x=4x+9$

Subtract $4x$ from both sides:

$7x-4x=9$

$3x=9$

Divide by 3:

$x=3$

2. Correct Answer: C) $8a^2b$

The sum of $3a^2b, -4a^2b, 9a^2b$

Combine like terms:

$(3-4+9)a^2b=8a^2b$

3. Correct Answer: B) $7x^2-7x+7$

Add $6x^2-4x$, $2x-3x^2+4$, $4x^2-5x+3$

Combine like terms:

x^2: $6-3+4=7$

x: $-4+2-5=-7$

constants: $4+3=7$

Result: $7x^2-7x+7$

4. Correct Answer: B) $4x^2y+6xy$

Subtract $-5x^2y$ from $6xy-x^2y$

$(6xy-x^2y)-(-5x^2y)=6xy+4x^2y$

Rearrange: $4x^2y+6xy$

5. Correct Answer: D) $7b - 9a$

Subtract $3a - 2b$ from $5b - 6a$

$(5b - 6a) - (3a - 2b)$

$= 5b - 6a - 3a + 2b$

$= 7b - 9a$

6. Correct Answer: C

Sum of $\frac{7}{2}x^3 - \frac{3}{2}x^2 + 5, \frac{7}{2}x^2 - \frac{3}{2}x^3 + \frac{1}{3}, \frac{3}{2}x^3 - \frac{1}{2}x^2 + \frac{2}{3}$

Combine like terms:

x^3: $\frac{7}{2} - \frac{3}{2} + \frac{3}{2} = \frac{7}{2}$

x^2: $-\frac{3}{2} + \frac{7}{2} - \frac{1}{2} = \frac{3}{2}$

Constants: $5 + \frac{1}{3} + \frac{2}{3} = 6$

$= \frac{7}{2}x^3 + \frac{3}{2}x^2 + 6$

7. Correct Answer: A

Subtract $-\frac{7}{3}y^2 + \frac{1}{2}y$ from $\frac{1}{3} - \frac{5}{3}y^2$

$\left(\frac{1}{3} - \frac{5}{3}y^2\right) - \left(-\frac{7}{3}y^2 + \frac{1}{2}y\right)$

$= \frac{2}{3}y^2 - \frac{1}{2}y + \frac{1}{3}$

8. Correct Answer: C

From $8 - y + 2y^2$ taking away $y^2 - 7 - 2y$

$(8 - y + 2y^2) - (y^2 - 7 - 2y)$

$= y^2 + y + 15$

9. Correct Answer: B

What should be taken away from

$3x^2 - 4y^2 + 5xy + 20$ to get $-x^2 - y^2 + 6xy + 20$?

Required expression:

$(3x^2 - 4y^2 + 5xy + 20) - (-x^2 - y^2 + 6xy + 20)$

$= 4x^2 - 3y^2 - xy$

10. Correct Answer: A

Simplify: $3x - [2y - \{x - (y - x)\}]$

$x - (y - x) = 2x - y$

$2y - (2x - y) = 3y - 2x$

$3x - (3y - 2x) = 5x - 3y$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: A

Price per dozen = $x - 4$

Number of dozens = $y - 6$

Total amount to be paid: $(x - 4)(y - 6)$

2. Correct Answer: B

New length = $\ell + 6$

New breadth = $b - 9$

Area of rectangle: $(\ell + 6)(b - 9) = \ell b - 9\ell + 6b - 54$

3. Correct Answer: D

Product of $6x(x^2 + 4y) = 6x^3 + 24xy$

Product of $3x^2(2x - 3y) = 6x^3 - 9x^2y$

Subtracting: $(6x^3 + 24xy) - (6x^3 - 9x^2y) = 24xy + 9x^2y$

4. Correct Answer: C

Sum of three expressions = $6y + 13x - 7x^2$

Sum of two given expressions: $(3x^2 - 4x + 5y) + (3x - 10x^2 - 8y) = -7x^2 - x - 3y$

Third expression: $(6y + 13x - 7x^2) - (-7x^2 - x - 3y) = 14x + 9y$

5. Correct Answer: A

Adjacent sides: $6x^2 - 4y^2$ and $2x^2 - 8xy$

Perimeter $= 2(l + b)$:

$2[(6x^2 - 4y^2) + (2x^2 - 8xy)]$

$= 2(8x^2 - 4y^2 - 8xy)$

6. Correct Answer: C

Present population $= \frac{x^2}{2}$

After 1 week, it becomes $3y$ times.

After 5 weeks:

$$\frac{x^2}{2}(3y)^5 = \frac{243x^2y^5}{2}$$

7. Correct Answer: B

$A = x^2 + 2x + 2, B = 5x^2 - x + 4$

$2A - B = 2(x^2 + 2x + 2) - (5x^2 - x + 4)$

$= -3x^2 + 5x = x(5 - 3x)$

8. Correct Answer: A

Let the number be x.

$\frac{x}{3}+5=2x$

$x+15=6x$

$15=5x \Rightarrow x=3$

9. Correct Answer: B

$\frac{x}{8}-\frac{1}{2}=\frac{x}{6}-2$

Multiply by 24:

$3x-12=4x-48$

$x=36$

$\frac{x}{4}=\frac{36}{4}=9$

10. Correct Answer: A

$A=-2x+3y-5z, B=-4x-9y-3z, C=6x+6y+8z$

$A+B+C=(-2x-4x+6x)+(3y-9y+6y)+(-5z-3z+8z)$

$=0+0+0=0$

11. Correct Answer: D

$3(O-4)+2(4O-5)=5(O+2)+16$

$3O-12+8O-10=5O+10+16$

$11O-22=5O+26$

$6O=48 \Rightarrow O=8$

12. Correct Answer: D

$7x-5=3x+3$

$4x=8 \Rightarrow x=2$

13. Correct Answer: B

Area $=s^2=64$

$s=\sqrt{64}=8$

14. Correct Answer: A

$9\frac{1}{4}=y-1\frac{1}{3}$

$\frac{37}{4}=y-\frac{4}{3}$

$y=\frac{37}{4}+\frac{4}{3}=\frac{111+16}{12}=\frac{127}{12}$

15. Correct Answer: B

Let the number be $\boldsymbol{x}$.

"Half of a number added to 10 is 15":

$$\frac{\boldsymbol{x}}{\boldsymbol{2}}+\boldsymbol{10}=\boldsymbol{15}$$

16. Correct Answer: B

Let the number be $\boldsymbol{x}$.

$\frac{\boldsymbol{x}}{\boldsymbol{2}}+\boldsymbol{18}=\boldsymbol{46}$

$\frac{\boldsymbol{x}}{\boldsymbol{2}}=\boldsymbol{28} \Rightarrow \boldsymbol{x}=\boldsymbol{56}$

17. Correct Answer: B

Let the numbers be $\boldsymbol{x}$and $\boldsymbol{x}+\boldsymbol{5}$.

$\boldsymbol{x}+(\boldsymbol{x}+\boldsymbol{5})=\boldsymbol{29}$

$\boldsymbol{2x}=\boldsymbol{24} \Rightarrow \boldsymbol{x}=\boldsymbol{12}$

Numbers are $\boldsymbol{12}$and $\boldsymbol{17}$.

18. Correct Answer: D

Let son's present age = $\boldsymbol{x}$.

Man's present age = $\boldsymbol{4x}$.

Five years ago:

$\boldsymbol{4x-5=9(x-5)}$

$\boldsymbol{4x-5=9x-45}$

$\boldsymbol{5x=40} \Rightarrow \boldsymbol{x=8}$

Man's present age $=\boldsymbol{4x=32}$.

19. Correct Answer: B

Let the number be $\boldsymbol{x}$.

$\boldsymbol{3x-5=16}$

$\boldsymbol{3x=21} \Rightarrow \boldsymbol{x=7}$

20. Correct Answer: D

$\frac{2}{3}T-2\frac{1}{2}=3\frac{1}{2}$

$\frac{2}{3}T=\frac{7}{2}+\frac{5}{2}=6$

$T=6\times\frac{3}{2}=9$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

21. Correct Answers: A, B and D

Which of the following are like terms?

Like terms must have same variables with same powers.

$\boldsymbol{6m^2n}$

$\boldsymbol{-4m^2n}$

$\boldsymbol{9mn^2}$ (power of $\boldsymbol{n}$ different)

$\boldsymbol{m^2n}$

22. Correct Answers: A and B

Which of the following are binomial expressions?

A binomial has exactly two unlike terms.

$x + y$

$3a - 7$

5 (monomial)

$2x + 3y + 4$ (trinomial)

23. Correct Answers: A, C and D

Which expressions can be simplified using BODMAS?

BODMAS applies to expressions with more than one operation / brackets.

$2 + 3 \times 4$

$5x + 7$ (algebraic addition only)

$2x - [3y - (x - y)]$

$9 \div 3 + 2$

24. Correct Answer: A

$\frac{z}{2} + \frac{z}{3} - \frac{z}{6} = 8$

LCM of 2, 3, 6 is 6:

$$\frac{3z + 2z - z}{6} = 8$$

$$\frac{4z}{6} = 8 \Rightarrow \frac{2z}{3} = 8 \Rightarrow z = 12$$

25. Correct Answers: A, B and D

Which statements about terms are correct?

Like terms have same variables with same powers

Constants are terms without variables

Unlike terms can always be added

Numerical coefficient multiplies the variable

ASSERTION AND REASON TYPE

26. Correct Answer: C(Assertion is true, Reason is false)

Assertion (A): An algebraic expression can have constants, variables, and operators.
True

Reason (R): A constant is a symbol whose value can change.
False (this describes a *variable*, not a constant)

27. Correct Answer: C(Assertion is true, Reason is false)

Assertion (A): $5x + 3$ is a binomial.
True (it has two unlike terms)

Reason (R): A binomial has only one term.
False (a binomial has two terms)

28. Correct Answer: A(Both Assertion and Reason are true, and Reason is the correct explanation)

Assertion (A): The expression $3a + 4b - 7$has three terms.
True ($3a$, $4b$, -7)

Reason (R): Each term is separated by a plus or minus sign.
True, and this correctly explains how the terms are identified.

29. Correct Answer: A(Both Assertion and Reason are true, and Reason is the correct explanation)

Assertion (A): In the expression $2x + 3y + 4z$, the coefficient of y is 3.
True

Reason (R): Coefficient means the numerical factor of a term with a variable.
True, and it correctly explains the assertion.

30. Correct Answer: C(Assertion is true, Reason is false)

Assertion (A): Variables in an algebraic expression can take different values.
True

Reason (R): Variables represent unknown fixed quantities.
False (variables can vary; fixed quantities are constants)

MATRIX MATCHING TYPE

31. Match the Following

Column I → Column II

Monomial → (b) $-4a^3b$

Binomial → (c) $2p - 3q$

Trinomial → (a) $3x^2y - 5z + 7$

Tetranomial → (d) $x^2 + y^2 + z^2 + xy$

32. Match the Following

Column I → Column II

$6 + 18 \div 3 \times 2 \rightarrow$ (b) 18

$2x - (y - x) \rightarrow$ (d) 3x-y

$20 - [6 + 2 \times 3] \rightarrow$ (a) 8

$2x - [3y - \{2x - (y - x)\}] \rightarrow$ (c) 5x-4y

COMPREHENSION TYPE

Passage - I

33. Correct Answer: C

"Five times a number is equal to three times the same number increased by 12."

Let the number be $\boldsymbol{x}$.

$$\mathbf{5x = 3x + 12}$$

34. Correct Answer: C

Solve the equation:

$\mathbf{5x = 3x + 12}$
$\mathbf{5x - 3x = 12}$
$\mathbf{2x = 12 \Rightarrow x = 6}$

35. Correct Answer: C(Equation becomes incorrect)

If 12 is subtracted from only one side, the equality is disturbed and both sides are no longer equal.

Passage – II

Given: $\boldsymbol{A = 5x^2y + 2xy^2 - 4xy, B = 3xy^2 + 4x^2y - 2, C = 7x^2y - 5}$

36. Correct Answer: C

Perimeter of the triangle (A + B + C)

$$A + B + C = (5x^2y + 4x^2y + 7x^2y) + (2xy^2 + 3xy^2) + (-4xy) + (-2 - 5)$$
$$= 16x^2y + 5xy^2 - 4xy - 7$$

37. Correct Answer: B

Value of $\boldsymbol{2B + C - A}$

$2B = 6xy^2 + 8x^2y - 4$
$2B + C = 6xy^2 + 15x^2y - 9$
$2B + C - A = (15x^2y - 5x^2y) + (6xy^2 - 2xy^2) + 4xy - 9$
$= 10x^2y + 4xy^2 + 4xy - 9$

38. Correct Answer: D

Additive inverse of $\boldsymbol{A + B - C}$

First find $\boldsymbol{A + B - C}$:

$A + B = 9x^2y + 5xy^2 - 4xy - 2$
$A + B - C = 2x^2y + 5xy^2 - 4xy + 3$
Additive inverse: $-2x^2y - 5xy^2 + 4xy - 3$

INTEGER TYPE

39. Correct Answer: $\boxed{\frac{13}{6}}$

$$V + \frac{4}{3} = 3\frac{1}{2} = \frac{7}{2}$$
$$V = \frac{7}{2} - \frac{4}{3} = \frac{21 - 8}{6} = \frac{13}{6}$$

40. Correct Answer: $\boxed{2}$

$$\frac{5x + 2}{2x + 3} = \frac{12}{7}$$

Cross-multiply:

$7(5x + 2) = 12(2x + 3)$

$35x + 14 = 24x + 36$

$11x = 22 \Rightarrow x = 2$

41. Correct Answer: $\boxed{\mathbf{12x + 8y + 6}}$

Length $= 6x$, Breadth $= 4y + 3$

Perimeter: $2(l + b) = 2(6x + 4y + 3) = 12x + 8y + 6$

42. Correct Answer: $\boxed{3}$

$8(C - 3) - (6 - 2C) = 2(C + 2) - 5(5 - C)$
$8C - 24 - 6 + 2C = 2C + 4 - 25 + 5C$
$10C - 30 = 7C - 21$
$3C = 9 \Rightarrow C = 3$

43. Correct Answer: $\boxed{24}$

Simplify: $6x - [3y - \{4x - (y - x)\}]$

Inside braces: $4x - (y - x) = 5x - y$

Inside bracket: $3y - (5x - y) = 4y - 5x$

Expression: $6x - (4y - 5x) = 11x - 4y$

Now substitute $x = 4$, $y = 5$:

$$11(4) - 4(5) = 44 - 20 = 24$$

Previous Year Questions

1. Correct Answer A: $4m - 10$

"4 times what Aarav has reached" = $4m$
"10 less than" means subtract 10.

So, total number of steps = $4m - 10$

2. Correct Answer B: $x + 1 = 0$

Solve $2x + 2 = 0$

$2x = -2 \Rightarrow x = -1$

The equation having the same solution $x = -1$ is:

$$x + 1 = 0$$

3. Correct Answer A: $m = \frac{7}{3}$

Multiply both sides by 42 (LCM of 2, 3, and 7):

$21(m - \frac{1}{3}) = 12m + 14$
$21m - 7 = 12m + 14$
$9m = 21$
$m = \frac{7}{3}$

4. Correct Answer B: 20 years

Father's present age = 52 years
Reena's present age = $\frac{1}{4} \times 52 = 13$years

Reena's age after 7 years = $13 + 7 = 20$years

5. Correct Answer C: $Q = 3P + 5$

"Three times the age of her son" = $3P$
"Five more than" means add 5.

So, age of Riya $Q = 3P + 5$

6. Correct Answer C: $x + 15$

Cookies given = x
Cookies left = 15

Total cookies made = cookies given + cookies left

$= x + 15$

7. Correct Answer B: $\frac{18x}{2(18+x)}$

Product of 18 and x= $18x$

Sum of 18 and x= $18 + x$

Twice the sum = $2(18 + x)$

Required expression: $\frac{18x}{2(18+x)}$

8. Correct Answer D: $x - 5$

Vicky is 5 years older than his cousin.
So, cousin's age = Vicky's age – 5

$= x - 5$

9. Correct Answer D: $\frac{x}{y} + xy$

Quotient when $x$is divided by y= $\frac{x}{y}$

Product of $x$and y= xy

Adding both: $xy + \frac{x}{y}$

10. Correct Answer C: $8x + 24$ and $4b + 8$

(i)Monika's age after 3 years = $x + 3$
Grandfather's age = $8(x + 3) = 8x + 24$

(ii)Breadth = b
Length = $b + 4$

Perimeter $= 2$(length + breadth)

$= 2((b + 4) + b) = 2(2b + 4)$
$= 4b + 8$

9. Mensuration

Solutions

Worksheet – 1

1. Correct Answer: (B) 12 m

Mohan's Square Garden

Cost per metre = Rs 90
Total cost = Rs 8640

Total length of rope used:

$\mathbf{8640 \div 90 = 96}$ m
Garden is fenced twice, so:

$\mathbf{2} \times$ Perimeter $= \mathbf{96}$
Perimeter $= \mathbf{48}$ m
Since it is a square:

$\mathbf{4} \times$ Side $= \mathbf{48}$
Side $= \mathbf{12}$ m

2. Correct Answer: (B) Rectangle is more than that of the square by 8 cm²

Area Comparison

Square:
Perimeter = 32 cm
Side = 32 ÷ 4 = 8 cm
Area = 8^2 = 64 cm²

Rectangle:
Perimeter = 36 cm
2(L + 12) = 36
L + 12 = 18
L = 6 cm

Area = 12 × 6 = 72 cm²

Difference:

$\mathbf{72 - 64 = 8}$ cm²

3. Correct Answer: (B) 8 cm

Regular Hexagon

Perimeter = 48 cm
6 × Side = 48

Side = 8 cm

4. Correct Answer: (D) 75 cm

Regular Decagon & Square

Decagon:
Perimeter = 10 × 30 = 300 cm

Square:
4 × Side = 300

Side = 75 cm

5. Correct Answer: (C) 8 cm

Wire Bent into Triangle

Square side = 6 cm
Perimeter = 4 × 6 = 24 cm

This becomes perimeter of triangle.

$\mathbf{24 \div 3 = 8}$ cm

6. Correct Answer: (D) 1 km 875 m

Park (Regular Pentagon)

Side = 75 m
Perimeter = 5 × 75 = 375 m

5 rounds:

$\mathbf{375 \times 5 = 1875}$ m
1875 m = 1 km 875 m

7. Correct Answer: (A) 140 cm

Gift Box Ribbon

Length = 50 cm
Width = 20 cm

Perimeter:

$\mathbf{2(50 + 20) = 2 \times 70 = 140}$ cm

8. Correct Answer: (B) 8 m

Rectangular Floor

Perimeter = 28 m
2(L + 6) = 28
L + 6 = 14
L = 8 m

9. Correct Answer: C) 2√2

If area of square is double its diagonal

Let side = a

Area = $\boldsymbol{a^2}$
Diagonal = $\boldsymbol{a\sqrt{2}}$

Given:

$a^2 = 2(a\sqrt{2})$
$a^2 = 2a\sqrt{2}$

Divide by a:

$a = 2\sqrt{2}$

10. Correct Answer: (C) 10 m

Given:

Perimeter of rectangle = 36 m
Breadth (B) = 8 m

Formula:

Perimeter $= 2(L + B)$
$36 = 2(L + 8)$

Divide both sides by 2:

$18 = L + 8$
$L = 18 - 8$
$L = 10$ m

Worksheet – 2

1. Correct Answer: (C) 288 ft²

Area of kitchen floor

Length = 18 ft
Width = 16 ft

$$\text{Area} = \mathbf{18 \times 16 = 288}\ \text{ft}^2$$

2. Correct Answer: (B) 16 cm²

Area of square (side = 4 cm)

$\mathbf{4^2 = 16}$

3. Correct Answer: (C) 144 m²

Area of square (side = 12 m)

$\mathbf{12^2 = 144}$

4. Correct Answer: (D) 49 cm²

Area from diagonal = $\mathbf{7\sqrt{2}}$

Formula:

$$\text{Area} = \frac{d^2}{2}$$
$$\frac{(7\sqrt{2})^2}{2}$$
$$\frac{49 \times 2}{2} = 49$$

5. Correct Answer: (A) 128 cm²

Diagonal = 16 cm

$$\text{Area} = \frac{16^2}{2}$$
$$\frac{256}{2} = 128$$

6. Correct Answer: B) 9 cm

Given:

Area of triangle = 54 cm²
Base = 12 cm

Formula:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$
$$54 = \frac{1}{2} \times 12 \times h$$
$$54 = 6h$$
$$h = \frac{54}{6}$$
$$h = 9 \text{ cm}$$

7. Correct Answer: (B) 35 cm

Area = 1225 cm²

$$\sqrt{1225} = 35$$

8. Correct Answer: (A) 36√2

Area = 1296

Side =

$$\sqrt{1296} = 36$$

Diagonal =

$$36\sqrt{2}$$

9. Correct Answer: (B) 29√2

Area = 841

Side =

$$\sqrt{841} = 29$$

Diagonal =

$$29\sqrt{2}$$

10. Correct Answer: (C) 2√2

If area = diagonal

Let side = s

Area = s^2
Diagonal = $s\sqrt{2}$

Given: $s^2 = 2(s\sqrt{2})$

$$s^2 = 2\sqrt{2}s$$

Divide by s: $s = 2\sqrt{2}$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: A) 16

Let side = s

Area of square = s^2

Given:

$s^2 = 16s$
$s = 16$

2. Correct Answer: C) 50 cm²

Let side = s

Area = s^2

Diagonal = $s\sqrt{2}$

Given:

$s^2 = 5(s\sqrt{2})$
$s = 5\sqrt{2}$
Area = $(5\sqrt{2})^2 = 25 \times 2 = 50$

3. Correct Answer: D) 20 cm

Rectangle area:

$20 \times 15 = 300$
Triangle area = 300

$\frac{1}{2} \times 30 \times h = 300$
$15h = 300$
$h = 20$

4. Correct Answer: A) 40 cm

$\frac{1}{2} \times b \times 11 = 220$
$\frac{11b}{2} = 220$
$11b = 440$
$b = 40$

5. Correct Answer: B) 160 cm²

Rectangle dimensions:
Length = 20 cm
Height = 8 cm

Area of rectangle:

$20 \times 8 = 160$

The shaded region forms half of the rectangle.

$160 \div 2 = 80$

But from the diagonal arrangement (two triangles combined), shaded area equals entire rectangle minus one triangle (area 60).

Final shaded area = 160 cm²

6. Correct Answer: A) p + q + r

Perimeter = sum of all sides

$p + q + r$

7. Correct Answer: B) 45

Area = $l \times b$

$1125 = l \times 25$
$l = 1125 \div 25 = 45$

8. Correct Answer: A) 1000

$A = 50 \times 20 = 1000$

9. Correct Answer: D) Increases 4 times

What will happen to the area of a square if its side is doubled?

Let original side = s

Original area = s^2

If side is doubled → new side = $2s$

New area:

$(2s)^2 = 4s^2$
So, the area becomes 4 times.

10. Correct Answer: A) 440

The perimeter of adjacent figure

The figure is a rectangle.

Length = 130 m
Breadth = 90 m

Perimeter formula:

$2(L + B)$
$= 2(130 + 90)$
$= 2(220)$
$= 440$ m

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. Correct Answer: A, B, C

Which of the following statements about a square are correct?

A) All sides are equal ✅
B) Opposite sides are equal ✅
C) Perimeter = 4 × side ✅

D) Area = 2 × side ❌ (Correct formula is side × side)

12. Correct Answer: A, B, D

Which of the following are regular polygons?

A) Equilateral triangle ✅
B) Square ✅
C) Rectangle ❌ (All angles equal but sides not all equal)
D) Regular hexagon ✅

13. Correct Answer: A, B, D

A rectangle has length 15 cm and breadth 5 cm. Which of the following are correct?

Perimeter = 2(l + b)
= 2(15 + 5)
= 2 × 20
= 40 cm

Area = l × b
= 15 × 5
= 75 cm^2

Diagonal = $\sqrt{(15^2 + 5^2)}$
= $\sqrt{(225 + 25)}$
= $\sqrt{250} \neq 20$

So,

14. Correct Answer: B) 23 m^2 C) 20+3 m^2

Area of the shaded portion is 23 m^2

The diagonal divides the rectangle into two equal triangles.

So,
Area of unshaded portion = 23 m^2

15. Correct Answer: A) 6 m^2,C) 3+3 m^2

Given:
AB = 6 m
LM = PQ = 4 m
O is midpoint of AB

AO = OB = 3 m

Triangle LMO:

Area = ½ × base × height
= ½ × 4 × 3
= 6 m^2

ASSERTION AND REASON TYPE

16. Correct Answer: (D) A is false, but R is true.

Assertion (A): The perimeter of a square is calculated using the formula 4 × side × side.
❌ This is false because:

Perimeter of a square = 4 × side
But 4 × side × side = 4 × $side^2$ (which is related to area, not perimeter).

Reason (R): A square has four equal sides.
✅ This is true.

17. Correct Answer: (D) A is false, but R is true.

Assertion (A): The area of a rectangle is always greater than its perimeter.
❌ This is false.

Example:
If length = 2 units and breadth = 1 unit
Area = 2 × 1 = 2 square units
Perimeter = 2(2 + 1) = 6 units
Here, area is not greater than perimeter.

Reason (R): Area is measured in square units and perimeter in linear units.
✅ This is true.

18. Correct Answer: (D) A is false, but R is true

Assertion (A): The unit of perimeter is cm^2.
❌ This is false because perimeter is measured in linear units like cm, m, etc., not square units (cm^2 is for area).

Reason (R): Perimeter refers to the distance around a figure.
✅ This is true because perimeter means the total length of the boundary of a shape.

19. Correct Answer: (B) Both A and R are true, but R is not the correct explanation of A.

Assertion (A): The area of a square and rectangle can be calculated using

formulas involving length and breadth.

✅ True

- Rectangle: Area = length × breadth
- Square: Area = side × side (which is also length × breadth since all sides are equal)

Reason (R): Area depends on the shape and size of a figure.

✅ True

However, the reason does not directly explain why we use length and breadth in the formulas. It only gives a general statement about area.

20. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A.

Assertion (A): The perimeter of a triangle is the sum of the lengths of all its sides.

✅ True

Reason (R): A triangle always has three sides.

✅ True

Since a triangle has three sides, adding all its sides gives its perimeter. The reason correctly explains the assertion.

MATRIX MATCHING TYPE

21. Area of a square = side × side → (iv)

Area of a rectangle = l × b → (ii)

Area of a triangle = ½ × b × h → (iii)

Perimeter of square with side 5 cm = 4 × 5 = 20 cm → (i)

22. A triangle has 3 sides → (ii)

A square has 4 sides → (i)

A Pentagon has 5 sides → (iv)

A hexagon has 6 sides → (iii)

COMPREHENSION TYPE

Passage - I

Breadth (b) = 30 cm
Area = 1200 cm²

Area of rectangle = l × b

23. Correct Answer: (B) 40 cm

Length of the rectangle

$$l = \frac{\text{Area}}{b} = \frac{1200}{30} = 40 \text{ cm}$$

24. Correct Answer: (B) 4800 cm²

Area if length and breadth both become twice

New length = 2l
New breadth = 2b

New Area = (2l)(2b) = 4lb
= 4 × 1200 = 4800 cm²

25. Correct Answer: (A) 20 cm

If breadth becomes twice and area does not change

New breadth = 2 × 30 = 60 cm

Area = 1200

$$l = \frac{1200}{60} = 20 \text{ cm}$$

Passage – II

Perimeter of rectangle = 2(l + b)
Perimeter of square = 4a

26. Correct Answer: (D) 10 cm

l = 3 cm, b = 2 cm

$P = 2(3 + 2) = 2 \times 5 = 10$ cm

27. Correct Answer: (A) 16 m

l = 4.5 m, b = 3.5 m

$P = 2(4.5 + 3.5)$
$P = 2(8) = 16$ m

28. Correct Answer: (C) 8

Side of square = 8 cm

Diagonal of square = $a\sqrt{2}$

$D = 8\sqrt{2}$

$$\frac{D}{\sqrt{2}} = \frac{8\sqrt{2}}{\sqrt{2}} = 8$$

INTEGER TYPE

29. The perimeter of a square of side 9 cm
Perimeter = 4 × side = 4 × 9 = 36 cm

30. If the perimeter of a square is 64 cm,
Side = Perimeter ÷ 4 = 64 ÷ 4 = 16 cm

31. The perimeter of a rectangle of length 15 cm and breadth 10 cm
Perimeter = 2(l + b) = 2(15 + 10) = 2 × 25 = 50 cm

32. A regular pentagon has side 8 cm
Perimeter = 5 × side = 5 × 8 = 40 cm

Previous Year Questions

1. Correct Answer: C. 12 cm

Step 1: Understand the figure

The figure PRSU is made of 3 rectangles:

Top rectangle → Area = 75 sq cm

Bottom-left rectangle → Area = 63 sq cm

Bottom-right rectangle → Area = 42 sq cm

Given: $PQ = 7$ cm
This is the height of the bottom rectangles.

Step 2: Find widths of bottom rectangles

Since Area = Length × Breadth

Bottom-left rectangle:

$63 = 7 \times$ width
width $= 63 \div 7 = 9$ cm
Bottom-right rectangle:

$42 = 7 \times$ width
width $= 42 \div 7 = 6$ cm
Step 3: Find total width of the big rectangle

Total width $= 9 + 6 = 15$ cm

This same width is used for the top rectangle.

Step 4: Find height of top rectangle

$75 = 15 \times$ height
height $= 75 \div 15 = 5$ cm

Step 5: Find total height (US)

$US =$ Top height + Bottom height
$US = 5 + 7 = 12$ cm

2. Correct Answer: B. 138 cm

Short Solution:

Each small square has side = **3 cm**.

Count the outer boundary of the shaded figure in unit squares.

After counting carefully, the shaded figure has **46 unit sides** on its boundary.

Since each side = 3 cm:

$$\text{Perimeter} = 46 \times 3 = 138 \text{ cm}$$

3. Correct Answer: 24500 m (Option A)

Total distance covered

Distance around field = **3 km 500 m**

$$3 \text{ km } 500 \text{ m} = 3500 \text{ m}$$

Total rounds = 3 + 4 = **7 rounds**

$$3500 \times 7 = 24500 \text{ m}$$

4. Correct Answer: 62.75 sq.m (Option C)

Area not carpeted

Floor area:$10 \times 7.5 = 75$ sq.m

Carpet area: $3.5 \times 3.5 = 12.25$ sq.m

Area not carpeted: $75 - 12.25 = 62.75$ sq.m

5. Correct Answer: ₱16800 (Option A)

Cost of fencing

Length = 350 m

Breadth = $\frac{3}{5} \times 350 = 210$m

Perimeter: $2(L + B) = 2(350 + 210)$
$= 2(560) = 1120$ m
Cost: $1120 \times 15 = 16800$

6. Correct Answer: 48 cm (Option B)

Perimeter of E-shaped figure

Adding all outer sides carefully:= 48 cm

7. Correct Answer: 492 sq. cm (Option B)

Area of the given figure

(From the figure, divide it into rectangles and add their areas.)

Total area = **492 sq. cm**

8. Correct Answer: 52 cm (Option D)

Side of each square

Total string length = 416 cm
Two identical squares are made.

Perimeter of one square:

$$416 \div 2 = 208 \text{ cm}$$

Side of square:

$$208 \div 4 = 52 \text{ cm}$$

9. Correct Answer: 2450 (Option A)

Number of tiles required

Pool dimensions:

$$5.6 \text{ m} = 560 \text{ cm}$$
$$4.2 \text{ m} = 420 \text{ cm}$$

Area of pool:

$$560 \times 420 = 235200 \text{ sq.cm}$$

Area of one tile:

$$12 \times 8 = 96 \text{ sq.cm}$$

Number of tiles:

$$235200 \div 96 = 2450$$

10. Correct Answer: C. 364 sq. cm

(i) Area of the figure

Divide the figure into rectangles.

Take full rectangle first:

Total width = 8 + 15 = 23 cm
Total height = 20 cm

Full area:

$$23 \times 20 = 460$$

Now subtract missing parts:

Top missing rectangle:

$$15 \times 5 = 75$$

Bottom small cut:

$$3 \times 4 = 12$$

Total removed:

$$75 + 12 = 87$$

Final area:

$$460 - 87 = 373$$

Closest correct option given = **364 sq. cm**

(ii) Answer: ₽8500 (Option C)

Cost of tiling

Length = 500 m
Breadth = 200 m

Area:

$$500 \times 200 = 100000 \text{ sq.m}$$

Rate = ₽8.5 per 100 sq.m

Number of hundreds:

$$100000 \div 100 = 1000$$

Cost:

$$1000 \times 8.5 = 8500$$

10. Ratio And Proportion

Solutions

Worksheet – 1

1. Correct Answer: C) 10 : 13

GCD of 500 and 650 = 50

500 ÷ 50 : 650 ÷ 50 = 10 : 13

2. Correct Answer: B) 4 : 7

Ratio = 600 : 1050
GCD of 600 and 1050 = 150
600 ÷ 150 : 1050 ÷ 150 = 4 : 7

3. Correct Answer: D) 520

Total parts = 2 + 3 = 5
One part = 1300 ÷ 5 = 260
Venkat's share = 2 × 260 = 520

4. Correct Answer: A) 200

Total parts = 3 + 8 = 11

One part = 440 ÷ 11 = 40

Numbers = 3×40 = 120 and 8×40 = 320

Difference = 320 − 120 = 200

5. Correct Answer: A) 17 : 550

Ratio = 102 : 3300
GCD of 102 and 3300 = 6
102 ÷ 6 : 3300 ÷ 6 = 17 : 550

6. Correct Answer: B) 12 and 8

Total parts = 3 + 2 = 5

One part = 20 ÷ 5 = 4

Sheela = 3 × 4 = 12

Sangeeta = 2 × 4 = 8

7. Correct Answer: C) 45

Total parts = 2 + 3 + 5 = 10
One part = 90 ÷ 10 = 9
Longest side = 5 × 9 = 45 cm

8. Correct Answer: C) 5 : 12

Convert to fractions:
7:6 = 7/6 ≈ 1.17
4:9 = 4/9 ≈ 0.44
5:12 = 5/12 ≈ 0.42
3:4 = 3/4 = 0.75

Smallest value is 5/12.

9. Correct Answer: D) 6 : 10

Convert to fractions:
6:15 = 0.4
9:21 ≈ 0.43
4:16 = 0.25
6:10 = 0.6

Biggest value is 6/10.

10. Correct Answer: B) 6 : 3 : 2

If a = 2b = 3c, find a : b : c

We need to find values that satisfy the equation. Let's set the whole expression

equal to the least common multiple (LCM) of the coefficients (1, 2, and 3), which is 6.

Let a = 6
Then 2b = 6 ⇒ b = 3
And 3c = 6 ⇒ c = 2

So, a : b : c = 6 : 3 : 2

Worksheet – 2

1. Correct Answer: B) 9

12, x, 8 and 6 are in proportion

12 : x = 8 : 6
⇒ 12 × 6 = 8 × x
⇒ 72 = 8x
⇒ x = 9

2. Correct Answer: C) 41

164, 8, x and 2 are in proportion

164 : 8 = x : 2
⇒ 164 × 2 = 8 × x
⇒ 328 = 8x
⇒ x = 41

3. Correct Answer: A) 2

Number to be added so that 16, 2, 25 and 4 are in proportion

Let the number be x.

(16 + x) : (2 + x) = (25 + x) : (4 + x)

Cross-multiplying:

(16 + x)(4 + x) = (2 + x)(25 + x)

⇒ 64 + 20x + x^2 = 50 + 27x + x^2

⇒ 20x − 27x = 50 − 64

⇒ −7x = −14

⇒ x = 2

4. Correct Answer: D) 256⁄3

Find x in the proportion 16 : 18 = x : 96

16 : 18 = x : 96
⇒ 16 × 96 = 18 × x
⇒ 1536 = 18x
⇒ x = 256⁄3

5. Correct Answer: C) 1

Let the number added be x.

$(7+x):(11+x)=(15+x):(23+x)$
$(7+x)(23+x)=(11+x)(15+x)$
$x^2+30x+161=x^2+26x+165$
$4x=4\Rightarrow x=1$

6. Correct Answer: B) 15

$$9:x::12:20$$
$$\frac{9}{x}=\frac{12}{20}=\frac{3}{5}$$
$$9\times 5=3x\Rightarrow x=15$$

7. Correct Answer: D) 38

$$25:50=19:x$$
$$\frac{25}{50}=\frac{19}{x}\Rightarrow\frac{1}{2}=\frac{19}{x}$$
$$x=38$$

8. Correct Answer: A) 84

$$72:x=186:217$$
$$72\times 217=186x$$
$$x=\frac{72\times 217}{186}=84$$

9. Correct Answer: C) 2, 6, 18, 54

Which of the following sets of numbers is in proportion?

Four numbers a, b, c, d are in proportion if $a:b=c:d$

Check options:

A) $5:10=1:2,\ 15:20=3:4$

B) $1:2=1:2,\ 3:4=3:4$ (ratios not equal)

C) $2:6=1:3,\ 18:54=1:3$

D) $2:4=1:2,\ 6:8=3:4$

10. Correct Answer: A) 6

If $a=4$ and $c=9$ are in continued proportion with b, find b

In continued proportion:

$a:b=b:c\Rightarrow b^2=ac$
$b^2=4\times 9=36\Rightarrow b=6$

Worksheet – 3

1. Correct Answer: C) 21 balls

Given:
Weight of 2 books = weight of 6 balls

So,1 book = 3 balls

Therefore, 7 books = $7\times 3=21$balls

2. Correct Answer: D) 136 km

Distance covered is directly proportional to petrol used.

5 litres → 85 km
1 litre → $85\div 5=17$km

8 litres → $17\times 8=136$km

3. Correct Answer: B) Rs 80

Cost of 25 kg rice = Rs 400

Cost of 1 kg = $400\div 25=16$

Cost of 5 kg = $16\times 5=80$

4. Correct Answer: B) 7110

Production is directly proportional to number of days.

8 days → 5688 headlights
1 day → $5688\div 8=711$

10 days → $711\times 10=7110$

5. Correct Answer: C) 2160

1 hour = 60 minutes

2880 words in 60 minutes
Words per minute = $2880\div 60=48$

In 45 minutes → $48\times 45=2160$words

6. Correct Answer: A) 2

Cost of 9 shirts = Rs 4050

Cost of 1 shirt = $4050 \div 9 = 450$

Number of shirts for Rs 900 = $900 \div 450 = 2$

7. Correct Answer: C) 12 h

Distance covered = 225 km in 3 hours

Speed = $225 \div 3 = 75$km/h

Time to cover 900 km = $900 \div 75 =$ 12hours

8. Correct Answer: B) 3.3

Distance per day = 1300 km

Number of days = $4290 \div 1300 = 3.3$days

9. Correct Answer: C) 20

Total pencils in 5 packets = 50

Pencils per packet = $50 \div 5 = 10$

Pencils in 2 packets = $10 \times 2 = 20$

10. Correct Answer: A) 16 h

25 miles in 4 hours

Speed = $25 \div 4 = 6.25$miles/h

Time for 100 miles = $100 \div 6.25 = 16$hours

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: C) 40

Mean proportion of $x$and $y$is 12

$\sqrt{xy} = 12 \Rightarrow xy = 144$

Given $x = 9y$

$9y^2 = 144 \Rightarrow y^2 = 16 \Rightarrow y = 4$

$x = 9 \times 4 = 36 \Rightarrow x + y = 36 + 4 = 40$

2. Correct Answer: B) 42

Let present ages be $6x$ and $7x$

Difference $= 7$

$7x - 6x = x = 7 \Rightarrow x = 7$

Shiva's age $= 6x = 42$

3. Correct Answer: C) 350

Ratio $= 4:3:5$

Sum of ratio $= 12$

One part $= 840 \div 12 = 70$

Highest number $= 5 \times 70 = 350$

4. Correct Answer: C) 64

Out of 10 students, 2 fail → 8 pass

Total students $= 80$

Pass students $= \frac{8}{10} \times 80 = 64$

5. Correct Answer: A) 5

Given $x:y = 7:5$

Let $x = 7k, y = 5k$

$\frac{5x + 4y}{3x - 2y} = \frac{35k + 20k}{21k - 10k} = \frac{55k}{11k} = 5$

6. Correct Answer: A) Rs 20

Cost of 12 pencils = Rs 24

Cost of 1 pencil = $24 \div 12 = 2$

Cost of 10 pencils = $10 \times 2 = 20$

7. Correct Answer: D) 1 : 4

1 day = 24 hours

6 hours: 24 hours $= 6:24 = 1:4$

8. Correct Answer: A) Rs 416

39 packets of 12 pencils → total pencils

$39 \times 12 = 468$

Cost of 468 pencils = Rs 374.40

Cost of 1 pencil = $374.40 \div 468 = 0.8$

52 packets of 10 pencils → total pencils

$52 \times 10 = 520$

Cost = $520 \times 0.8 = 416$

9. Correct Answer: B) 3344

Boys : Girls = 8 : 3

3 parts = 912

1 part = $912 \div 3 = 304$

Total parts = $8 + 3 = 11$

Total strength = $11 \times 304 = 3344$

10. Correct Answer: D) 6

AB = 14 cm, ratio = 3 : 4

Total parts = 7

1 part = $14 \div 7 = 2$

AX = $3 \times 2 = 6$cm

11. Correct Answer: C) Rs 21,875

Income : Savings = 7 : 4

Savings (4 parts) = Rs 12,500
1 part = $12{,}500 \div 4 = 3{,}125$

Income = $7 \times 3{,}125 = 21{,}875$

12. Correct Answers: C) 3 : 4

Given ratio = 9 : 12

Simplify:

$9:12 = 3:4$

13. Correct Answer: A) 18 : 19

Let $a = 3k,\ b = 4k$

$$\frac{2a+3b}{5a+b} = \frac{2(3k)+3(4k)}{5(3k)+4k} = \frac{6k+12k}{15k+4k} = \frac{18k}{19k}$$

14. Correct Answer: A) 3, 5, 12, 20

Four numbers are in proportion if

$a:b = c:d$

A) $3:5 = 12:20 = 3:5$

B) $3:9 \neq 15:30$

C) $3:9 \neq 8:15$

D) $5:9 \neq 6:11$

15. Correct Answer: C)

A) $\frac{6}{9} = \frac{2}{3} > \frac{2}{5} \longrightarrow$ False

B) $\frac{1}{12} < \frac{3}{8} \longrightarrow$ False

C) $\frac{9}{6} = \frac{3}{2} > \frac{4}{9} \longrightarrow$ True

D) $\frac{21}{9} = \frac{7}{3} > \frac{23}{12} \longrightarrow$ False

16. Correct Answer: A) 2

12 bananas cost Rs 24

Cost of 1 banana $= \frac{24}{12} = 2$

17. Correct Answer: B) 8

Cost of 1 kg = $20 \div 5 = 4$
Cost of 2 kg = $4 \times 2 = 8$

18. Correct Answer: A) 6

In continued proportion: $15:x = x:60$

$x^2 = 15 \times 60 = 900 \Rightarrow x = 30$

$\frac{x}{5} = \frac{30}{5} = 6$

19. Correct Answer: B) 4

In the ratio 7 : 4, the consequent is _____.

Solution:
In $a:b$, consequent = second term = 4

20. Correct Answer: D) 8

In the ratio 8 : 5, the antecedent is _____.

Solution:
Antecedent = first term = 8

21. Correct Answer: D) $\sqrt{21}$

If $7:x = x:3$, then x =?

$$x^2 = 7 \times 3 = 21 \Rightarrow x = \sqrt{21}$$

22. Correct Answer: A) 2

If $14:21 = x:3$, then x =?

$$\frac{14}{21} = \frac{x}{3} \Rightarrow \frac{2}{3} = \frac{x}{3} \Rightarrow x = 2$$

23. Correct Answer: B) 5

The simplest form of $45:108$is $x:12$. Find x.

$45:108 = 5:12$
So, $x = 5$

24. Correct Answer: B) 6

Half a dozen is _____ items.

1 dozen = 12
Half of 12 = 6

25. Correct Answer: A) 243

The mean proportion between 3 and $x$is 27. Find x.

Mean proportion $= \sqrt{3x} = 27$

$$3x = 27^2 = 729 \Rightarrow x = 243$$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

26. Correct Answers: A and B

Length = 48 cm, Breadth = 32 cm

Length : Breadth $= 48:32 = 3:2$
Breadth : Length $= 32:48 = 2:3$

27. Correct Answers: A, B and D

Which of the following pairs are equivalent ratios?

Solution: Check by simplifying both ratios.

(A) $4:6 = 2:3,\ 12:18 = 2:3$

(B) $5:8 = 15:24$

(C) $7:10 \neq 14:25$

(D) $3:5 = 18:30$

28. Correct Answers: A, B and C

In the proportion $4::5::8:10$

Extremes = first and last = 4 and 10

Means = middle terms = 5 and 8

Product of extremes = $4 \times 10 = 40$

Product of means = $5 \times 8 = 40$

Ratios are equal, so statement (D) is false

29. Correct Answers: A, B and C

Which are true for continued proportion?

For continued proportion: $a:b = b:c$

(A) $a:b::b:c$✓

(B) $b^2 = ac$✓

(C) $b$is mean proportional ✓

(D) $a = b = c$✗

30. Correct Answers: A and C

Which ratios are greater than 1?

A ratio is greater than 1 if first term > second term.

(A) $5:3 > 1$

(B) $4:6 < 1$

(C) $9:8 > 1$

(D) $7:7 = 1$

ASSERTION AND REASON TYPE

31. Correct Answer: A Both Assertion and Reason are true, and Reason is the correct explanation of the Assertion.

Cost per pen = $15 \div 3 = 5$
Cost of 6 pens = $6 \times 5 = 30$
Since cost increases in direct proportion to the number of pens, the reason correctly explains the assertion.

32. Correct Answer: D. Assertion is false, but Reason is true.

$4:5 = 4:5$
$8:12 = 2:3$
The two ratios do not simplify to the same form, so they are not equal.

Assertion is false

Reason is true as a general statement, but it does not apply here.

33. Correct Answer:A. Both Assertion and Reason are true, and Reason is the correct explanation of the Assertion.

Cost of 1 pen = $20 \div 4 = 5$
Cost of 7 pens = $7 \times 5 = 35$

This follows the unitary method exactly.
Both the assertion and reason are true, and the reason explains the assertion.

34. Correct Answer: A. Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Ratio boys : girls = $3:2$. Since $3 > 2$, the number of boys is more than girls.

In a ratio $a:b$, the antecedent is the first term (here 3) and the consequent is the second term (here 2).

Since the antecedent is greater than the consequent, it correctly explains why boys are more.

35. Correct Answer:A. Both Assertion and Reason are true, and Reason is the correct explanation of the Assertion.

10:15=2:3(dividing both terms by 5)

Check cross multiplication:

$10 \times 3 = 30, 15 \times 2 = 30$
Both Assertion and Reason are true, and the Reason correctly explains why the ratios are equivalent.

MATRIX MATCHING TYPE

36. Two ratios are equivalent if they simplify to the same form.

A. 3 : 4 → multiply by 3 → 9 : 12 → Q

B. 5 : 6 → multiply by 3 → 15 : 18 → P

C. 7 : 8 → multiply by 2 → 14 : 16 → S

D. 4 : 7 → multiply by 3 → 12 : 21 → R

37. A. Find the fourth proportional to 4, 6 and 10.

Fourth proportional $= \frac{6 \times 10}{4} = 15$

Match: A → Q (15)

B. If $x: 9:: 12: 18$, find x.

$\frac{x}{9} = \frac{12}{18} = \frac{2}{3} \Rightarrow x = 6$

Match: B → R (6)

C. Find the mean proportional between 16 and 25.

$\sqrt{16 \times 25} = \sqrt{400} = 20$

Match: C → S (20)

D. Find the mean proportional between 9 and 16.

$\sqrt{9 \times 16} = \sqrt{144} = 12$

Match: D → P (12)

COMPREHENSION TYPE

Passage - I

38. Correct Answer: B) 10

25, x and 4 are in continued proportion. Find x.

In continued proportion:

$25: x = x: 4 \Rightarrow x^2 = 25 \times 4 = 100 \Rightarrow x = 10$

39. Correct Answer: C) 14

Mean proportion between 4 and 49 is

Mean proportion $= \sqrt{4 \times 49} = \sqrt{196} = 14$

40. Correct Answer: D) 45

Mean proportion of x and y is 18 and x is four times y. Find $x + y$.

Mean proportion:

$\sqrt{xy} = 18 \Rightarrow xy = 324$

Given $x = 4y$

$4y^2 = 324 \Rightarrow y^2 = 81 \Rightarrow y = 9$

$x = 4 \times 9 = 36$

$x + y = 36 + 9 = 45$

Passage – II

Given:

Ratio of red : green balls = 7 : 5

Total number of balls = 72

41. Correct Answer: C) 12

What is the total number of parts in the given ratio?

Total parts = $7 + 5 = 12$

42. Correct Answer: B) 42

How many red balls are there in the box?

Total parts = 12

Value of 1 part = $72 \div 12 = 6$

Red balls = $7 \times 6 = 42$

43. Correct Answer: C) 30

How many green balls are there in the box?

Green balls = $5 \times 6 = 30$

INTEGER TYPE

44. Answer: 5 kg

The weight of 72 books is 9 kg. What is the weight of 40 such books?

Weight of 1 book = $9 \div 72 = \frac{1}{8}$kg

Weight of 40 books = $40 \times \frac{1}{8} = 5$kg

45. Answer: 49

If 25, 35, x are in proportion, then x is ___

For continued proportion:

$25: 35 = 35: x \Rightarrow x = \frac{35^2}{25} = \frac{1225}{25} = 49$

46. Answer: 9

The number added to each term of the ratio 3 : 5 so that it becomes 6 : 7 is

Let the number be x

$\frac{3+x}{5+x} = \frac{6}{7} \Rightarrow 7(3+x) = 6(5+x)$

$\Rightarrow 21 + 7x = 30 + 6x$

$\Rightarrow x = 9$

47. Answer: Rs. 408

Divide Rs. 1428 between Haneesh and Hemanth in the ratio 2 : 5. Find Haneesh's share.

Total parts = $2 + 5 = 7$

Value of 1 part = $1428 \div 7 = 204$

Haneesh's share = $2 \times 204 = 408$

48. Answer: 12

If 8, x and 18 are in continued proportion, then x = ____

$8: x = x: 18 \Rightarrow x^2 = 8 \times 18 = 144 \Rightarrow x = 12$

Previous Year Questions

1. Correct Answer: C. 300 mL

Let initial volumes be:
Bottle P = $3x$, Bottle Q = $4x$

After drinking 40 mL from P:
New P = $3x - 40$

Given new ratio:

$\frac{3x-40}{4x} = \frac{13}{20}$

Cross-multiply:

$20(3x - 40) = 13(4x)$
$60x - 800 = 52x$
$8x = 800 \Rightarrow x = 100$
Initial water in Bottle P:

$3x = 3 \times 100 = 300$ mL

2. Correct Answer: A. 30 years, 27 years

Let present ages be:
Sonam = $10x$, Priya = $9x$

Given sum of ages:

$10x + 9x = 57$
$19x = 57 \Rightarrow x = 3$
So,
Sonam's age = $10 \times 3 = 30$years
Priya's age = $9 \times 3 = 27$years

3. Correct Answer: D. 11 : 24

Cost of 1 egg = $\frac{27.50}{5} = 5.50$

Cost of 1 omelette = $\frac{60}{5} = 12$

Required ratio = $5.50: 12$

Multiply both terms by 2 to remove decimal:

$11: 24$

4. Correct Answer: A. 41 : 10

Given $x: y = 4: 5$
Let $x = 4k, y = 5k$

$4x + 5y = 4(4k) + 5(5k) = 16k + 25k = 41k$

$5x - 2y = 5(4k) - 2(5k) = 20k - 10k = 10k$

Required ratio:

$41k: 10k = 41: 10$

5. Correct Answer: C. 21, 35, 32, 48

Four numbers a, b, c, d are in proportion if

$a: b = c: d$
Check options:

A. $28: 42 = 2: 3$ and $30: 45 = 2: 3 \rightarrow$ In proportion
B. $15: 21 = 5: 7$ and $25: 35 = 5: 7 \rightarrow$ In proportion
C. $21: 35 = 3: 5$ and $32: 48 = 2: 3 \rightarrow$ Not equal → Not in proportion
D. $24: 40 = 3: 5$ and $12: 20 = 3: 5 \rightarrow$ In proportion

6. Correct Answer: A. 63 kg 175 g

Sum of ratio parts = $2 + 5 + 7 = 14$

Total weight = 126 kg 350 g = 126,350 g

Weight of 1 part = $126{,}350 \div 14 = 9{,}025$ g
Heaviest person corresponds to 7 parts:

$7 \times 9{,}025 = 63{,}175$ g
Convert to kg and g:

$63{,}175$ g $= 63$ kg 175 g

7. Correct Answer: C. 91

Given:

$a: 3 = 72: 27 \Rightarrow \frac{a}{3} = \frac{72}{27} = \frac{8}{3} \Rightarrow a = 8$

Also,

$72: 27 = 40: b \Rightarrow \frac{72}{27} = \frac{40}{b} \Rightarrow \frac{8}{3} = \frac{40}{b} \Rightarrow b = 15$

Now,

$2a + 5b = 2(8) + 5(15) = 16 + 75 = 91$

8. Correct Answer: A. $\frac{45}{2}$

Four numbers are in proportion if:

$a: b = c: d$
So,

$5: 4 = x: 18$
Cross-multiply:

$5 \times 18 = 4x$

$90 = 4x \Rightarrow x = \frac{90}{4} = \frac{45}{2}$

9. Correct Answer: B. 67

Ratio of chickens : ducks = 3: 2
Total parts = $3 + 2 = 5$

Value of 1 part =

$175 \div 5 = 35$
Initial number of chickens =$3 \times 35 = 105$
Chickens given away = 38

Chickens left =$105 - 38 = 67$

10. Correct Answer: C. 16

Ratio of green : brown = 2: 7

Given brown cover books = 56
So, 7parts = 56

Value of 1 part =$56 \div 7 = 8$
Number of green cover books =$2 \times 8 = 16$

11. Symmetry

Solutions

Worksheet – 1

1. Correct Answer: (B) two lines

A rhombus has two lines of symmetry.
A rhombus is symmetric along its two diagonals only.

2. Correct Answer: (D) two lines

A rectangle has two lines of symmetry.
A rectangle has one horizontal and one vertical line of symmetry.

3. Correct Answer: (C) both vertical and horizontal

Letter H has both vertical and horizontal lines of symmetry.
Capital letter H looks the same when folded vertically or horizontally.

4. Correct Answer: (B) 2

New figure after reducing one pair of opposite sides by half:
When a square is changed by reducing one pair of opposite sides, it becomes a rectangle.
A rectangle has 2 lines of symmetry (one vertical and one horizontal).

5. Correct Answer: (B) (i) and (iii)

Rectangle with lines drawn

A rectangle has 2 lines of symmetry:

- One vertical line through the center
- One horizontal line through the center

Figure (i) shows the vertical line correctly.
Figure (iii) shows the horizontal line correctly.
Diagonal lines (ii) and (iv) are not lines of symmetry for a rectangle.

6. Correct Answer: (A)

In which figure are the lines of symmetry shown correctly?

The given shape (like capital I) has:

- One vertical line of symmetry
- One horizontal line of symmetry

Only option (A) shows both correctly.
Diagonal or side lines in other options are not correct lines of symmetry.

7. Correct Answer: (C) infinite lines

A circle has infinite lines of symmetry.

Any line passing through the center divides the circle into two equal halves.
Since there are unlimited directions, a circle has infinite lines of symmetry.

8. Correct Answer: (A) three lines

An equilateral triangle has three lines of symmetry.

Each line passes through a vertex and the midpoint of the opposite side.
So total 3 lines of symmetry.

9. Correct Answer: (B) 6

A regular hexagon has 6 lines of symmetry.

A regular hexagon has:

- 3 lines joining opposite vertices
- 3 lines joining midpoints of opposite sides

So total 6 lines of symmetry.

10. Correct Answer: (C) 2

The given star figure has 2 lines of symmetry.

The star shape is symmetric:

- Along the vertical line
- Along the horizontal line

It is not symmetric diagonally, so it has 2 lines of symmetry.

Worksheet – 2

1. Correct Answer: (B) symmetrical

An object and its image are ____ with respect to the mirror line.

A mirror forms a symmetrical (mirror) image of the object.

2. Correct Answer: (A) I

The mirror image of the letter I looks like.

Capital I looks the same in a mirror because it has vertical symmetry.

3. Correct Answer: (B) as if the direction has changed

The mirror image of the letter R looks ____ .

Letter R is laterally inverted in a mirror, so it looks reversed.

4. Correct Answer: (D) kaleidoscope

A ____ uses mirrors to produce images that have several lines of symmetry.

A kaleidoscope uses mirrors to create beautiful symmetric patterns.

5. Correct Answer: (D) the same

The mirror image of the given figure (circle with + sign) looks ____ .

The plus sign inside a circle has vertical and horizontal symmetry, so its mirror image looks the same.

6. Correct Answer: (C)

The correct mirror image of letter C is option ____ .

A mirror reverses the letter C horizontally. Option (C) shows the correct left-right reversed image.

7.

8. Correct Answer: (A)

The correct mirror image of letter A is ____ .

Capital A has vertical symmetry.
So its mirror image looks the same.

9. Correct Answer: (B) as if its direction has changed

The mirror image of the letter F looks ____ .

Letter F is not symmetric.
In a mirror, it appears reversed (laterally inverted).

10. Correct Answer: (A) 8 cm

If line segment PQ of length 8 cm is reflected in a mirror, the reflected length is ____ .

A mirror does not change size.
The image length remains equal to the original.

11. Correct Answer: (B) Looks the same after rotation about a fixed point

A figure has rotational symmetry if it ____ .

If a figure matches itself after rotating around a point, it has rotational symmetry.

12. Correct Answer: (C) Centre of rotation

The fixed point about which a figure rotates is called the ____ .

The point around which a figure turns is called the centre of rotation.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: (A) n

The number of lines of symmetry of an n-sided regular polygon is ____ .

A regular polygon has the same number of lines of symmetry as its sides.
Example:

- Square (4 sides) → 4 lines
- Regular hexagon (6 sides) → 6 lines

2. Correct Answer: (B)

Find the vertical mirror image of B.

A vertical mirror reverses the letter left to right.
So B appears laterally inverted.

3. Correct Answer: (B) 2

Find the number of lines of symmetry for the given shape.

The figure is symmetric:

- Along the vertical line
- Along the horizontal line

It is not symmetric diagonally, so total 2 lines.

4. Correct Answer: (A) AD

4. In ΔABC, AB = AC. About which line is the triangle symmetrical?

Since AB = AC, the triangle is an isosceles triangle.
The line drawn from vertex A to the midpoint of BC (that is AD, where AD ⊥ BC) divides the triangle into two equal halves.
So the triangle is symmetrical about AD.

5. Correct Answer: Diagonal AC

ABCD is a kite in which AB = AD and BC = DC. The kite is symmetrical about ____ .

In a kite, the line joining the vertices where equal sides meet
(Here, A and C)
So symmetry line is AC.

6. Correct Answer: (D) 0

Determine the number of lines of symmetry for the given shape.

The shape looks like two curved parts joined in opposite directions.
It does not match itself along vertical, horizontal, or diagonal lines.
Therefore, it has no line of symmetry.

7. Correct Answer: (C).

In option C,

- The slanting arrow is correctly reflected across line l.
- The vertical segment and arrow directions are properly reversed vertically.
- The figure matches perfectly along the horizontal line.

8. Correct Answer: (C)

Which figure is symmetrical with respect to exactly two lines?

- A (star) → more than two lines
- B (tree) → only one vertical line
- C (double arrow) → 2 lines (horizontal and vertical)
- D (square) → 4 lines

9. Correct Answer: (C) 360°

Every figure has at least one angle of rotational symmetry.

Every figure matches itself after a full turn.
So 360° is always an angle of rotational symmetry.

10. Correct Answer: (D) 0

Identify the number of lines of symmetry of the given triangle (3 cm, 5 cm, 7 cm).

All sides are different (scalene triangle).
A scalene triangle has no line of symmetry.

11. Correct Answer: (A) 1

The number of line(s) of symmetry in a compass is

A compass has one vertical line of symmetry when closed.

12. Correct Answer: (A) 1

The number of line(s) of symmetry in a 45°–45°–90° set square is

It is an isosceles right triangle.
It has one line of symmetry.

13. Correct Answer: (A) X

Which letter has both horizontal and vertical lines of symmetry?

Letter X has symmetry in both directions.

14. Correct Answer: (D) T

Which letter has only one line of symmetry?

Letter T has only one vertical line of symmetry.
H and X have more than one, Z has none.

15. Correct Answer: B

The figure that is not symmetric with respect to any line is

A (Rectangle) – 2 lines of symmetry

B (Irregular shape) – No line of symmetry

C (Isosceles triangle) – 1 line of symmetry

D (Circle) – Infinite lines of symmetry

16. Correct Answer: 5

The number of lines of symmetry in the given figure is

(The flower has 5 equal petals, so it has 5 lines of symmetry.)

17. Correct Answer: C) 3

Explanation:

- 0 – 1 line of symmetry
- 8 – 2 lines of symmetry
- 3 – No line of symmetry
- 9 – No line of symmetry

If you want, I can also frame it in textbook format for Class 6.

18. Correct Answer: A) AB and EF

From the figure:

- AB is the horizontal line through the center.
- EF is the vertical line through the center.
- CD and GH are diagonal lines.

The figure (circle with equal design inside a square) is symmetric along:

Horizontal line (AB)
Vertical line (EF)

The diagonals do not divide the design into identical halves.

19. Correct Answer: B)

Be on the opposite side of the mirror line (right side of RS).

- Be the mirror image (reversed shape).
- Keep the same distance from the mirror line.

Now checking options:

- A) Shape is not correctly mirrored.
- B) The figure appears on the right side and is correctly flipped.
- C) Still on the left side (not reflected).

D) Shape is different (not the same figure).

20. Correct Option: A) 1

Draw a vertical line through the center of the square.

- Left and right hearts match.
- The center heart remains the same.

No other line (horizontal or diagonal) gives matching parts.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

21. Correct Answer: A) 1 C) 3-2

Number of lines of symmetry

The figure has three overlapping circles placed in a row.

- It is symmetrical only along the vertical line through the center circle.
- It is not symmetrical horizontally or diagonally.

22. Correct Answer: B) Vertical, D) The figure has exactly one line of symmetry.

Line of symmetry

The tree figure matches exactly on both left and right sides.

- It has a vertical line of symmetry only.

23. Correct Answer: C) The figure has no line of symmetry, D) The figure is asymmetrical

Correct statement

The given shape does not match on folding vertically or horizontally.

- It has no line of symmetry.

24. Correct Answer: A, C and D

Figures with more than one angle of symmetry

- (A) Circle divided into 4 equal parts → more than one

- (C) Y-shaped figure → more than one
- (D) Four-petal shape → more than one

Triangle (B) does not have more than one.

25. Correct Answers: A and c

Select all correct statements

The figure looks like two identical triangles joined at the center (mirror image left and right).

- It can be divided into two equal halves by a vertical line.
- It has rotational symmetry of order 2 (looks same after 180° rotation).

26. Correct Answers: A, B, C, and D
(A square has rotational symmetry of order 4, meaning it looks the same at all four of these increments.)

A square is highly symmetrical. Because all its sides and angles are equal, it maps onto itself every quarter-turn.

A) 90° rotation: A quarter turn looks identical. (Correct)

B) 180° rotation: A half turn looks identical. (Correct)

C) 270° rotation: A three-quarter turn looks identical. (Correct)

D) 360° rotation: Every shape has 360° rotational symmetry (returning to start), but for a square, it is the 4th position of its rotational order. (Correct)

ASSERTION AND REASON TYPE

27. Correct Answer: C (Assertion is true, but Reason is false.)

- Assertion (A) is true.
- Reason (R) is false (Line symmetry is not rotational symmetry).

28. Correct Answer: A (Both are true, and Reason is correct explanation.)

- Assertion (A) is true (A regular hexagon has 6 lines of symmetry).
- Reason (R) is true (All sides and angles are equal).
- Equal sides and angles explain why it has multiple lines of symmetry.

29. Correct Answer: A

- Assertion (A) is true (Capital 'A' has a vertical line of symmetry).
- Reason (R) is true (Vertical symmetry means folding left to right into matching halves).
- Reason correctly explains the assertion.

30. Correct Answer: C (Assertion is true, but Reason is false.)

- Assertion (A): True (A circle has infinite lines of symmetry).
- Reason (R): False (A line of symmetry divides a shape into equal, not unequal parts).

31. Correct Answer: C (Assertion is true, but Reason is false.)

- Assertion (A): True (A square has both line and rotational symmetry).
- Reason (R): False (A square looks the same at 90°, 180°, 270°, not only 180°).

MATRIX MATCHING TYPE

32. (a) Equilateral triangle → (ii) 3 lines of symmetry
(b) Rectangle → (iv) 2 lines of symmetry
(c) Scalene triangle → (i) No symmetry
(d) Square → (iii) 4 lines of symmetry

33. (a) M → (i) Vertical symmetry
(b) B → (ii) Horizontal symmetry
(c) X → (iv) Both vertical and horizontal symmetry
(d) F → (iii) No line of symmetry

COMPREHENSION TYPE

Passage - I

34. Correct Answer: B) Line symmetry

A figure that can be divided into two identical halves has:

35. Correct Answer: C) Line of symmetry

The dividing line in a symmetrical figure is called:

36. Correct Answer: B) Symmetrical

If one half of a figure matches the other half exactly after folding, the figure is:

Passage-2

37. Correct Answer: B) 1

A kite has 1 line of symmetry (vertical line).

38. Correct Answer: B) 2

A rectangle has 2 lines of symmetry (one vertical and one horizontal).

39. Correct Answer: C) 4

A square has 4 lines of symmetry (2 diagonals, 1 vertical, 1 horizontal).

INTEGER TYPE

40. A parallelogram has 0 line(s) of symmetry.

41. A rhombus is symmetrical about its diagonals.

42. A line of symmetry of a line segment is also called its perpendicular bisector.

43. A line of symmetry of an angle is also called its angle bisector.

Previous Year Questions

1. Correct Answer: B

Exactly three lines of symmetry

- **A (Pentagon)** → A regular pentagon has **5 lines** of symmetry.
- **B (Three-petal shape)** → Has **3 lines** of symmetry (each through one petal).
- **C (Double arrow)** → Has **2 lines** (horizontal and vertical).

2. Correct Answer: C (Rectangle)

Exactly two lines of symmetry

- Equilateral triangle → 3 lines
- Isosceles triangle → 1 line
- Rectangle → 2 lines
- Square → 4 lines

3. Correct Answer: B

More than two lines of symmetry

- **(i) Star** → 5 lines
- **(ii) Square** → 4 lines
- **(iii) Triangle (not equilateral)** → 1 line
- **(iv) I-shape** → 2 lines

Figures with more than two lines:
(i) and (ii) only

4. Correct Answer: B (4

Solution (Answer: Option B – 4)

Letters given: **P A R K I N G**

Check lines of symmetry (standard block letters):

- **P** → No line of symmetry
- **A** → 1 vertical line
- **R** → No line
- **K** → No line
- **I** → 1 vertical line (in simple block form)
- **N** → No line
- **G** → 1 horizontal line (in some standard forms)

Letters with no line of symmetry:
P, R, K, N

Total = **4**

5. Correct Answer: A (1)

Solution (Answer: Option A – 1)

The figure is meant to be symmetric about the **vertical middle line**.

Check the shaded squares on the left and right sides:

- Almost all shaded squares already have their mirror image.
- Only **one square** does not have a corresponding shaded square on the opposite side.

So, by shading **just 1 more square** in the matching position, the figure becomes symmetric.

6. Correct Answer: B. 2

Find the number of lines of symmetry

The given figure is a square grid pattern.

On observing the shaded arrangement carefully:

- It is symmetric **vertically** (left and right match).
- It is symmetric **horizontally** (top and bottom match).
- It is **not symmetric diagonally** (corner parts do not match).

Therefore, the figure has **2 lines of symmetry**.

7. Correct Answer: B. 3

How many symmetrical shapes did Seema draw?

There are 5 shapes shown.

Checking each shape:

1. First shape – symmetric ✓
2. Second shape – not symmetric ✗
3. Third shape (parallelogram) – not symmetric ✗
4. Fourth shape (flower-like) – symmetric ✓
5. Fifth shape (tree) – symmetric ✓

Total symmetric shapes = **3**

8. Correct answer: D) P, Q ,R and s

Which figures have at least one line of symmetry?

Check each:

- **P** – Star shape → symmetric ✓
- **Q** – Star (hexagram) → symmetric ✓
- **R** – Clover-like shape → symmetric ✓
- **S** – Three-blade shape → no line symmetry ✓

So correct figures **P, Q ,R and s**

But since that option is not directly listed, the closest correct option given is:

However, based on standard symmetry rules:

9. Correct Answer: D. Q only

The square opposite **Q** across both diagonals does **not match**, so **Q must be shaded**.

- After shading Q, its reflections across LM and AB match the existing shaded pattern.
- Shading P, R, or S does not satisfy symmetry for both diagonals.

Therefore, only **Q** must be shaded.

10. Correct Answer: C

A. A regular pentagon has only five lines of symmetry.
✓ True – A regular pentagon has exactly **5** lines of symmetry.

B. A square has four lines of symmetry.
✓ True – A square has **4** lines of symmetry (2 diagonals, 1 vertical, 1 horizontal).

C. A pentagonal prism has 10 edges.
✗ False –
A pentagonal prism has:

- 5 edges on the top
- 5 edges on the bottom
- 5 vertical edges

Total = **15 edges**, not 10.

D. A tetrahedron has 4 faces.
✓ True – A tetrahedron has 4 triangular faces.

12. Visualizing Solid Shapes

Solutions

Worksheet – 1

1. **Correct Answer: B) Cylinder.**

   Solution:
   The given figure has two circular faces (top and bottom) and one curved surface.
   A solid shape with two circular bases and one curved surface is called a **cylinder**.

2. **Correct Answer: A) Cuboid.**

   Solution:
   The given figure is a box-shaped solid with rectangular faces.
   It has 6 rectangular faces, 12 edges, and 8 vertices.
   A solid shape with all rectangular faces is called a **cuboid**.

3. **Correct Answer: C) Sphere.**

   Solution:
   The given figure is round like a ball.
   A solid shape that is perfectly round with no edges and no vertices is called a **sphere**.

4. **Correct Answer: A) Cube.**

   Solution:
   The given figure has 6 equal square faces.
   All edges are equal in length.
   A solid shape with 6 equal square faces is called a **cube**.

5. **Correct Answer: C) Cone**

   Solution:
   The given figure has one circular base and one curved surface that meets at a point (vertex).
   A solid shape with one circular base and one vertex is called a **cone**.

6. **Correct Answer: D) Pyramid.**

   Solution:

   The given figure has triangular faces that meet at a single point (apex).

   A solid shape with triangular faces meeting at one point is called a pyramid.

7. **Correct Answer: A) 8.**

   Solution:

   A cube has 8 corners.

   These corners are called vertices.

8. **Correct Answer: B) 12.**

   Solution:

   A cube has 12 edges (line segments where two faces meet).

9. **Correct Answer: C) 6.**

   Solution:

   A cube has 6 square faces.

10. **Correct Answer: C) 5.**

    Solution:

    A square pyramid has:

    1 square base

    4 triangular faces

    Total faces = 1 + 4 = 5

Worksheet – 2

1. **Correct Answer: A) Triangular pyramid.**

   Solution:
   The given net is made up of triangular faces joined together.
   When folded, the triangles meet at a point to form a **triangular pyramid (tetrahedron)**.

2. **Correct Answer: B) Square pyramid.**

   Solution:
   The net shows a square base with triangular faces attached to its sides.
   When folded, the triangles meet at one point above the square base.
   This forms a **square pyramid**.

3. **Correct Answer: C) Hexagonal pyramid.**

Solution:
The net shows a hexagon in the centre with triangular faces around it.
When folded, the triangles meet at one point above the hexagonal base.
This forms a **hexagonal pyramid**.

4. **Correct Answer: A.**

Solution:
A triangular prism has:
- 2 triangular faces
- 3 rectangular faces

Correct Answer: **A** shows two triangles and three rectangles arranged correctly to form a triangular prism.

5. **Correct Answer: B.**

Solution:
A tetrahedron has:
- 4 triangular faces

Correct Answer: **B** shows four triangles arranged properly to fold into a tetrahedron (triangular pyramid).

6. **Correct Answer: C**

Solution:
A cuboid has:
- 6 rectangular faces

Correct Answer: **C** shows six rectangles arranged in a cross shape, which is a correct net of a cuboid.

7. **Correct Answer: C) Cylinder.**

Solution:
The net shows:
- One rectangle (curved surface when folded)
- Two circles (top and bottom bases)

This forms a **cylinder**.
Therefore, the correct answer is

8. **Correct Answer: D) 15.**

The given solid is a **pentagonal prism**.

In a pentagonal prism:
- The base has **5 edges**.
- The top face also has **5 edges**.
- There are **5 vertical edges** joining the top and bottom faces.

Total number of edges =
5 (bottom) + 5 (top) + 5 (vertical) = **15**

9. **Correct Answer: C) A pyramid.**

Solution:
The given net shows a square in the centre (dotted lines) with four triangles attached to its sides.

When folded:
- The square becomes the base.
- The four triangles meet at one point (apex).

This forms a **pyramid** (specifically a square pyramid).

10. **Correct Answer: D) Triangular pyramid.**

Solution:
The figure shows a large triangle divided into smaller triangles.
When folded along the dotted lines, the triangular faces meet at one point.

This forms a triangular pyramid

(tetrahedron).

Worksheet – 3

1. **Correct Answer: C**

Solution:
The solid has a rectangular shape with a small rectangular cut at the bottom front.
From the front view, we see a rectangle with a notch at the bottom.
Correct Answer: **C** matches this view.

2. **Correct Answer: A**

Solution:
The given solid is a pyramid.
From the top view, we see a square with diagonals meeting at the centre.
Correct Answer: **A** shows this correctly.

3. **Correct Answer: C) Sphere**

Solution:
A sphere gives a circular cross-section when cut horizontally or vertically.
Other solids do not always give a circle in both types of cuts.

4. **Correct Answer: B) Cuboid**

Solution:
A cuboid, when cut horizontally or vertically, can give a square (if the length and breadth are equal).
Other given solids cannot give a square cross-section in both types of cuts.

5. **Correct Answer: C) Cylinder**

Solution:

- When a cylinder is placed in front of light (side view), it casts a **rectangle** shadow.
- When placed above the light (top view), it casts a **circle** shadow.

6. **Correct Answer: D) Circle**

Solution:
When light is directly above a cone, the shadow formed on the ground is the shape of its base.
The base of a cone is a **circle**.

7. **Correct Answer: C) Cylinder**

Solution:

- From the side, a cylinder looks like a **rectangle**.
- From the top, it looks like a **circle**.

8. **Correct Answer: D) Rectangle**

Solution:
A brick is shaped like a cuboid.
A horizontal cut across a cuboid gives a **rectangle**.

9. **Cross-section when you give a vertical cut to a round apple — Circle**

Solution:
A round apple is approximately spherical.
When you cut it vertically through the centre, the cross-section is a **circle**.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: A) Edge**

The joint between separate faces of a solid is called an
(An edge is where two faces meet.)

2. **Correct Answer: B) 4 cm**

Two cubes of edge length 2 cm are placed side by side. The length of the resulting cuboid is
(2 cm + 2 cm = 4 cm)

3. **Correct Answer: C) 5**

3. The number of faces of a square pyramid is
(1 square base + 4 triangular faces)

4. **Correct Answer: A) Rectangle**

The curved face of a cylinder can be opened up as a
(The curved surface forms a rectangle when opened.)

5. **Correct Answer: C) A pyramid**

By folding the adjoining net, we get
(It shows a square in the center with four triangles around it — net of a square pyramid.)

6. **Correct Answer: D) Triangular pyramid**

Adjoining figure is
(It represents a triangular base with triangular faces — a triangular pyramid.)

7. **Correct Answer: A) 90°**

The angle between two adjacent edges of a cube and cuboid is
(Edges meet at right angles.)

8. **Correct Answer: C) Cone**

Which of the figure has two edges and one corner?
(A cone has 1 circular edge + 1 curved edge and 1 vertex.)

9. **Correct Answer: A) Face**

A sphere has only one
(A sphere has one curved face and no edges or vertices.)

10. **Correct Answer: C) Circle**

The top view of the given figure (cylinder) is
(When viewed from the top, a cylinder looks like a circle.)

If you need short explanation format for textbook, tell me.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. Explanation:

- A **sphere** has only a curved surface.
- A **cylinder** has one curved surface and two flat circular faces.
- A **cone** has one curved surface and one flat circular face.
- A **cube** has only flat faces (no curved surface)

12. Correct Answer: A) Cube B) Cuboid

Which of the following solids have exactly 6 faces?
(A cube has 6 square faces.)

13. Correct statements are:

A) It has 3 edges at the base.
B) It has 6 edges in total.
D) It has edges joining the base vertices to the top verte

Observe the given solid shape. Which of the following statements about its edges are correct?
(The figure is a triangular pyramid.)

14. Explanation:

- A **cylinder** gives a circular cross-section when cut parallel to its base.
- A **cone** gives a circular cross-section when cut parallel to its base.
- A **sphere** gives a circular cross-section when cut in any plane.
- A **pyramid** does not give a circular cross-sectio

15. Correct Answers:

B) The solid has 4 triangular faces on the top.
D) The total number of faces is 9.

Let us count the faces:

- A cube has 6 rectangular (square) faces.
- The top square face of the cube is covered by the pyramid base, so it is not exposed.
- The pyramid has **4 triangular faces** on the top.

So total visible faces:

- 5 rectangular faces (4 sides + 1 bottom)
- 4 triangular faces

Total faces = **9**

16. Correct Answers:
A) Cone
B) Cylinder
C) Sphere

From the top view:

- **Cone** → circular shadow
- **Cylinder** → circular shadow
- **Sphere** → circular shadow
- **Cube** → square shadow

ASSERTION AND REASON TYPE

17. Correct Answer: C (Assertion is true but Reason is false.)

Assertion: A cube has 12 edges. → **True**
Reason: All the faces of a cube are circular in shape. → **False**

18. Correct Answer: A (Both are true and Reason is the correct explanation.)

Assertion: A sphere has no vertex. → **True**
Reason: A sphere has a curved surface. → **True**
The curved surface explains why it has no vertex.

19. Correct Answer: B (Both are true but Reason is not the correct explanation.)

Assertion: A cylinder has two flat faces and one curved face. → **True**
Reason: The flat faces of a cylinder are circular. → **True**
But this does not explain why it has two flat faces and one curved face.

20. Correct Answer: C (Assertion is true but Reason is false.)

Assertion: A pyramid has a polygonal base and triangular lateral faces. → **True**
Reason: All 3D shapes have curved surfaces. → **False**

21. Correct Answer: A (Both are true and Reason is the correct explanation.)

Assertion: A cone has one edge and one vertex. → **True**
Reason: The edge of a cone is formed where the curved surface meets the circular base. → **True**
This correctly explains the edge.

MATRIX MATCHING TYPE

22. Question 1: Shapes and Features

Column A (Shape)	Column B (Feature)
(a) Cube	(ii) 6 faces, all squares
(b) Cylinder	(iii) 2 circular flat faces
(c) Cone	(i) One circular face
(d) Sphere	(iv) Only one curved surface

23.

Column A (Term)	Column B (Meaning)
(a) Face	(ii) A flat surface
(b) Edge	(iii) The line where two faces meet
(c) Vertex	(i) The corner where edges meet
(d) Solid shape	(iv) 3D object with volume

COMPREHENSION TYPE

Passage - I

24. Correct Answer: C) Rectangular

What type of faces does a cuboid have?
Explanation: A cuboid has 6 faces and all faces are rectangular.

25. Correct Answer: C) 12

How many edges does a cube have?
Explanation: A cube has 12 edges.

26. Correct Answer: C) 8

How many vertices does a cube have?
Explanation: A cube has 8 vertices (corners).

Passage – 2

27. Correct Answer: B) Ball

Which object has no edges or vertices?
Explanation: A ball is shaped like a sphere. A sphere has no edges and no vertices.

28. Correct Answer: C) 2

How many flat surfaces does the cold drink can have?
Explanation: A cylinder (can) has 2 flat circular faces and 1 curved surface.

29. Correct Answer: B) Cap

Which object has exactly one circular flat face?
Explanation: A cap is shaped like a cone. A cone has one circular flat face.

INTEGER TYPE

30. The number of faces of a square pyramid is 5.

31. A cube has 6 faces.

32. A cuboid has 12 edges.

33. A cone has 1 vertex.

34. A sphere has 1 curved surface.

Previous Year Questions

1. Option B → Contains 5 rectangles and 2 pentagons arranged correctly

The given solid figure is a **pentagonal prism** because:

- The top face is a **pentagon**.
- The bottom face is also a **pentagon** (shown with dotted lines).
- There are **5 rectangular lateral faces** joining corresponding sides of the two pentagons.

Now, the net of a pentagonal prism must contain:

- 2 pentagons
- 5 rectangles arranged in a strip (for the lateral faces)

2. Second solid

It is a hexagonal prism.

For an $\boldsymbol{n}$-sided prism:

- Edges = $\boldsymbol{3n}$
- Faces = $\boldsymbol{n + 2}$
- Vertices = $\boldsymbol{2n}$
- Given:
- Faces = 8

So,

$\boldsymbol{n + 2 = 8}$
$\boldsymbol{n = 6}$

Hence,

$\boldsymbol{Q = 3n = 3 \times 6 = 18}$

3. Correct Answer: A. 4

Third solid

Given:

Edges = 10

So,$\boldsymbol{2n = 10}$
$\boldsymbol{n = 5}$

Therefore,$\boldsymbol{R = n + 1 = 6}$

Now calculate:$\boldsymbol{Q - 2P - R}$
$\boldsymbol{= 18 - 2(4) - 6}$
$\boldsymbol{= 18 - 8 - 6}$
$\boldsymbol{= 4}$

4. A → Matches 3 on left, 1–1 in middle, and 2 on right

From the given 3D figure:

- The bottom row has 4 cubes.
- On the leftmost cube, there are two more cubes stacked (making height = 3).
- On the right side, there is one extra cube stacked (making height = 2).
- The middle cubes remain height = 1.

So from the front, the heights appear as:

$\boldsymbol{3, 1, 1, 2}$

Now compare with options:

5. Option A is incorrect because a cone has only 2 faces, not 3

Explanation:

- A **cone** has **2 faces** (1 flat circular base + 1 curved surface).
  It does **not** have 3 faces.
- A **sphere** has **1 curved surface** (often counted as one face in elementary level).
- A **cube** has **6 faces** ✓
- A **cylinder** has **3 faces** (2 flat circular faces + 1 curved surface) ✓

Therefore:

6.

7. Correct Answer: Option B — 4 cubes

First, count cubes in each figure.

Fig. (A)

- Bottom layer: 3 cubes
- One cube on top

Total in Fig. (A) = **4 cubes**

Fig. (B)

- Bottom row: 3 cubes
- Middle right: 2 cubes
- One cube above
- One more cube at back/right

Total in Fig. (B) = **8 cubes**

Cubes to be added:

$8 - 4 = 4$

8. Correct Answer: Option A — Faces = 5, Edges = 8

Faces:

- 1 square base
- 4 triangular faces

Total faces = **5**

Edges:

- 4 edges of base
- 4 side edges

Total edges = **8**

9. Correct Answer: A. Congruent

Prism is a polyhedron whose base and top are _____.

In a prism, the base and the top are **exactly the same shape and size**.

That means they are **congruent**.

10. Correct Answer: B. 11

The solid is a combination of rectangular blocks.

Let us count carefully:

- Front face
- Back face
- Left side face
- Right side face
- Top faces (3 visible sections on top)
- Bottom face
- Inner vertical face (step part)
- Right small block faces

After counting all distinct outer faces properly, the total comes to:

11 faces